

Supplementary Information

Permafrost active-layer thickness prediction under sparse observations using physics-based pseudo-label augmentation and machine learning

[DECISION: 저자, 소속, 교신 저자]

Contents. Supplementary Methods 1–9; Supplementary Notes 1–2; Supplementary Tables S1–S18; Supplementary Figs. S1–S9; Supplementary References. Values in the Supplementary Tables are copied from the evidence tables of the claims folders (paper/claims/*/README.md) or from the source tables they cite; no value was recomputed for this document. Blue bracketed text marks items that are not yet filled.

Supplementary Methods

Supplementary Methods 1. Label parsing rules and processing by data source

Labels. ALT labels of the main regions come from three compilations: GTN-P/CALM¹, the ALLena thaw-depth compilation for the Lena River Delta² and the ABoVE soil moisture and ALT data set for Alaska and Canada, version 2³. Records were aggregated to cells before modelling. For ABoVE and ALLena, a cell is a unique coordinate rounded to four decimal places, and the label is the mean of all observations at that coordinate. For CALM sites, a cell is one site and the label is the multi-year mean. ABoVE records were retained when $0 < \text{ALT} < 300$ cm. The version-3 cell table contains 17,572 rows, of which the analysis uses the 17,467 cells with direct labels. In version 3, direct labels are mechanical probing and GPR. CALM sites added in version 2 whose event metadata name thaw tubes or ground temperature were moved to a separate class and excluded. Twenty cells that the source event metadata identify as temperature-derived remained in the direct class (15 in Alaska, 4 in Canada, 1 on the Tibetan Plateau). The table was frozen before this was found. These cells are flagged in the version-4 label table and are handled like the probe-only sensitivity variant. Most Alaskan labels are GPR-derived (86 % of the 219,089 ABoVE point observations in Alaska and 7.7 % probe; GPR dominates in 10,835 of the 13,606 Alaskan cells; label-definition sensitivity E4.4 of the project log). Label years span 1990 to 2024 and are not matched to the covariate reference period. The resulting maps are static and refer to the 2015–2020 climatology.

Label unit. A label row has a different meaning in different regions. In Russia W and Russia E a row is the multi-year mean of one CALM site. In the Lena River Delta a row is a point observation at a rounded coordinate, usually from a single visit. In Alaska and Canada most rows are point observations and some are CALM site means. In the additional regions (below) a row is the mean over an approximately 1 km cell. The label count n in all figures and tables is the number of rows drawn. Table 1 reports for each region the number of rows, the number of distinct 1 km locations, the number of distinct ERA5-Land climate values and the number of 0.5° blocks. Statements that compare regions at equal n state the unit.

Scoring mask. A cell is scored only if its label, its CCI ALT value and its soil thawing degree-days are valid. The same scoring cells are used for all methods.

Supplementary Methods 2. Assembly and eligibility of additional regions (LGD, XF)

Additional labels were collected from public archives for regions at least 100 km from existing labelled regions (Supplementary Tables S9 and S11). The label rules were fixed before any model was fitted on these regions. A label enters as a target label only if it is a direct measurement (probe, thaw tube, frost tube, GPR, or soil pit, core or auger) of the end-of-season thaw depth. The end of season is established by the data protocol (CALM, thaw tubes), by the maximum over repeated visits after 1 August, by an observation date in August or September for single visits, or by an explicit statement in the data description. Thaw tubes therefore count as direct measurements in the additional regions, whereas in version 3 only the CALM sites added in version 2 with thaw-tube or ground-temperature event methods were excluded; among the other version-3 direct cells, one Alaskan and one Canadian cell match thaw-tube events and seven Canadian cells are ambiguous between probe and thaw tube (data README D, caveat 3). Values estimated from degree-day ratios or models are not used. Disturbed sites (burned areas, water bodies, thermo-erosional landforms) and censored probe readings are excluded. Only annual values from 1990 onward with $0 < \text{ALT} \leq 600$ cm are used. New records are aggregated to approximately 1 km cells, indexed by $\text{ky} = \lfloor \text{lat}/0.009 \rfloor$ and $\text{kx} = \lfloor \text{lon} \cdot \cos \varphi / 0.009 \rfloor$ with $\varphi = (\text{ky} + 0.5) \cdot 0.009^\circ$. New cells within 0.01° of an existing direct-label cell are removed.

Three target regions were defined geographically: Tibet (Tibetan Plateau and Qilian Mountains, 26–40°N, 73–105°E), NAtlantic (Greenland, Svalbard and mainland Norway, Sweden and Finland) and Russia_C (Russia between 90°E and 140°E, excluding the Lena River Delta window 71.5–73.6°N, 123.3–130.1°E). The North Atlantic grouping was fixed before any result because none of its parts was expected to meet the block criterion alone. Extended versions of Russia W, Russia E and Canada were built as sensitivity cases. A region is eligible for confidence intervals if at least one split is valid and the union of scoring blocks over the used splits contains at least 8 blocks. The regime is shallow if the mean ALT of the target cells is below 150 cm and deep otherwise. Eligibility and regime were determined from the split structure before any model was fitted on these regions. They were recorded with the hash of the input table (18,088 rows = 17,572 version-3 rows + 516 new rows; sha256 4c387b2b...). Tibet (132 cells, 34 blocks, deep regime), NAtlantic (41 cells, 13 blocks, shallow) and Russia_C (57 cells, 13 blocks, shallow) are eligible. Some NAtlantic splits have only three scoring blocks, and NAtlantic results carry a few-blocks flag. All Tibetan target cells lie above the source-cell elevation range. Contrasts against P0 in Tibet are therefore labelled as predicted outcomes and are not interpreted. Temperature-derived labels in Tibet (39 cells) are used only to test the effect of the label definition (L39).

Three pools are compared. P4 contains the four main regions (version-3 cells, main-run shards). PE1 adds the eligible shallow new regions, and PE2 adds the deep region. Hypotheses L1e, L4e and L8e repeat L1, L4 and L8 on PE1 and PE2 with the same decision rules. Hypotheses L38–L42 test region-level replication, the label definition, the extended versions of existing regions, label-set variants and the source composition. For PE2, stratified means are reported in cm and in units of each region’s P0 RMSE, because the deep region can dominate a mean in cm.

Licence status.

Licence status. Several sources carry no redistribution licence or ask to be consulted before publication (CALM web files of two sub-sites, the CUSP v1.1 synthesis, NSIDC GGD353 and ten report-derived rows of the Yamal transect; Data availability). A licence-verified version of every affected run table was defined before the pooled aggregation (LG revision 15 (m)). It removes all cells flagged as licence-unverified. Counted over target cells, it retains 19 of the 38 new NAtlantic cells, 37 of the 75 new cells of the Canada extension and 7 of the 8 new cells of the Russia W extension. The eligibility rules are applied unchanged to the licence-verified version. If permission for any source has not been obtained at submission, the primary PE1, PE2 and L38–L42 verdicts are reported from the licence-verified version, and the full version is placed in the Supplementary Information with a flag. This choice was fixed before results. No permission has been recorded (5 October 2026), so the primary verdicts are those of the licence-verified version.

XF.

The additional-region analysis XF applies the same label rules and eligibility rules, unchanged and without access to label values, to public data sets found after the main runs (survey in Supplementary Table S18), and repeats the core contrasts in each eligible region. New regions are labelled blind (independent-region confirmation); the North Atlantic licence-verified edition enters as a sensitivity case. Pending: registered data deadline 11 October 2026 (XF).

Supplementary Methods 3. Extension contrasts (LGX) and details of the combination structures

The extension refits the eight base methods with the same splits, draws and seeds, so that every extension contrast is closed within its own shards. A reproduction check compares the refitted base methods with the main-run shards (per-key cell-weighted RMSE; tolerance 1e-9 cm for physics methods and 0.02 cm for CatBoost methods). The extension covers physics outputs as input features (F1a, F1k, F1n), log-ratio regression (V2) and multiplicative correction (RM). It covers pseudo-label controls: shuffled, target-level constant, source-level constant and linear-TDD pseudo-labels, pseudo-label ratios $r \in 1, 3, 30$ and shuffled target labels. It covers a validation ladder on the pooled data: random cells, coarse covariate groups, 0.05° cells, 0.5° blocks, leave-one-cluster-out with 0, 100 and 500 km buffers, and region holdout. It also covers within-region block cross-validation, a proximity ladder at fixed scoring cells, cell-level predictions with distance strata, learner equivalence over five splits and a physics-baseline ladder (affine Stefan, a Kudryavtsev-type model⁴, a soil-property Stefan variant, the CCI ALT product and their ensemble). Finally, it covers an error floor from within-group variance, learner capacity (CatBoost with 600 iterations and depth 6, a random forest with 500 trees, and CatBoost tuned by leave-one-source-region-out validation) and sensitivity variants (κ , exclusion of June–July observations, probe-only labels, alternative anchors, inputs without the CCI product, year-matched TDD). The confirmatory

hypotheses are L10 (residual structure versus physics-input structure), L12 (additive versus multiplicative correction), L15 (Stefan pseudo-labels versus four placebo pseudo-labels), L19 (conclusions under random splits versus region holdout), L29 (existence of a physics or product baseline better than P0 at $n = 0$) and L30 (direct ML at $n = 0$ with higher-capacity learners). If L29 identifies a better baseline P^* , the contrasts of L4, L8 and L10 are repeated against P^* . The manuscript does not state that a learner improves on the physics baseline when it improves on P0 but not on P^* . Holm-adjusted p values within each item are reported as auxiliary columns.

Supplementary Methods 4. Prediction intervals (LGU)

The uncertainty experiment evaluates predictive distributions in held-out regions. Every method outputs 99 quantiles per scoring cell. Intervals, the continuous ranked probability score (CRPS, grid approximation) and pinball losses are computed from these quantiles⁵. The interval score of a central $(1 - \alpha)$ interval $[L, U]$ is

$$\text{IS}_\alpha = (U - L) + \frac{2}{\alpha}(L - y)_+ + \frac{2}{\alpha}(y - U)_+ \quad (\text{cm}). \quad (\text{S1})$$

Experiment A is a ladder of hierarchical predictive distributions for $u = \log y - \log(E_0 s)$, modelled as $u = a_r + b_{r,k} + \varepsilon$ with a region offset a_r , a block offset $b \sim N(0, \tau_b^2)$ and a cell residual ε . Hyperparameters are estimated from the source only. Each source region's offset is computed against the coefficient estimated without that region, so that source regions mimic a new target. The confirmatory stage (iii) uses a Student- t prior with 4 degrees of freedom for a_r and a half-normal hyperprior on its scale, integrated on a grid. The posterior of a_r given the n labels yields the predictive distribution in unlabelled blocks. Other stages use a normal prior, heteroscedastic residuals from a CatBoost scale model, a normalizing-flow residual distribution that is run only if it passes a registered gate, and a conformal post-correction. The main baseline B4 is centred on $E_n s$ with a constant log half-width. B4 is calibrated by treating each source region in turn as a pseudo-target with the same draw procedure. Experiment B compares normalizers of a hierarchical conformal interval at $n = 0$: constant width, normalizing flow, CatBoost multi-quantile regression, conditional flow matching and a physics-propagated width. Each normalizer is compared with a placebo that permutes it among cells. Calibration groups are source macro-regions with at least 20 cells, pooled with equal weight⁶. Experiment C reports spatial-information diagnostics without hypotheses.

Confidence intervals use a two-stage bootstrap. Blocks are resampled within macro-regions on both the calibration side and the scoring side, and the calibration quantities are recomputed in each replicate. The registration specified 2,000 replicates and a timing rule for reducing them; the rule was applied before the main run and set the count to 1,000. The equivalence margin for interval scores is 5 % of the baseline score (auxiliary 10 %). If the half-width of an interval exceeds the margin, equivalence is recorded as not testable. A lower interval score is used as support only if the 90 % coverage of the method is at least 0.80. The confirmatory hypotheses are LGU-A1 (stage (iii) versus B4 at $n \in 10, 40$, pooled over the Lena River Delta, Canada and Alaska) and LGU-B2 (flow normalizer versus its placebo at $n = 0$ in the four main regions). A statement that conditional interval width transfers between regions is made only if LGU-B2 meets its four registered conditions. With two to four calibration groups there is no finite-sample coverage guarantee, and coverage is reported as an empirical quantity.

Supplementary Methods 5. Tabular foundation models and source cross-validation tuning of neural networks (LGT, LGF)

TabPFN v2⁷ (package tabpfn 8.0.7 with the public v2 regression weights) and TabICL v2⁸ (package tabicl 2.0.2) were run on the same 27 targets, splits and draws with default settings and without fine-tuning. Both receive a context of at most 10,000 rows, the pre-training range of TabPFN v2. All n target labels enter the context. Source rows fill the remaining budget of $10,000 - \max(T, 1,000)$ rows, where T is the number of target rows, by a region-proportional nested subsample. The source context is therefore identical for all $n \leq 1,000$. E_0 is always computed from the full source set. Each foundation model is paired with CatBoost fitted on the identical context in the same process. TabICL is additionally run with the full source set as context. The methods are D0, R0 and R1. The augmentation methods (D1, R2, R3) are omitted because the context limit does not allow the pseudo-label ratio of the main experiment.

The tuning experiment re-selects the hyperparameters of the MLP, the multi-head MLP, the reduced FT-Transformer-type model and RealMLP using source data only. For each target and learner, a random search⁹ over 33 configurations including the default (34 for the FT-Transformer-type model, which adds an approximation of its published default; 32 random trials each after extension E1 of LGF revision 5) is scored by leave-one-source-region-out cross-validation with equal region weight. The best candidates are refitted with two

new seeds. A non-default configuration is selected only if it improves the fold-wise score consistently over the default, which limits selection bias¹⁰. Target labels and scoring cells are never used for selection. Default and tuned versions are both refitted on the local server, so that each comparison is closed on one platform. The confirmatory hypotheses are LGF-F1 (TabICL direct prediction at $n = 0$ under both context conditions versus P0) and LGF-N1 (tuned direct prediction of each of the four neural learners at $n = 0$ versus P0). Both hypotheses have the form “not superior”. Support is split into two classes: inferior or equivalent in all contrasts, and no evidence of superiority when a contrast is indeterminate. Only the first class enters statements that a conclusion holds across learner types. Statements about tabular foundation models as a class are limited to the two models tested.

Supplementary Methods 6. Cross-environment rules and registration times

Two execution environments were used (Supplementary Table S13). The main experiment and the extension runs LGX-A and LGX-B ran on cloud nodes (Rescale iolite-4: 64 CPU cores and four NVIDIA T4 GPUs). LGT, LGF, LGU, LGD, the local continuation of LGX, the LGX aggregation, the WRAPUP aggregation and the map were run on a shared local server with NVIDIA RTX 3090 GPUs (24 GB). The LG aggregation is taken from the cloud job if it completed without error and is otherwise repeated locally; the main-run job completed on 30 September 2026 (12:25 KST) with every step, including its aggregation, ending without error, so its aggregation was used. New work was moved to the local server following an instruction on 30 September 2026 to avoid further cloud cost. The extension run LGX-A was stopped at 09:02 KST on 30 September. At that time all CPU shards were complete, and 58 of 572 GPU shards, all of the FT-Transformer-type learner, were missing. Primary verdicts are computed from the cloud shards alone. Learner-equivalence verdicts that require the missing splits are recorded as not determined over five splits, and the three-split verdict of LG is retained. All GPU shards of the main run were complete (231 of 231, no RealMLP unit missing), so no local continuation was needed and the RealMLP rows are judged over complete splits.

Four comparison rules were fixed before results. First, both sides of a contrast $\Delta(A - B)$ come from shards of the same platform. Second, verdicts that combine split distributions use shards of one platform; if platforms are mixed, the verdict is labelled “cross-environment” and the single-platform verdict is reported alongside. Third, pooled means that combine regions run on different platforms use only within-region contrasts and carry the cross-environment label; PE1, with P4 from the cloud and the new regions from the local server, is such a pool. Fourth, cross-platform comparison tables are descriptive and are not used for verdicts. Contrasts involving V1r are never formed across platforms.

Three cross-environment checks were defined. Check (i) refits four CPU units of the cloud pre-check and smoke runs on the local server with the same settings. It compares per-key cell-weighted RMSE with a tolerance of $1\text{e-}9$ cm for physics methods and 0.02 cm for CatBoost and ridge methods. Physics keys agreed to 2.8×10^{-14} cm (46 keys) and CatBoost keys without the V1r anchor to 1.4×10^{-14} cm (398 keys), but 11 of 39 V1r keys differed by more than 0.02 cm (at most 0.46 cm) and one of 83 ridge keys exceeded the tolerance (at most 0.0205 cm); the check passed for physics methods and failed for CatBoost and ridge methods (LG revision 14). Check (ii) reruns the extension aggregation of the pre-check shards on the local server. It requires point estimates and interval limits identical to within $1\text{e-}6$ cm and identical verdict strings. It passed: in the nine tables with values the largest absolute difference over 204 numeric columns was 1.4×10^{-14} , and none of 121 string columns, verdict strings included, differed; the path that builds the LG auxiliary columns is not run in the pre-check and was not covered. Check (iii) refits four GPU units of the FT-Transformer-type learner twice on the local server and compares them with the cloud shards. The local continuation of the missing shards proceeds only if the physics keys agree to within $1\text{e-}9$ cm and one of three registered rules holds for the neural keys. Rule (a) requires that no key differs from the cloud value by more than 0.5 cm. Rule (b) applies when the two local refits differ (median absolute difference above $1\text{e-}6$ cm) and requires the median absolute difference between local and cloud values to be at most three times the median difference between the two local refits. Rule (c) applies when the local refits are identical to within $1\text{e-}6$ cm and requires a median absolute difference between local and cloud values of at most 0.1 cm, with at most 5 % of keys differing by more than 0.5 cm. Physics keys agreed to 8.9×10^{-15} cm; the two local refits of the 288 FT-Transformer keys were identical, and local and cloud values differed by a median of 0.059 cm and at most 5.61 cm, with 43 keys (14.9 %) above 0.5 cm. None of rules (a) to (c) held, so the local continuation was not run (LG revision 15 (u)) and the FT-Transformer rows keep the three-split verdict of LG. Verdicts that use locally continued shards are auxiliary, carry the cross-environment label and are not used in confirmatory statements or in the abstract. GPU refits were identical within the local platform but differed between platforms (check (iii)). CatBoost did not pass check (i), so pooled rows that combine platforms carry a cross-environment flag (Methods, deviations).

Registered experiments.

The table below lists the experiments of the label-grid family with their registration commits (committer times, Korea Standard Time; commits were not time-stamped by a third party). The workflow experiments were registered as WF0–WF5 in 4168331 (1 October 2026), WF6–WF8 in 1b42b4d (1 October 2026), WF9 and WF10 in 8180632 (2 October 2026) and the soil-grouping sensitivity in 7da0064 (2 October 2026), and the analyses XA to XJ in 2678100 (4 October 2026, 14:22:39) (Methods).

Experiment Question		Execution environment	Registration commit (KST)
LG	Error of 12 methods as a function of n in 27 targets (hypotheses L1–L8)	Cloud (Rescale, NVIDIA T4)	40be64c, 2026-09-29 15:45:46; run submitted with 19f3f93, 17:04:45
LGX	Combination forms, pseudo-label controls, validation ladder, learner equivalence, sensitivity (L9–L31; confirmatory L10, L12, L15, L19, L29, L30)	Cloud; aggregation and auxiliary continuation local	8513cc2, 2026-09-29 18:17:29; operational definitions f29bfc8, 21:28:41
LGD	Additional independent regions (L1e, L4e, L8e, L38–L42)	Local CPU	6799b74, 2026-09-29 22:43:12; input tables and eligibility 896a15e, 2026-09-30 02:35:11; run and aggregation rules 028900a, 04:19:23
LGT	TabPFN v2 on the label-count grid (L32–L37)	Local, NVIDIA RTX 3090	6799b74, 2026-09-29 22:43:12; operational definitions 45ef3b6, 2026-09-30 01:39:42
LGU	Predictive distributions and interval evaluation (LGU-A1–A7, B1–B6, C1–C4; confirmatory A1, B2)	Local	a112cef, 2026-09-29 23:30:02
LGF	TabICL v2 and source-tuned neural networks (LGF-F1–F6, N1–N4; confirmatory F1, N1)	Local, NVIDIA RTX 3090	3a4fe84, 2026-09-30 01:44:27; revision 1 f208516, 02:22:24
WRAPUP	Reporting rules, abstract bundle AB1–AB10, scenarios SC1w–SC3w, placement L43, AK1w, map	Local	316714c, 2026-09-30 03:02:42

Registration events and later amendments.

Plan (section)	Hypotheses	Registered	Later amendments (no change of hypothesis wording)
LG §4	L1–L8	40be64c 2026-09-29 15:45:46 (initial text with revisions 1–2)	revision 3 249005c 16:35:29; revision 4 19f3f93 17:04:45 (ZovWo submitted); revision 6 d8355d2 18:18:36 (D0 description “no physics anchor”)
LG 6A (LGX)	L9–L31	8513cc2 2026-09-29 18:17:29 (revision 5)	revisions 7–8 f29bfc8 21:28:41; revision 11 (first committed as “revision 9” in 6799b74 22:43:12, renumbered in cec5719 23:29:45); revision 15 (h) ff1be02 2026-09-30 11:44:30
LG 6C (LGT)	L32–L37	6799b74 2026-09-29 22:43:12 (revision 9; section written 21:45)	revision 12 45ef3b6 2026-09-30 01:39:42
LG 6B (LGD)	L1e, L4e, L8e, L38–L42	6799b74 2026-09-29 22:43:12 (revision 10; section written 22:42)	revision 13 896a15e 02:35:11; revision 14 028900a 04:19:28; revision 15 (m) ff1be02 11:44:30
LGU	A1–A7, B1–B6, C1–C4	a112cef 2026-09-29 23:30:02 (written 22:45)	revisions 1–3 582be9a 2026-09-30 03:36:00; revision 4 ff1be02 11:44:30
LGF	F1–F6, N1, N2, N2s, N3, N4	3a4fe84 2026-09-30 01:44:27 (written 01:30)	revision 1 f208516 02:22:24; revisions 2–4 43df2f8 04:02:52; revision 5 b1a4349 05:00:40; revision 6 ff1be02 11:44:30
WRAPUP	L43, SC1w–SC3w, AK1w, AB1–AB10	316714c 2026-09-30 03:02:42 (initial draft with revision 1)	revision 2 ff1be02 11:44:30

Supplementary Methods 7. Label unit and label year sensitivity (S-a, S-b) and the sequential stopping rule (SC3w)

S-a, label unit.

S-a compared recalibration with point labels and with 1 km location means in Lena, Canada and Alaska (mode x). We used split seeds 1–200 with the split rules of the label grid (valid splits: Lena 197, Canada 200, Alaska 200), five draws per split and $n \in 3, 10, 40$. In the row condition, n label rows were drawn from the target half A as in the label grid. In the location-mean condition, the rows of A were first averaged within 1 km locations (mean ALT and mean $\sqrt{\text{TDD}}$), and n locations were drawn. In both conditions, E_ls was the slope through the origin of the drawn labels and $E_n = (n E_ls + 10 E0)/(n + 10)$. Both conditions were scored on the same B-half rows. For each split we averaged $\Delta(P1 - P0)$ over draws and computed $\Delta_unit = \Delta_location\text{-mean} - \Delta_row$. In Canada at $n = 40$, the location-mean condition was skipped in the 100 splits where A had at most 40 locations. Median Δ_unit [10th, 90th percentile] was +0.31 [−0.16, +1.37] cm ($n = 3$) and +0.34 [−0.26,

+1.99] cm ($n = 10$) in Lena, $-0.47 [-1.76, +0.81]$ cm and $-0.86 [-3.48, +1.05]$ cm in Canada, and $+0.07 [-0.19, +0.41]$ cm and $+0.09 [-0.36, +0.58]$ cm in Alaska. At $n = 40$ (descriptive), the medians were $+0.56$ cm (Lena), -0.96 cm (Canada, 100 splits) and $+0.08$ cm (Alaska). Following the registered rule, the Canadian median at $n = 10$ (below -0.5 cm) gives the sentence: “With point labels, the gain from few-label recalibration was smaller than with 1 km location means (Canada, $n = 10$, -0.86 cm)”. The deployment guideline therefore states its label unit as “1 km location mean”. Lena and Alaska stayed within ± 0.5 cm at $n \in 3, 10$ with the opposite sign, so the sentences are written per region. No median exceeded $+0.5$ cm at $n \in 3, 10$. At $n \in 3, 10$, the 10–90 % intervals of Lena and Canada did not overlap the Russia W reference interval in either condition, so the sentence “part of the between-region difference in recalibration gain is explained by the label unit” is not written. With B also averaged to 1 km locations (auxiliary), the medians at $n = 10$ were -0.65 cm (Canada), -0.17 cm (Lena) and -0.02 cm (Alaska).

S-b, label year.

S-b replaced the multi-year CALM site means of Russia W and Russia E by single-year values. From the CALM raw data (PANGAEA 972777; 3,819 rows), we removed values with inequality signs (24 “>” and 10 “<”), inactive rows (117) and empty values (122), leaving 3,663 rows (the conditions can overlap). Label cells were matched to CALM sites by coordinates within 0.005° in latitude and longitude. Of the 31 Russia W cells, 27 matched one site, 4 matched several sites and none was unmatched. Of the 30 Russia E cells, the counts were 27, 3 and 0. Cells with multiple matches kept their multi-year means. For the 54 single-match cells, the multi-year mean of the valid years reproduced the label value (maximum absolute difference below $1e-13$ cm). The median number of valid years was 13 (Russia W) and 20 (Russia E). For split seeds 1–200, five draws and $n \in 3, 10$, one observation year was drawn for each drawn site, E_{ls} and E_n were computed from the single-year values, and scoring used the multi-year means of the B half. $\Delta_{year} = \Delta(P1 - P0)_{single-year} - \Delta(P1 - P0)_{multi-year}$ was paired within split and draw. The share of drawn labels replaced by a single-year value was 0.88 (Russia W) and 0.90 (Russia E). Median Δ_{year} [10th, 90th percentile] was $+0.02 [-0.62, +0.63]$ cm ($n = 3$) and $+0.05 [-0.48, +0.52]$ cm ($n = 10$) in Russia W, and $+0.01 [-0.19, +0.21]$ cm and $+0.04 [-0.25, +0.28]$ cm in Russia E. Both medians stayed within ± 0.5 cm at both n , so the registered sentence applies: “The recalibration gain from 3–10 labels was maintained with single-year labels.”

SC3w, sequential stopping rule.

SC3w tested a sequential rule that decides from the target labels alone whether to stop at 3, 10 or 40 labels. For each valid split (1–5) and each of 20 repetitions, a random permutation of A fixed nested label sets of 3, 10 and 40. At 3 and 10 labels, the jackknife standard error of log E (leave-one-label-out least-squares slopes before shrinkage) was compared with $\tau = 0.05$. The rule stopped when the standard error was at most τ and otherwise continued, with 40 labels as the cap. The stopped stage gave E_n ($\kappa = 10$) and P1. The comparison was P1 from the first 40 labels of the same permutation. $\Delta_{SC3} = \text{RMSE}(P1_{rule}) - \text{RMSE}(P1_{40})$ was scored on the B half with 10,000 block bootstrap resamples, and the label ratio was the mean number of labels used divided by 40. Support required (i) label ratio ≤ 0.5 , (ii) a three-region mean point estimate ≤ 0.3 cm, (iii) both CI upper limits of the three-region mean < 0.5 cm and (iv) no inferior region. A label ratio ≥ 0.9 was classified as “rule not operating”. Values $\tau \in 0.025, 0.10, 0.20$ were descriptive sensitivities. At $\tau = 0.05$ the rule was classified as not operating. The three-region mean label ratio was 0.909 (condition (i) failed; conditions (ii)–(iv) held). The registered sentence is: “At the main τ of 0.05 the stopping rule rarely operated (label ratio 0.91)”. By the registered rule, Δ_{SC3} is descriptive and the rule is not part of the deployment guideline. The shares of repetitions stopping at 3, 10 and 40 labels were 5, 0 and 95 % in Lena, 7, 0 and 93 % in Canada, and 11, 8 and 81 % in Alaska. The three-region mean Δ_{SC3} was -0.048 cm $[-0.079, -0.002]$ (block-equal -0.077 cm $[-0.105, -0.049]$), no region was inferior, and each region was equivalent. For $\tau = 0.025$ the label ratio was 0.988 (not operating). For $\tau = 0.10$ it was 0.542, and only condition (i) failed. For $\tau = 0.20$ it was 0.185 and all four conditions held ($\Delta_{SC3} -0.677$ cm $[-0.909, -0.353]$); this value is descriptive and does not change the verdict. In descriptive targets, Russia W (cap equal to its A size, 14–16 labels) was inferior at $\tau = 0.05$ ($+0.29$ cm $[+0.16, +0.40]$), and among the ten sub-regions in mode x only AL-2 was inferior ($+0.04$ cm $[+0.02, +0.18]$).

Supplementary Methods 8. Placement strategies and the learned placement policy (XD)

Candidates are the cells of the label half A of a target (A pool); label values are not used for selection except by S6. Sources: WF plan section 2 (WF2) and revision items 2 and 3 of 1 October 2026; `scripts/3_deep_learning/h54_workflow.py` (`strategy_sets`, `kcenter_order`, `s4_order`) and `scripts/3_deep_learning/h40_label_grid.py` (`draw_blocks`).

S1, random cells. `h40.draw_cells`; the seed is fixed by target, mode, split, n and draw.

S2, block stratification. The 0.5° blocks of A are put in random order, and one cell per block, in random order within the block, is drawn in turn until n cells are drawn (`h40.draw_blocks`).

S3, covariate-cluster stratification. k -means clustering ($k = 8, 10$ initializations) of the A pool in the standardized x25 covariate space, allocation proportional to cluster size by largest remainders, random cells within clusters.

S4, covariate max-min distance. The x25 covariates of the A pool are median-imputed and standardized with the A-pool mean and standard deviation (no variable weights). The first cell is chosen at random with the draw seed; then the cell with the largest minimum Euclidean distance to the chosen set is added in turn (greedy k -centre; ties to the smaller index).

S5, source-extrapolation first. Cells in descending order of the dissimilarity index of the area of applicability: the standardized x25 nearest-neighbour distance to the source cells (mode x) divided by the mean pairwise distance within the source. Ties are shuffled with the draw seed; one to five draws were stored per (split, n). The number of ties per cell was not counted.

S6, sequential active selection. Start from 20 random cells (the S1 draw at $n = 20$) and add at most 20 cells at a time without crossing the next budget (20, 40, 50, 70, 90, 100, ..., 500). At each step, choose the cells with the largest prediction variance of a virtual ensemble (10 members) of the residual CatBoost fitted to the current labels.

S7, $\sqrt{\text{TDD}}$ first. Cells in descending order of $\sqrt{\text{TDD}}$, where a coefficient error propagates most into ALT. Cells in the same ERA5-Land grid cell tie; ties are shuffled with the draw seed (eight cells of the grid have fewer than five stored draws).

Algorithm P (Methods) is the pre-specified placement procedure tested in the workflow analysis; its setting was chosen by a mechanical rule on Alaskan targets (FINAL_BATCH appendix XC-0). The candidates were S2, S4 and the S8 settings (S8a, γ 0 with farthest-first block order and the cell nearest the block centre first; S8b, farthest-first choice within blocks; S8c, γ 0.5; S8p, γ 1). Rule 2 excluded all candidates, so Algorithm P is S1 (cell-random sampling); the workflow contrasts XC-5b, XC-5c and S-XC3 were therefore not tested and remain in the XC Holm family with $P = 1$. Design-weighted (S8w) and block random-effect (S8r) coefficients reuse the S8a labels.

Learned placement policy (XD-learn).

Two policies were developed on Alaskan tasks only (Alaska with its parent region excluded and the sub-regions AL-1 to AL-6, splits 1 to 5, $n \in \{10, 20, 40\}$), with 200 label sets per task, split and n (S1 draws 40%, S2, S4 and S8 sets with other seeds 30%, and those sets with 20 to 50% of cells replaced at random 30%). The utility of a label set is the RMSE of residual ML (λ 0.25) relative to the mean over random draws, averaged over the two weightings. Set features use no label values: coverage of the candidates in standardized covariate space, energy distance, block allocation, $\sqrt{\text{TDD}}$ statistics, the unweighted dissimilarity index and the task context. S9-GBM is a CatBoost regressor; S9-DS is a DeepSets network (element features: 25 standardized covariates, $\sqrt{\text{TDD}}$, block share, dissimilarity index and distance to the nearest chosen cell; ranking loss; three seeds). Both policies add five cells at a time, choosing among the 300 farthest-first and 200 random candidates the cells with the smallest predicted utility. The variant with the smaller development cross-validation utility (one Alaskan sub-region left out together with Alaska) was frozen as S9* = S9-DS (commit f4a89ea, 5 October 2026, 03:17:28 KST; freeze record sha256 2751fecf49b8bcbd) before any test task was run. The test tasks outside Alaska (LE-1, LE-2, CA-2, CA-3 and the Tibetan Plateau, mode x, splits 1 to 5, n 20 and 40, policy seeds 0 to 4) were evaluated once against Algorithm P. Because the five tasks form three families (Lena Delta, Canada, Tibetan Plateau), the smallest possible family-level sign-test P is 0.125, so no P values are reported. The test fits ran on the local server (operational deviation). S9* had lower error than Algorithm P in 2 of 5 test tasks (0 of 3 families) and higher error in 2 of 5 tasks (2 of 3 families), in both weightings and without a test (Supplementary Table S2). The candidate pool consists of already measured cells, so gains are not guaranteed to transfer to field candidates. XD-5 (descriptive, built after the XD-4 tables were opened): against the best of 200 candidate label sets, S9* had lower regret than Algorithm P in 3 of 5 test tasks and higher regret in 2; leaving a family out, the cross-validation utility was negative in 2 of 5 tasks for S9-DS and in 5 of 5 for S9-GBM (post hoc comparison). The S9-DS utility is sensitive to the minibatch-order seed, which includes the fold name; this is not a determinism failure, because the frozen model gave identical selections on two GPUs (Supplementary Table S2).

Supplementary Methods 9. Additional registered analyses XA to XM

The ten analyses were registered in commit 2678100 (4 October 2026, 14:22:39) before any fit (docs/EXPERIMENT_PLAN_FINAL_BATCH_2026-10-04.md). Their placement in the main text or the Supplementary Information was fixed before results, and results are reported in Supplementary Table S2.

XA, gain decomposition (main text, post hoc).

The gain of the selection rule W over the source-coefficient model with all labels, $G_{\text{tot}} = P0 - W$, was split into the recalibration share $G_{\text{recal}} = P0 - P1$ and the share added by the rule $G_W = P1 - W$; the ML share beyond recalibration was $G_{\text{ML2}} = \min(P1, P2) - R1(\lambda 0.25)$ (auxiliary $G_{\text{ML}} = P1 - R1$ and the shrinkage remainder $G_{\text{shr}} = P1 - P2$). Coefficient error was the absolute ten-label bias (practical, main) or the absolute log ratio of the all-label target coefficient to the source coefficient (oracle). Spearman rank correlations used the 28 targets of WF4-c. The main interval resampled target families (clusters) jointly with blocks under the same resample index; the auxiliary interval kept the targets fixed. A result was 'positive' or 'negative' only if the main interval excluded zero in both weightings and the subset with at least 100 cells in half A (23 targets) did as well; otherwise it was 'not established'. When the main and auxiliary categories differed, both were recorded and the weaker one was reported. The analysis refitted nothing: it read the stored WF4 shards (132 units), and two reproduction gates (WF4-c, ρ 0.38 [0.17, 0.55]; WF0) passed before the sealed tables were opened. The registered family list had seven families; the North Atlantic target (point estimates only) was added as an eighth single-target family in the implementation. All three registered hypotheses were 'not established' (Supplementary Table S2).

XB to XJ.

The designs are summarized in Methods (Additional registered analyses). XB learns convex weights of physics models and ALT products from the target labels by cross-fitting; XC applies Algorithm P, the ten-label diagnostic and the method rule in sequence on new splits (seeds 201–210); XD selects the setting of Algorithm P on Alaskan targets and tests a learned placement policy; XE adds public inputs at 10–500 m to within-region residual ML; XF adds new regions from public data (Supplementary Methods 2); XG scores existing ALT maps on the same blocks after masking blocks near product training sites; XH describes the validation ladder in three regions; XI repeats the climate-extrapolation test on the licence-verified expanded Canadian table; XJ tests temperature-derived labels as low-weight auxiliary rows.

Execution and deviations (XB to XJ).

XB, XD-alg, XH, XI and XJ ran as one cloud job on 5 October 2026 (02:18 to 02:38 KST); XE stage 1 (xh0) and the XD-4 test fits ran on the local server; XG ran locally after a second product extraction (02:13 to 02:15) and a re-check of the downloaded archives against their recorded md5 sums. Each analysis passed its reproduction gate against the earlier shards before its sealed tables were opened (first openings 03:18 to 03:45 KST; FINAL_BATCH section 8). Deviations: XG compares products under a registration deviation (WRAPUP 10); the XE gate was first scored at the cloud-to-cloud tolerance and rescored at the registered local-to-cloud tolerance (a correction of the gate level, not of the design); the sealed XH reuse table of region-holdout rows was empty, so those values were read from the earlier source table without new fits; the XJ source-side rows were not tested because their licence basis had not been recorded before submission of the job. Flags: all analyses were designed after earlier results were viewed; XB, XD-1 to XD-3, XE (xh0), the CCI rows of XG and XJ-3 are partly unblinded; the Alaska rows of XH and XI are unblinded replications; XI uses warm blocks of the same period as a spatial proxy and is not a true extrapolation test; XD is exploratory and XH descriptive. XC ran on the local server on 5 October 2026 (03:16 to 05:19 KST; 208 units, no failure) after its reproduction gate passed (128 keys), and its sealed tables were first opened at 05:20:39 KST (FINAL_BATCH 8.10). Because Algorithm P = S1 (Supplementary Methods 8), the placement step of the workflow was random placement and contributed nothing; at ten labels the workflow and random placement with the fixed recipe use the same labels and model, and at 40 and 160 labels the workflow contrast against random placement with the fixed recipe compares the selection rule with the fixed recipe on the same labels (equal to the auxiliary contrast S-XC1). XC-5b, XC-5c, XC-F3, S-XC0 and S-XC3 were therefore not tested (Holm family $m = 8$, $P = 1$). All XC statements refer to new random splits of the same regions (re-test in reused regions, partly unblinded: WF2, WF4, WF8, LGX-N4). Pending: registered data deadline 11 October 2026 (XC-F3 appendix, after XF). Later local runs on 5 October 2026 completed XD-5 (03:30 to 05:37 KST; opened 05:38:08), a sensitivity edition of XB with the year-matched CCI v5 map added as an eighth candidate (05:20 to 05:38; opened 05:43:31; the registered XB verdicts remain those of the main run, and XB-3 was not retained in this edition) and the XE-e diagnostic with on-site soil moisture (05:41; opened 05:42:35; changes no verdict). Pending: registered data deadline 8 October 2026 (XE stage 2: XE-a, XE-b, XE-a0 and the xt2 table). Pending: registered data deadline 11 October 2026 (XF).

XK and XL (added registration).

Both were registered on 5 October 2026 (commit d721d24) before computation, were designed after earlier results were viewed and are descriptive (no verdict words). XK aggregates within-region out-of-block predictions (0.5° block five-fold cross-validation; coefficient refitted on the training folds; λ chosen by inner block cross-validation) and labels to grid cells of 0.05° , 0.1° and 0.25° , dropping grid cells with fewer than three label cells, and compares RMSE of P1, R1 and four ALT products at the same grid size (Supplementary Fig. S6). In Alaska the RMSE difference between P1 and R1 was 0.50 cm at 1 km and 1.54 to 1.67 cm at 0.05° , 0.1° and 0.25° (cell-weighted). At label cells, CCI v5 exceeded the labels on average by 35.63 cm in Alaska, 58.60 cm in Canada and 0.27 cm in the Lena Delta, whereas CCI v4 differed by -5.09 , $+10.35$ and -6.04 cm (product minus label, 1 km, cell-weighted); no cause is asserted. XL maps ALT at 1 km in the Alaskan permafrost zone and the Lena Delta with the recalibrated anchor plus residual ML trained on all labels and compares the maps with the recalibrated Stefan model and with CCI v5, Wei 2026, Aalto 2018 and Yi–Kimball (Supplementary Figs S1 to S5). The 1 km map is a display resolution; its information resolution is bounded by the climate (about 10 km) and soil (about 5 km) inputs. The maps carry no accuracy claim; accuracy is compared only at label cells (XG, XK). Values: Supplementary Table S2, parts m and n.

XM (added registration).

XM was registered on 5 October 2026 (commit ff7c97f) before feature extraction, was designed after the XE results had been viewed, and is exploratory and partly unblinded; it adds to the 25 covariates the fractions of 11 ESA WorldCover classes (10 m) in each 1 km label cell and three Sentinel-2 summer index columns (20 m) and compares the within-grid RMSE of residual ML with that of the 25-covariate model (Supplementary Table S2, part q). The label-free features came from 51 WorldCover tiles and 534 Sentinel-2 L2A scenes (89 tiles, six scenes each, July and August 2019–2023, no read failure); the registered reflectance offset was corrected as a registration deviation recorded before fitting (commit 31e19f7; the Earth Search image values carry no offset, so reflectance is $\text{DN} \times 0.0001$ in all scenes), the feature table was fixed by its sha256 before the fits (148 units, no failure), and the sealed tables were first opened at 14:20:47 KST after the reproduction gate of the re-used 25-covariate shards passed.

Supplementary Notes

Supplementary Note 1. Physical meaning of the Stefan coefficient E

What E represents.

D1. What E represents. The Stefan coefficient E is a lumped parameter. In the classical Stefan solution, the thaw depth is $Z = (2 k_t I_s / L)^{0.5}$ ¹¹. Here k_t is the thermal conductivity of the thawed layer ($\text{W m}^{-1} \text{K}^{-1}$), I_s is the ground-surface thawing index ($\text{K}\cdot\text{s}$), and $L = \rho_w L_f \theta$ is the volumetric latent heat of the thawed layer (J m^{-3}). ρ_w is $1,000 \text{ kg m}^{-3}$, L_f is $3.34 \times 10^5 \text{ J kg}^{-1}$, and θ is the volumetric water or ice content. When the thawing index is taken from air temperature, $I_s = n_t I_a$, where n_t is the thaw n -factor¹². In the form $\text{ALT} = E \cdot \sqrt{\text{TDD}_{\text{air}}}$, the coefficient becomes $E = (2 k_t n_t \cdot 86,400 \text{ s d}^{-1} / (\rho_w L_f \theta))^{0.5}$, expressed in $\text{cm } (^\circ\text{C}\cdot\text{d})^{-0.5}$. E thus combines the thermal conductivity of the soil, the latent heat set by moisture and ground-ice content, and the surface energy transfer summarized by n_t .

Site factors that act through E .

D2. Site factors that act through E . Organic surface layers lower k_t when dry and raise θ when wet. Both effects reduce E . Vegetation, moss and shading lower n_t . Late-lying snow delays the start of ground thaw, so the air thawing index overstates the surface thawing index and the fitted E decreases. Soil texture and bulk density change both k_t and θ . Edaphic factors of this kind were the basis of earlier regional Stefan mapping with land-cover specific coefficients^{13,14}.

Non-physical terms absorbed by E .

D3. Non-physical terms absorbed by E . E also absorbs terms that are not soil physics. The air thawing index is computed from monthly means of a 0.1° reanalysis grid for 2015–2020. E therefore absorbs the grid-to-site temperature bias, the elevation mismatch within the grid cell, the difference between monthly and daily degree-day sums, and the difference between the climatology period and the observation years. The label definition enters as well. Probe, ground-penetrating radar (GPR) and temperature-derived labels differ, and a label measured before the end of the thaw season underestimates the seasonal maximum. Finally, E absorbs the error of the square-root law itself, which neglects sensible heat, layered soils and unfrozen water¹⁵.

Magnitude.

D4. Magnitude. For illustration, $k_t = 1.0 \text{ W m}^{-1} \text{ K}^{-1}$, $n_t = 1.0$ and $\theta = 0.40$ give $E \approx 3.6 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$. The combination $k_t = 0.5 \text{ W m}^{-1} \text{ K}^{-1}$, $n_t = 0.8$ and $\theta = 0.60$ gives $E \approx 1.9 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$. The median label ratios $\text{ALT}/\sqrt{\text{TDD}}$ in the input table are $1.62 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$ in Alaska, $1.28 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$ in Lena and $1.37 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$ in Canada (LG plan 6B.7, label statistics). These medians lie below the second illustrative value. This is consistent with low thermal conductivity, high moisture content, n_t below 1, or a combination of these. The data used in this study do not separate these factors.

Why E transfers poorly between regions.

D5. Why E transfers poorly between regions. A coefficient estimated in one set of regions carries that set's mixture of soils, ground ice, vegetation, snow regimes and label types. Air temperature explains the thawing index but not these controls. Three observations in the input data show the size of the variation. First, the standard deviation of ALT among labels in the same 1 km cell is 11.3 cm in Alaska, 13.0 cm in Lena and 17.7 cm in Canada (WRAPUP 2.2). Second, across Canadian 0.5° blocks with at least 3 labels, the coefficient of variation of the block-mean ratio $\text{ALT}/\sqrt{\text{TDD}}$ is 0.22 (unit metadata viewed in WRAPUP revision 1; WRAPUP 0.1). Third, the label ratio in the Tibetan candidate data is $11.2 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$ for GPR labels and $12.9 \text{ cm } (^{\circ}\text{C}\cdot\text{d})^{-0.5}$ for temperature-derived labels, about 7–9 times the Arctic values (LG plan 6B.7). The label type also differs by region. Alaskan labels are mostly ABoVE point observations, many from GPR, with some CALM site means. Canadian labels are ABoVE point observations with some CALM site means. Lena labels are mostly single-visit points. Russia W and E labels are multi-year CALM site means (WRAPUP 2.1–2.2).

What the source-coefficient model tests in this design.

D6. What P0 tests in this design. E0 is pooled over source cells without regional weights. For each of the four main targets, 78.0–94.3 % of the source cells are Alaskan (Table M3-1). P0 therefore tests how well an Alaska-dominated coefficient transfers. It does not test a generic physical coefficient. The registered covariate-dependent coefficient models test whether covariates recover the between-region variation of E: V1 under LG L7 (rejected: in Canada and CA-3 all V1 and V1r minus P0 point estimates at $n \geq 40$ were positive, +2.9 to +8.1 cm; partly unblinded) and V2 under LGX L13 (not supported: V2 had higher error than P1, +3.94 cm, and than R1, +4.12 cm). The wrap-up plan records earlier negative results for models that predict E (WRAPUP 11.1, H27 and V1) and adds no new E model.

Relation of the two baselines.

D7. Relation of the two baselines. P0 gives the error of a coefficient transferred from other regions with no target label. P1 gives the error after the mean level of E in the target is estimated from n labels, with E0 as a prior worth 10 labels. P1 changes one multiplicative factor. It corrects the regional level of E but not the variation of E within the target. The contrast $P1 - P0$ at $n = 10$ (AB4) measures the value of a few labels for the coefficient level (AB4: -2.45 cm , 95% CI -3.04 to -1.88 ; lower error, Holm-adjusted $P = 0.001$; Supplementary Table S14). The contrasts $R1 - P1$ at $n = 10$ (AB5) and at full labels (AB9), and the minimum n of LG L4, measure what residual learning adds after this level correction (AB5: -0.18 cm , lower error before Holm correction only, $P = 0.136$ after it, dependent on the split-independence assumption; AB9: -0.40 cm , lower error, Holm-adjusted $P = 0.001$; L4: minimum $n = 10$ for the four-region mean). Two pre-result sensitivity analyses concern the label unit and the label year (M5.4, M5.5). A sequential stopping rule based on the jackknife precision of the target coefficient rarely stopped early at the registered threshold (M5.6).

Limitation.

D8. Limitation. The construction of E0 is a design choice with two consequences. First, P0 and every method anchored on E0 (R0, D1 pseudo-labels at $n = 0$) inherit the Alaskan dominance of the source pool. Second, their performance relative to direct ML may differ under another pooling rule. We did not evaluate a region-equal-weight E0, so the direction of such a change is unknown. The dependence on E0 decreases as n grows, because E0 carries a fixed weight of 10 labels in E_n .

Supplementary Note 2. Differences between the earlier contest test and the tests reported here

An earlier within-Alaska analysis (a contest entry of this project) and the tests reported here answer different questions, and the word ‘Alaska result’ referred to three different tests. In the contest test (six-fold 0.5° block cross-validation within Alaska; five of six block folds for training), the physics residual model reached 13.33 cm

against 14.46 cm for the Stefan model (7.8% lower RMSE), the best of 56 settings. In the within-region retest of this study (WF6; half of the blocks for training, the setting chosen by cross-validation within the labels, 25 splits, comparator the Stefan model recalibrated on the same labels), residual ML reached 13.87 cm against 14.41 cm (3.75%). In the transfer test (Alaska treated as a new region, trained on the other regions, mostly the Lena Delta, before Alaskan labels were added), the values with all labels were 14.26 cm and 14.54 cm and no difference was established. The two within-region results agree in direction; the smaller gain of the retest follows from the smaller training share (one half instead of five sixths), the absence of selection among several settings and the recalibrated comparator. Relative to the error floor of 11.31 cm (error not reducible with the present covariates), the contest setting reduced the reducible squared error by 38.6% and the stricter setting by 19.2% (79.71 to 64.42 cm²; Supplementary Table S6). Under four scoring schemes in Alaska (means of three seeds), the median distance from test cells to the nearest training cell was 0.004 km under random splits and 53.5 km under k -fold nearest-neighbour distance matching (kNNDM) cross-validation, which mimics the 81.6 km from map cells to the nearest label; the RMSE of direct ML rose from 11.6 to 17.9 cm, that of the Stefan model stayed at 14.2 to 14.5 cm, and the physics residual model was always below the Stefan model (12.77–13.70 cm; Supplementary Fig. S7). Source: docs/QA_FINAL_REVIEW_2026-10-02.md, Q14 and Q15, and paper/claims/C4_sufficient_labels/README.md 2.6 (values from data/processed/m1/cv_scheme_comparison.csv and the WF6 tests); the distance and RMSE values of the scoring schemes were moved here from the Results at the length revision of 5 October 2026.

Supplementary Table S1. Register of registered hypotheses.

Role: C confirmatory; P primary; A auxiliary; D descriptive or diagnostic; N direction-neutral report; abstract, contrast of the abstract bundle. Blinding: B blind; PU partially unblinded; RU replication (unblinded); RE result existed on disk at registration, not viewed; Sa S-a viewing label; NS not listed. Registered: first commit containing the hypothesis text (hash, date, Korea Standard Time). The verdict column gives the first clause of the registered verdict string of each plan (scope = verdict rows of the test tables cited in the claims READMEs); contrasts, Δ , both 95% CIs, uncorrected and Holm-adjusted P values are in Supplementary Tables S2 to S10, S14 and S15. Rows LG to WRAPUP: frame of docs/MANUSCRIPT_DRAFT_SUPPORT_2026-09-30.md M4.3 (88 rows); WF and X rows added at assembly.

ID	Plan	Role	Statement (abridged)	Registered	Blinding	Registered verdict
L1	LG §4	P	Direct ML (D0) does not outperform the source-coefficient Stefan model (P0) at $n \leq 40$ (at most 1 of 4 main regions with both CI upper bounds < 0)	‘40be64c’ 09-29 15:45:46	PU (M1 $n = 0$ direct ML; P2/W3)	supported (regions with both CI upper bounds below 0: 0/4)
L2	LG §4	P	Augmented combinations R2 or R3 have lower error than R1 at $n \in \{10, 40, 160\}$; otherwise "augmentation is for $n = 0$ only"	‘40be64c’ 09-29 15:45:46	B	rejected (augmentation is for zero labels only); R3 partial supported
L3	LG §4	P	Target weighting $\alpha \in \{10, 100, \text{continued training}\}$, chosen by within-A cross-validation, gives $n^* \leq 40$ in ≥ 2 of 4 structure targets	‘40be64c’ 09-29 15:45:46	NS	
L4	LG §4	P	Residual ML (R1) outperforms the recalibrated Stefan model (P1); report the minimum n	‘40be64c’ 09-29 15:45:46	PU (a2, h25b)	net value of ML with sparse labels (minimum n 10, four-region mean)
L5	LG §4	P	Neural and generative learners are on par with CatBoost in R1 (manuscript uses L26, WRAPUP 1.5)	‘40be64c’ 09-29 15:45:46	NS	learner differences (partial: region count short at n 40, 160)
L6	LG §4	P	Combinations that pass in Alaskan sub-regions (mode i) also pass in the main 4 regions at the same n (share ≥ 0.5)	‘40be64c’ 09-29 15:45:46	NS	not decidable (no passing combination)
L7	LG §4	P	The covariate-dependent coefficient V1 (or V1r) removes the degradation in Canada and CA-3 ($n \geq 40$, point estimate)	‘40be64c’ 09-29 15:45:46	NS	rejected
L8	LG §4	P	With all labels, R1 outperforms P0 (main-4 stratified mean, both weightings)	‘40be64c’ 09-29 15:45:46	PU (M1 H13)	supported (-2.64 [$-3.47, -1.90$]; regions 4/4)
L9	LGX X1	A	F1a (D0 plus E0's input) and D0 are indistinguishable	‘8513cc2’ 09-29 18:17:29	B	see the Supplementary Table of its family supported
L10	LGX X1	C	Residual structure has lower error than physics-input structure (four contrasts at $n = 0$ and $n = 10$)	‘8513cc2’ 09-29 18:17:29	B	
L11	LGX X1	A	Direct ML with process-model inputs (F1k) does not outperform P0 at $n = 0$	‘8513cc2’ 09-29 18:17:29	RU (M1 $n = 0$ direct ML)	see the Supplementary Table of its family rejected
L12	LGX X1	C	Multiplicative correction RM and additive residual R1 are equivalent ($n \in \{10, \text{all}\}$, λ 0.25 and 1.0)	‘8513cc2’ 09-29 18:17:29	B	
L13	LGX X1	A	V2 (log-ratio model) is better than P1 and not different from R1 at $n = 10$	‘8513cc2’ 09-29 18:17:29	RU for Canada and CA-3	see the Supplementary Table of its family
L14	LGX X1	N	Effect of removing physics-related inputs from D0 (D0t, D0c, D0m)	‘8513cc2’ 09-29 18:17:29	B	see the Supplementary Table of its family
L15	LGX X2	C	Stefan pseudo-labels (D1) have lower error than four placebos at $n \in \{0, 10\}$	‘8513cc2’ 09-29 18:17:29	RU at $n = 0$ (M1 H3)	mixed (described by n)
L16	LGX X2	A	Verdicts are insensitive to the pseudo-label ratio r ; D1 approaches the physics model as r grows	‘8513cc2’ 09-29 18:17:29	B	see the Supplementary Table of its family
L17	LGX X2	A	R1 and the label-shuffle control R1s are equivalent ($n \in \{10, \text{all}\}$)	‘8513cc2’ 09-29 18:17:29	B	see the Supplementary Table of its family
L18	LGX X2	A	With target weighting, the augmented combination is not better than R1	‘8513cc2’ 09-29 18:17:29	B	see the Supplementary Table of its family
L19	LGX X3a	C	Direct ML beats the physics model under random splits (V-R) but not under region hold-out (V-G)	‘8513cc2’ 09-29 18:17:29	RU (2026-06 split leakage table, M1)	not supported (no reversal wording; degradation size reported)

ID	Plan	Role	Statement (abridged)	Registered	Blinding	Registered verdict
L20	LGX X3a	A	D0 RMSE increases in the order V-R < V-B < V-C100 < V-C500	‘8513cc2‘ 09-29 18:17:29	RU	supported
L21	LGX X3a	A	Degradation from V-R to V-G is smaller for anchor plus residual than for D0	‘8513cc2‘ 09-29 18:17:29	B	supported (λ 0.25)
L22	LGX X3b	A	Within a region, R1w outperforms P1w (count of regions out of 3)	‘8513cc2‘ 09-29 18:17:29	RU for Alaska (M1 X-ak)	lower error in 2 of 3 regions (Lena Delta, Canada)
L23	LGX X3c	A	Labels near scored cells reduce error more than the same number of distant labels	‘8513cc2‘ 09-29 18:17:29	B	see the Supplementary Table of its family
L24	LGX X5	A	Residual-learning net value concentrates near labels; recalibration gain does not depend on distance	‘8513cc2‘ 09-29 18:17:29	B	see the Supplementary Table of its family
L25	LGX X5	N	R1 90 % interval coverage and width versus the R0 n = 0 interval (6 cells)	‘8513cc2‘ 09-29 18:17:29	B	see the Supplementary Table of its family
L26	LGX X6	A	Each learner minus CatBoost (R1) lies within the 0.5 cm margin at n $\in$ {0, 10, all}, 5 splits	‘8513cc2‘ 09-29 18:17:29	B	see the Supplementary Table of its family
L27	LGX X6	A	The sign of R0 – P0 in Canada does not depend on the learner ($\geq$ 6 of 8 learners)	‘8513cc2‘ 09-29 18:17:29	B	see the Supplementary Table of its family
L28	LGX X9	A	4-way verdicts of core contrasts are unchanged under κ , label-definition, anchor, input-set and matched-year variants	‘8513cc2‘ 09-29 18:17:29	RU for κ , λ and anchors	see the Supplementary Table of its family
L29	LGX N1	C	No physics or product baseline is better than P0 at n = 0	‘8513cc2‘ 09-29 18:17:29	PU for B:ens – P0 (LG revision 15 (h))	a baseline better than P0 exists (P* = year-matched Stefan)
L30	LGX N3	C	The n = 0 direct ML result does not depend on learner capacity (catboost, rf, catboost_tuned)	‘8513cc2‘ 09-29 18:17:29	RU (M1 n = 0 direct ML)	supported
L31	LGX N4	A	The full-label R1 – P0 improvement is robust to split composition (50 splits)	‘8513cc2‘ 09-29 18:17:29	RU for Canada, CA-3, AL-5; Sa	see the Supplementary Table of its family
L32	LGT	A	R1[TabPFN] – R1[CatBoost, same context] lies within the 0.5 cm margin at n $\in$ {0, 10, 40, 160, all}	‘6799b74‘ 09-29 22:43:12	RU at n $\in$ {10, 40} (b4); B at n = 0, 160, all	see the Supplementary Table of its family
L33	LGT	A	TabPFN residual (R1) outperforms P1; report the minimum n	‘6799b74‘ 09-29 22:43:12	RU at n $\in$ {3, 10, 40} (b4); B otherwise	see the Supplementary Table of its family
L34	LGT	A	(a) TabPFN direct prediction does not outperform P0 at n = 0; (b) R0[TabPFN] – P0 reported	‘6799b74‘ 09-29 22:43:12	B; CatBoost rows RU	see the Supplementary Table of its family
L35	LGT	A	With all labels, TabPFN R1 outperforms P0	‘6799b74‘ 09-29 22:43:12	B; CatBoost rows RU	see the Supplementary Table of its family
L36	LGT	N	D0[TabPFN] – R1[TabPFN] at five n	‘6799b74‘ 09-29 22:43:12	B	see the Supplementary Table of its family
L37	LGT	A	Core-contrast verdicts are stable under context variants (d3, d10, rid, tgt)	‘6799b74‘ 09-29 22:43:12	RU for d3, rid	see the Supplementary Table of its family
L1e	LGD	A	L1 recomputed on the extended pools PE1 and PE2	‘6799b74‘ 09-29 22:43:12	as L1 for P4; RE under revision 15 (m)	supported (P4, PE1, PE2)
L4e	LGD	A	L4 recomputed on PE1 and PE2 (minimum n with region count k/N)	‘6799b74‘ 09-29 22:43:12	as L4 for P4; RE under revision 15 (m)	P4 and PE2: net value with sparse labels; PE1 partial (2/5)
L8e	LGD	A	L8 recomputed on PE1 and PE2	‘6799b74‘ 09-29 22:43:12	Tibet unblinded (label statistics); RE under revision 15 (m)	supported (P4, PE1, PE2)
L38	LGD	A	In each new eligible region, the region-level verdicts follow the hypothesis directions	‘6799b74‘ 09-29 22:43:12	Tibet unblinded; NAtlantic, Russia C blind	diverging regions: Central Russia, Tibetan Plateau
L39	LGD	A	In Tibet, changing the label definition leaves core verdicts unchanged	‘6799b74‘ 09-29 22:43:12	unblinded (label statistics)	robust

ID	Plan	Role	Statement (abridged)	Registered	Blinding	Registered verdict
L40	LGD	A	Adding new cells to Russia W, Russia E and Canada leaves region-level verdicts unchanged	‘6799b74’ 09-29 22:43:12	RU	weakened
L41	LGD	A	New-region verdicts are robust to label-set variants (a)–(c)	‘6799b74’ 09-29 22:43:12	as L38	robust or weakened by region
L42	LGD	A	New-region verdicts are unchanged when other new regions join the source	‘6799b74’ 09-29 22:43:12	as L38	robust or weakened by region
L43	WRAPUP 4	A	Block-dispersed label placement gives lower P1 error than cell-random placement ($n \in \{10, 40\}$)	‘316714c’ 09-30 03:02:42	PU (a2_spread, h29b)	placement effect not established
SC1w	WRAPUP 3.3	A	At T3 and T10, recipe R1 is non-inferior to P0 (0.5 cm) in all 5 independent regions (10 cells)	‘316714c’ 09-30 03:02:42	PU; SC1w-P: RE and Sa	safety not confirmed (non-inferior cells 4/10)
SC2w	WRAPUP 3.4	N	Number of regions (Lena, Canada, Alaska x) where R1 beats P1 at $n = 40$	‘316714c’ 09-30 03:02:42	PU	sentence A (2 of 3 regions at $n = 40$)
SC3w	WRAPUP 3.5	A	A sequential stopping rule (jackknife SE of $\log E \leq \tau = 0.05$) halves labels versus always 40 with error increase ≤ 0.3 cm and no inferior region	‘316714c’ 09-30 03:02:42	PU; computed before results (M5.6)	rule not operating (label ratio 0.91)
AK1w	WRAPUP 5	A	Combinations that pass in Alaskan sub-regions with a within-Alaska source (mode i) also pass with an outside-only source (mode x); $r \geq 0.5$	‘316714c’ 09-30 03:02:42	PU (h25b, h25x)	majority ($r = 0.57$; 68 pairs, 5 targets)
AB1	WRAPUP 1.1	abstract	D0 – P0, $n = 0$, $\lambda = 1.0$, main 4 (basis L1)	‘316714c’ 09-30 03:02:42	PU	see Supplementary Table S14
AB2	WRAPUP 1.1	abstract	B:ens – P0, $n = 0$ (basis L29)	‘316714c’ 09-30 03:02:42	PU	see Supplementary Table S14
AB3	WRAPUP 1.1	abstract	D1 – D1@shuffle, $n = 0$, $\lambda = 1.0$ (basis L15)	‘316714c’ 09-30 03:02:42	RU	see Supplementary Table S14
AB4	WRAPUP 1.1	abstract	P1 – P0, $n = 10$ (curve core, L28 core contrast)	‘316714c’ 09-30 03:02:42	PU; RE (LGT P0, P1); Sa	see Supplementary Table S14
AB5	WRAPUP 1.1	abstract	R1 – P1, $n = 10$, $\lambda = 0.25$ (basis L4)	‘316714c’ 09-30 03:02:42	PU	see Supplementary Table S14
AB6	WRAPUP 1.1	abstract	R2 – R1, $n = 10$, $\lambda = 0.25$ (basis L2)	‘316714c’ 09-30 03:02:42	B	see Supplementary Table S14
AB7	WRAPUP 1.1	abstract	R1 – F1k, $n = 10$ (basis L10)	‘316714c’ 09-30 03:02:42	B	see Supplementary Table S14
AB8	WRAPUP 1.1	abstract	R1 – P0, all labels, $\lambda = 0.25$ (basis L8)	‘316714c’ 09-30 03:02:42	PU	see Supplementary Table S14
AB9	WRAPUP 1.1	abstract	R1 – P1, all labels, $\lambda = 0.25$ (basis L4)	‘316714c’ 09-30 03:02:42	PU	see Supplementary Table S14
AB10	WRAPUP 1.1	abstract	Interval score ($\alpha = 0.1$), rung (iii) – B4, $n = 10$, Lena, Canada, Alaska x (basis LGU-A1)	‘316714c’ 09-30 03:02:42	per LGU §6; RE (LGU revision 4)	see Supplementary Table S14
LGU-A1	LGU 3.6	C	The hierarchical predictive interval (rung iii) has a lower interval score than the constant-width calibrated interval B4 at $n \in \{10, 40\}$	‘a112cef’ 09-29 23:30:02	NS; RE under revision 4	see the Supplementary Table of its family
LGU-A2	LGU 3.6	D	Share of infinite intervals at $n = 3$ is 0 (implementation check)	‘a112cef’ 09-29 23:30:02	RU (C2 wconf)	see the Supplementary Table of its family
LGU-A3	LGU 3.6	N	Median RMSE, rung (iii) – P1, $n \in \{3, 10\}$, $\delta = 0.5$ cm	‘a112cef’ 09-29 23:30:02	RU (B2 h32 offsets)	see the Supplementary Table of its family
LGU-A4	LGU 3.6	gate	CRPS (log scale), rung (v) – (iv), per target, leave-one-region-out	‘a112cef’ 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-A5	LGU 3.6	D	$n = 0$ coverage and interval score under hyperprior variants (exploratory)	‘a112cef’ 09-29 23:30:02	RU	see the Supplementary Table of its family

ID	Plan	Role	Statement (abridged)	Registered	Blinding	Registered verdict
LGU-A6	LGU 3.6	D	Russia W at $n \in \{3, 10\}$: interval score, coverage, width	'a112cef' 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-A7	LGU 3.6	A	Ladder contrasts (ii)–(i), (iii)–(ii), (iv)–(iii), (vi)–(iii), (v)–(iv)	'a112cef' 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-B1	LGU 4.4	A	Pooled generative quantile intervals (nflow, cfm) cover less than 0.85 in new regions (limit check)	'a112cef' 09-29 23:30:02	unblinded (M1 H15, H17)	see the Supplementary Table of its family
LGU-B2	LGU 4.4	C	nflow-normalized conformal intervals have a lower interval score than their σ -shuffled placebo at $n = 0$ (4 conditions)	'a112cef' 09-29 23:30:02	NS; RE under revision 4	see the Supplementary Table of its family
LGU-B3	LGU 4.4	A	Interval score nflow – cbq (5 % margin)	'a112cef' 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-B4	LGU 4.4	A	Each normalizer – constant, raw and width-matched	'a112cef' 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-B5	LGU 4.4	D	Coverage and width by instrument, season, width quintile and source distance	'a112cef' 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-B6	LGU 4.4	A	nflow – const and cbq – const at $n \in \{10, 40, 160\}$	'a112cef' 09-29 23:30:02	NS	see the Supplementary Table of its family
LGU-C1	LGU 5	D	Variogram of the centred log ratio (no verdict)	'a112cef' 09-29 23:30:02	RU	see the Supplementary Table of its family
LGU-C2	LGU 5	D	Within-block correlation (no verdict)	'a112cef' 09-29 23:30:02	RU	see the Supplementary Table of its family
LGU-C3	LGU 5	D	Share of scored cells with a label within d km (no verdict)	'a112cef' 09-29 23:30:02	RU	see the Supplementary Table of its family
LGU-C4	LGU 5	D	Alignment of SAR products with probe labels (no verdict)	'a112cef' 09-29 23:30:02	RU	see the Supplementary Table of its family
LGF-F1	LGF 3.6	C	TabICL direct prediction does not outperform P0 at $n = 0$ (context cap and full source)	'3a4fe84' 09-30 01:44:27	B; CatBoost rows RU	supported (not superior to P0)
LGF-F2	LGF 3.6	A	$R1[\text{TabICL}] - R1[\text{CatBoost, same context}]$ lies within 0.5 cm at five n	'3a4fe84' 09-30 01:44:27	B	see the Supplementary Table of its family
LGF-F3	LGF 3.6	N	TabICL versus TabPFN (six contrasts)	'3a4fe84' 09-30 01:44:27	B; TabPFN column RE at revision 1	see the Supplementary Table of its family
LGF-F4	LGF 3.6	A	TabICL residual (R1) outperforms P1; full-label $R1 - P0$	'3a4fe84' 09-30 01:44:27	B; TabPFN rows at $n \leq 40$ RU	see the Supplementary Table of its family
LGF-F5	LGF 3.6	N	$n = 0$ $R0[\text{TabICL}] - P0$	'3a4fe84' 09-30 01:44:27	B	see the Supplementary Table of its family
LGF-F6	LGF 3.6	N	Effect of the context cap (full source minus 10,000 rows)	'3a4fe84' 09-30 01:44:27	B	see the Supplementary Table of its family
LGF-N1	LGF 4.5	C	Direct ML from four tuned discriminative networks does not outperform P0 at $n = 0$	'3a4fe84' 09-30 01:44:27	PU (default-version direction, M1)	supported (not superior to P0; four networks)
LGF-N2	LGF 4.5	A	Tuned and default networks differ by less than the 0.5 cm margin (9 contrasts per learner)	'3a4fe84' 09-30 01:44:27	B	see the Supplementary Table of its family
LGF-N2s	LGF 4.5	A	Core-contrast verdicts are the same for default and tuned networks	'3a4fe84' 09-30 01:44:27	B	see the Supplementary Table of its family
LGF-N3	LGF 4.5	A	$R1[\text{tuned network}] - R1[\text{tuned CatBoost}]$ lies within 0.5 cm at $n \in \{0, 10, \text{all}\}$	'3a4fe84' 09-30 01:44:27	B; default rows RU	see the Supplementary Table of its family
LGF-N4	LGF 4.5	D	Selection tables and tuning diagnostics (no hypothesis)	'3a4fe84' 09-30 01:44:27	NS	see the Supplementary Table of its family
WF0	WF	D	Zero-label risk table and misspecification diagnostics over 30 targets (post hoc description)	4168331 10-01	post hoc	descriptive (Supplementary Table S3)

ID	Plan	Role	Statement (abridged)	Registered	Blinding	Registered verdict
WF1-a	WF	P	Within-region R1 and R2 (λ by CV) have lower error than the best physics calibration at $n \geq 1000$ (Alaska, five splits)	4168331 10-01	designed after viewing	rejected
WF1-b	WF	A	Nine SAR covariates add value to within-region R1	4168331 10-01	designed after viewing	rejected
WF1-c	WF	A	Recipes with fewer higher-error n than direct ML	4168331 10-01	designed after viewing	partially supported (R2, D1)
WF2-a	WF	P	Placement strategies S2 to S7 have lower error than random cells (S1) within regions	4168331 10-01	designed after viewing	supported (S3 at n 100; size below 0.5 cm)
WF2-b	WF	A	The best Alaskan strategy (S3) transfers to the Lena Delta and Canada	4168331 10-01	designed after viewing	rejected
WF3-a	WF	A	A covariate-dependent coefficient (Pc) has lower error than P1 at $n \geq 500$	4168331 10-01	designed after viewing	rejected
WF4-a	WF	P	Selection rule W is non-inferior to the fixed recipe R1(λ 0.25) (margin 0.5 cm)	4168331 10-01	designed after viewing	partial (regions 2/4): non-inferior at 40, 160 and all labels
WF4-b	WF	A	W has lower error than P1 in more targets than R1(0.25)	4168331 10-01	designed after viewing	rejected
WF4-c	WF	P	The ten-label bias is rank-correlated with the all-label gain of W over P0	4168331 10-01	designed after viewing	supported (ρ 0.38 [0.17, 0.55])
WF6-a	WF	P	Within-region recipes (λ by CV) have lower error than P1 at n 200 to all (Alaska; Lena Delta; Canada)	1b42b4d 10-01	designed after viewing	Alaska supported; Lena Delta rejected; Canada partial, rejected
WF6-b	WF	A	Stratified mean of three targets, R2 versus P1 at $n \geq 500$	1b42b4d 10-01	designed after viewing	partial (regions 2/3): rejected
WF8-a	WF	P	In transfer, S4 has lower error than S1 for R1 at n 40	1b42b4d 10-01	designed after viewing	supported (Canada)
WF8-b	WF	A	In transfer, S2 and S4 are not worse than S1 for R1 at n 10	1b42b4d 10-01	designed after viewing	supported
WF9-a	WF	P	Within-grid gain of R1 (λ by CV) over P1 within regions	8180632 10-02	designed after viewing	partial (regions 2/3): rejected
WF9-c	WF	P	Within-grid gain of R1 (λ 0.25) over P1 in transfer	8180632 10-02	designed after viewing	partial (regions 2/4): partially supported
WF9 soil	WF	A	Soil- $\sqrt{\text{TDD}}$ grouping sensitivity of WF9	7da0064 10-02	designed after viewing	see Supplementary Table S7
WF10-a to c	WF	P	Climate extrapolation (warm variant)	8180632 10-02	designed after viewing	not decidable (contrasts 0/2)
XA	FINAL_BATCH 2.1	post hoc	Rank correlation of the recalibration share, the selection-rule share and the ML share beyond recalibration with coefficient error	2678100 10-04	replication (unblinded)	not established (XA-1, XA-2, XA-3; Supplementary Table S2)
XB	FINAL_BATCH 2.2	see plan	Label-weighted stacking of physics models and products	2678100 10-04	PU	XB-1, XB-2 rejected (undetermined); XB-3 partially supported (lower error at n 500 and 1,000; not retained when CCI v5 was added as a candidate, sensitivity edition); XB-4 supported (non-inferior)

ID	Plan	Role	Statement (abridged)	Registered	Blinding	Registered verdict
XC	FINAL_ BATCH 2.3	see plan	End-to-end workflow on new splits of the same regions	2678100 10-04	PU (XC-1, XC-2, XC-5); B (XC-F3)	Algorithm P = S1; XC-1b undetermined; XC-1c lower error (Canada; Lena Delta undetermined); XC-2a, XC-2b, XC-2c non-inferior for the pooled mean (XC-2s lower error at n 40 and 160; regions over 0.5 cm worse: E Russia, Alaska, Lena Delta); XC-5b, XC-5c not tested; XC-F3 not tested (appendix pending; registered data deadline 11 October 2026)
XD	FINAL_ BATCH 2.4	exploratory	Placement algorithm and learned placement policy	2678100 10-04	PU (XD-1 to XD-3); B (XD-4)	Algorithm P = S1; XD-1 rejected; XD-2 rejected (higher error in Alaska and Canada); XD-3 non-inferior 8, higher error 2, not established 6; XD-4, XD-5 descriptive
XE	FINAL_ BATCH 2.5	see plan	Sub-grid public covariates within regions	2678100 10-04	PU (xh0)	XE-c (xh0, auxiliary) equivalent; XE-e diagnostic (no verdict); XE-a, XE-b pending; registered data deadline 8 October 2026 (XE stage 2)
XF	FINAL_ BATCH 2.6	see plan	New independent regions from public data	2678100 10-04	B (new regions)	Pending; registered data deadline 11 October 2026 (XF).
XG	FINAL_ BATCH 2.7	see plan	Existing ALT products on the same blocks	2678100 10-04 (WRAPUP 10 deviation)	B (Wei rows); PU (CCI rows)	XG-1c higher error; XG-2c, XG-3c lower error; XG-1w to XG-3w undetermined
XH	FINAL_ BATCH 2.8	descriptive	Validation ladder	2678100 10-04	RU (Alaska); B (Lena Delta, Canada)	descriptive (no verdict words)
XI	FINAL_ BATCH 2.9	see plan	Climate-extrapolation retest (warm-block spatial proxy)	2678100 10-04	B (Canada); RU (Alaska)	XI-a, XI-b rejected (undetermined); XI-c supported (lower error, non-inferior)
XJ	FINAL_ BATCH 2.10	see plan	Temperature-derived auxiliary labels	2678100 10-04	PU (L39)	XJ-1, XJ-2 not tested (licence basis not recorded); XJ-3 lower error (six rows)
XK	addendum XK	descriptive	Error by grid size within regions	d721d24 10-05	designed after viewing	descriptive (no verdict words)
XL	addendum XL	descriptive	1 km maps compared with ALT products	d721d24 10-05	designed after viewing	descriptive (no verdict; no accuracy claim)
XM	addendum XM	exploratory	Open 10 to 20 m land-cover fractions and summer vegetation indices of the 1 km label cell added to residual ML within regions (within-grid RMSE)	ff7c97f 10-05	PU; designed after viewing	XM-a partially supported (lower error at n 1,000 and all, size below 0.5 cm; equivalent at 500)

Supplementary Table S2. Additional registered analyses XA to XM.

XA was registered in commit 2678100 (4 October 2026, 14:22:39) and run locally without any new fit (sealed tables first opened 21:07:28). Both reproduction gates passed. Categories: positive or negative only if the main CI (family-cluster and block joint resampling) excludes zero in both weightings and the $|A| \geq 100$ subset (23 targets) does too; otherwise not established. Where the main and auxiliary categories differ, both are given and the weaker is reported. The registered family list had seven families; the North Atlantic target (point estimate only) was added as an eighth single-target family (implementation deviation, sealed/xa_hyp.csv). All five hypotheses were robust to the scale variants and to the oracle definition of coefficient error (absolute log ratio of the all-label target coefficient to the source coefficient). Holm family: XA-1 to XA-3. Source: paper/claims/C2_bias_diagnosis/XA_result_2026-10-04.md sections 4 and 6 (data/processed/xbatch/XA_c2_gain_decomposition/sealed/). Parts c to q: XB to XJ were registered in the same commit (2678100) and run on 5 October 2026; sealed tables were first opened between 03:18 and 05:21 KST (XC 05:20:39) after their reproduction gates passed (docs/EXPERIMENT_PLAN_FINAL_BATCH_2026-10-04.md section 8). XK and XL were added on 5 October 2026 (commit d721d24) before computation (docs/EXPERIMENT_PLAN_FINAL_BATCH_ADDENDUM_XK_XL_2026-10-05.md, results), and XM on 5 October 2026 (commit ff7c97f) before feature extraction (docs/EXPERIMENT_PLAN_FINAL_BATCH_ADDENDUM_XM_2026-10-05.md, results; sealed tables first opened 14:20:47 KST). All were designed after earlier results were viewed; further flags are given with each part. Δ is the RMSE of the first method minus that of the second (cm; negative, the first has lower error), cell-weighted and block-equal with block-bootstrap 95% CIs (10,000 resamples unless stated). Registered interpretation sentences are rendered in English in the notes. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: |x parent region excluded from the source, |i rest of the parent kept, |r within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. XA gain decomposition (post hoc; all labels; coefficient error = absolute ten-label bias; 28 targets, 8 families; 10,000 resamples)

Hypothesis	Gain	ρ cell / block	Main CI (family clusters) cell / block	Auxiliary CI (targets fixed) cell / block	$ A \geq 100$ CI cell / block	P_{one} cell, block	Holm P	Category main / auxiliary / final
XA-1	recalibration share P0 – P1	0.37 / –0.04	[–0.11, 0.93] / [–0.41, 0.86]	[0.14, 0.48] / [–0.08, 0.18]	[–0.19, 0.30] / [–0.54, –0.17]	0.06, 0.41	0.82	not established / not established / not established
XA-2	selection-rule share P1 – W	0.18 / 0.01	[–0.40, 0.66] / [–0.60, 0.46]	[–0.08, 0.32] / [–0.20, 0.19]	[–0.07, 0.37] / [–0.18, 0.30]	0.34, 0.57	0.82	not established / not established / not established
XA-3	ML share min(P1, P2) – R1(0.25)	0.28 / 0.27	[–0.19, 0.80] / [–0.51, 0.59]	[0.18, 0.54] / [0.07, 0.40]	[0.08, 0.47] / [0.09, 0.50]	0.09, 0.22	0.66	not established / positive / not established (two categories, weaker reported)
XA-3a (aux.)	ML share P1 – R1(0.25)	0.39 / 0.32	[–0.08, 0.82] / [–0.44, 0.63]	[0.27, 0.58] / [0.12, 0.44]	[0.08, 0.47] / [0.09, 0.50]	0.04, 0.17	outside the family	not established / positive / not established (two categories, weaker reported)
XA-4 (aux.)	shrinkage remainder P1 – P2	–0.14 / –0.48	[–0.64, 0.64] / [–0.81, 0.47]	[–0.46, –0.01] / [–0.70, –0.18]	[–0.63, –0.25] / [–0.82, –0.50]	0.69, 0.88 (negative direction: $P(\hat{\rho}^b \geq 0)$ at most 0.31)	outside the family	not established / negative / not established (two categories, weaker reported)
descriptive	total P0 – W	0.38 / 0.08	[–0.07, 0.91] / [–0.31, 0.80]	[0.17, 0.54] / [–0.02, 0.28]	[–0.13, 0.42] / [–0.40, 0.03]	0.07, 0.26		not established $\times 3$

b. XA-6 decomposition, four main regions, stratified mean (cm, positive = improvement; post hoc description)

n	Pool	Recalibration share (cell / block)	Shrinkage remainder	Selection-rule share
All	4/4	2.23 [1.33, 3.05] / 2.13 [1.19, 3.01]	–0.02 [–0.87, 0.57] / –0.43 [–1.23, 0.33]	1.14 [0.22, 2.22] / 1.43 [0.52, 2.32]
10	4/4	2.45 [1.86, 3.04] / 2.62 [1.88, 3.29]	–0.84 [–1.91, –0.03] / –1.36 [–2.33, –0.42]	0.43 [–0.49, 1.36] / 0.51 [–0.45, 1.45]

n	Pool	Recalibration share (cell / block)	Shrinkage remainder	Selection-rule share
40	2/4 (Lena Delta, Canada)	$-1.99 [-2.71, -0.89] / -2.01 [-2.63, -1.37]$	$-0.76 [-0.90, -0.57] / -0.84 [-0.98, -0.70]$	$1.43 [0.73, 1.90] / 1.55 [1.03, 2.12]$
160	2/4	$-2.14 [-2.93, -0.95] / -2.12 [-2.79, -1.44]$	$-0.19 [-0.23, -0.14] / -0.21 [-0.24, -0.17]$	$1.96 [1.18, 2.51] / 2.06 [1.47, 2.69]$

c. XB label-weighted stacking of physics models and products (transfer: XB-1, XB-2; within region: XB-3, XB-4)

Hypothesis and contrast	n	Pool	Δ cell [95% CI]	Δ block-equal [95% CI]	Fallback	Holm P	Verdict
XB-1 Stack – P1 (transfer)	10	four main regions, partial (3/4; no stacking in E Russia, fallback 0.52)	$-0.28 [-1.11, +1.16]$	$+1.42 [+0.46, +2.39]$	0.36	1.00	undetermined
	40	Lena Delta, Canada (2/2)	$-1.86 [-2.44, -0.55]$	$+0.10 [-0.44, +0.64]$	0.14	1.00	undetermined
	160	Lena Delta, Canada (2/2)	$-2.73 [-3.38, -1.35]$	$-0.46 [-1.14, +0.22]$	0.10	1.00	undetermined
	all	four main regions (4/4)	$-1.54 [-2.28, -0.47]$	$-0.39 [-1.13, +0.35]$	0.20	1.00	undetermined; hypothesis rejected
XB-2 StackR(0.25) – R1(0.25)	10	partial (3/4)	$-0.33 [-1.10, +0.99]$	$+1.16 [+0.33, +2.00]$	0.36	1.00	undetermined
	40	Lena Delta, Canada (2/2)	$-1.65 [-2.15, -0.49]$	$-0.01 [-0.47, +0.44]$	0.14	1.00	undetermined
	160	Lena Delta, Canada (2/2)	$-2.27 [-2.82, -1.05]$	$-0.35 [-0.94, +0.23]$	0.10	1.00	undetermined
	all	four main regions (4/4)	$-1.38 [-2.00, -0.41]$	$-0.43 [-1.05, +0.20]$	0.20	1.00	undetermined; hypothesis rejected
XB-3 StackR – R1 (CV λ), within region	200	three targets (3/3)	$-0.57 [-0.71, -0.20]$	$+0.42 [+0.24, +0.60]$	0.08	0.004	undetermined (the two CIs exclude zero on opposite sides)
	500	partial (2/3; Canada $ A < 500$)	$-0.55 [-0.69, -0.35]$	$-0.33 [-0.47, -0.19]$	0.05	0.0012	lower error
	1,000	partial (2/3)	$-0.498 [-0.62, -0.34]$	$-0.29 [-0.44, -0.15]$	0.01	0.0012	lower error (distinguishable, size below 0.5 cm)
	all	three targets (3/3)	$-0.99 [-1.19, -0.43]$	$+0.19 [-0.05, +0.43]$	0.01	1.00	undetermined; hypothesis partially supported
XB-4 Stack – P1, non-inferiority (margin 0.5 cm), Alaska within region	200	Alaska	$-0.47 [-0.61, -0.28]$	$-0.73 [-0.92, -0.55]$	0.16	<0.0001	non-inferior (four-way lower error, size below 0.5 cm)
	500	Alaska	$-0.63 [-0.77, -0.44]$	$-0.86 [-1.06, -0.67]$	0.09	<0.0001	non-inferior (four-way lower error)
	1,000	Alaska	$-0.72 [-0.84, -0.51]$	$-0.85 [-1.05, -0.65]$	0.03	<0.0001	non-inferior (four-way lower error)
	all	Alaska	$-0.75 [-0.89, -0.54]$	$-0.90 [-1.12, -0.69]$	0.04	<0.0001	non-inferior (four-way lower error); hypothesis supported

Flags: designed after earlier results were viewed; registered before running; partly unblinded as in the registered hypothesis table. Holm families: XB-1 to XB-3 ($m = 12$), XB-4 non-inferiority ($m = 4$); verdicts use the uncorrected CIs. Fallback: share of draws in which stacking was replaced by P1. Registered sentences (rendered): XB-1, ‘No difference between stacking and the recalibrated Stefan model was established’ (40 and 160 labels: Lena Delta and Canada, two regions); the statement that the gain from combining physics models and products is region-conditional (WF7, L29) is retained; at ten labels the stacking fallback rate was 0.36, so XB-1 at ten labels may in effect compare P1 with P1. XB-2, ‘No difference between stacked residuals and recalibrated-anchor residuals was established’ (same qualifiers). XB-3, ‘Within label-rich regions, stacked residuals had 0.55 cm (cell-weighted) lower error than recalibrated-anchor residuals with 500 labels (partial, 2 of 3 regions)’ and ‘... 0.498 cm lower error with 1,000 labels (partial, 2 of 3 regions); the difference is statistically distinguishable but below 0.5 cm in size’; ‘no difference was established with 200 and all labels (3 of 3 regions)’. XB-4, ‘Within Alaska, where the physics model is close to the error floor, the error increase of stacking relative to the recalibrated Stefan model was within 0.5 cm (non-inferior)’ XB-3 was not retained when CCI v5 was added as a candidate (sensitivity edition, part c continued); the registered verdict remains the main-run one. Regional rows: no higher-error row. XB-3 Canada rows had opposite signs in the two weightings ($n = 200$ $-0.95 [-1.29, +0.14] / +1.68 [+1.24, +2.13]$; all $-2.05 [-2.60, -0.47] / +1.05 [+0.41, +1.69]$; undetermined). Weights are reported descriptively only (no causal sentence): mean weight with all labels, Alaska soil-thaw-index Stefan 0.82, Canada Kudryavtsev 0.57, Lena Delta year-matched Stefan 0.77, Tibetan Plateau affine-calibrated CCI v4 0.92; Spearman ρ of the CCI weight with the P1 RMSE of the target 0.31 ($P = 0.094$, 31 stores). Auxiliary XB-5 (not counted): Stack0 – P0 without labels, four main regions $+0.51 [+0.09, +0.69] / +0.09 [-0.14, +0.31]$, undetermined. Source:

data/processed/xbatch/XB_multisource_stacking/sealed/ (xb_tests.csv, xb_weights_summary.csv, xb_fallback.csv); reproduction gate passed (xb_gate.csv).

c (continued). XB sensitivity edition with CCI v5 added as an eighth candidate (registered verdicts remain those of the main run)

Hypothesis	n	Main-run verdict	CCI v5 edition: cell / block-equal [95% CI]	Verdict, CCI v5 edition	Holm P (CCI v5 edition)
XB-1 Stack – P1	10	undetermined	+0.27 [−0.78, +1.84] / +1.64 [+0.68, +2.67]	undetermined (fallback 0.37)	1.00
	40	undetermined	−2.17 [−2.79, −0.41] / +0.70 [+0.06, +1.33]	undetermined (CIs exclude zero on opposite sides)	0.33
	160	undetermined	−2.79 [−3.43, −1.06] / +0.38 [−0.43, +1.19]	undetermined	1.00
	all	undetermined	−1.57 [−2.24, −0.44] / −0.04 [−0.82, +0.75]	undetermined	1.00
XB-1 hypothesis		rejected		rejected	
XB-2 StackR(0.25) – R1(0.25)	10	undetermined	+0.16 [−0.84, +1.64] / +1.39 [+0.55, +2.28]	undetermined	1.00
	40	undetermined	−2.02 [−2.60, −0.36] / +0.65 [+0.07, +1.22]	undetermined (opposite sides)	0.33
	160	undetermined	−2.35 [−2.92, −0.74] / +0.53 [−0.21, +1.28]	undetermined	1.00
	all	undetermined	−1.42 [−1.98, −0.38] / −0.04 [−0.71, +0.64]	undetermined	1.00
XB-2 hypothesis		rejected		rejected	
XB-3 StackR – R1 (CV λ), within region	200	undetermined	−1.24 [−1.34, −0.67] / +0.32 [+0.11, +0.54]	undetermined (opposite sides)	0.031
	500	lower error	−0.50 [−0.62, −0.19] / −0.06 [−0.24, +0.11]	undetermined (margin-dependent)	1.00
	1,000	lower error	−0.36 [−0.54, −0.03] / +0.08 [−0.14, +0.30]	undetermined (margin-dependent)	1.00
	all	undetermined	−1.44 [−1.59, −0.74] / +0.17 [−0.10, +0.43]	undetermined	1.00
XB-3 hypothesis		partially supported		rejected	
XB-4 Stack – P1, non-inferiority, Alaska	200 to all	non-inferior at all four n	all −0.70 [−0.82, −0.49] / −0.92 [−1.13, −0.73]; n 200 −0.46 / −0.75	non-inferior at all four n (four-way lower error)	<0.0001
XB-4 hypothesis		supported		supported	

Sensitivity edition registered in plan 2.2 as a second-stage candidate: the year-matched CCI v5 map (affine-calibrated; missing cells take the P1 value) added as an eighth candidate, not as a replacement ($K = 8$ in all 219 store-by- n rows). Local run on 5 October 2026, 05:20 to 05:38 KST (211 units, no failure); the keys shared with the main run agreed within 6.0×10^{-14} cm; sealed tables first opened 05:43:31 KST. Flags as in the main run; designed after earlier results were viewed. The registered verdicts remain those of the main run (XB-1 and XB-2 rejected, XB-3 partially supported, XB-4 supported). In this edition XB-1, XB-2 and XB-4 kept their verdicts, and XB-3 was not retained: the lower error at 500 and 1,000 labels became undetermined (block-equal point estimates -0.06 and $+0.08$ cm). Regional rows: XB-1 Lena Delta at n 10 changed from undetermined to higher error ($+0.31$ / $+1.98$ cm), XB-1 Canada at n 160 and all labels from undetermined to lower error (-5.54 / -1.59 and -6.03 / -1.71 cm), XB-3 Alaska and Lena Delta within region at n 1,000 from lower error to undetermined and Alaska with all labels from equivalent to undetermined; XB-2 and XB-4 regional rows did not change. Weights (descriptive, no causal statement): mean CCI v5 weight with all labels, Canada 0.21, Lena Delta 0.23, Tibetan Plateau 0.32 and Alaska 0.03 in transfer, Canada 0.46, Lena Delta 0.10 and Alaska 0.10 within regions; Spearman ρ of the total CCI weight with the P1 RMSE 0.35 (P 0.051, 31 transfer stores) and 0.50 (P 0.67, three within-region targets). Source: data/processed/xbatch/XB_multisource_stacking/sealed/ (xb_tests_cci5y.csv, xb_weights_summary_cci5y.csv, xb_cci_relation_cci5y.csv); plan 8.12.

d. XG existing ALT maps scored on the same blocks (registration deviation)

Hypothesis and contrast	n	Pool	5 km block mask: cell / block-equal	25 km block mask: cell / block-equal	Holm P (5 / 25 km)	Verdict	Product values imputed (5 / 25 km)
XG-1w B:wei – P0	0	Lena Delta, Canada (2/2)	+1.44 [−0.02, +4.30] / +4.09 [+2.10, +6.12]	+0.97 [−0.41, +3.87] / +4.01 [+1.99, +6.12]	0.156 / 0.383	undetermined	47.4% / 26.5%
XG-2w R1(0.25) – P1@wei	10	Lena Delta, Canada	+1.34 [+0.00, +1.83] / −0.49 [−1.36, +0.37]	+1.25 [−0.07, +1.75] / −0.52 [−1.43, +0.35]	0.542 / 0.504	undetermined	47.4% / 26.5%
XG-3w R1(0.25) – P1@wei	all	Lena Delta, Canada	+0.87 [−0.41, +1.37] / −0.39 [−1.11, +0.33]	+0.61 [−0.61, +1.19] / −0.43 [−1.17, +0.30]	0.542 / 0.504	undetermined	47.4% / 26.5%

Hypothesis and contrast	n	Pool	5 km block mask: cell / block-equal	25 km block mask: cell / block-equal	Holm P (5 / 25 km)	Verdict	Product values imputed (5 / 25 km)
XG-1c B:cci5y – P0	0	four main regions (4/4)	+23.79 [+15.34, +30.16] / +19.17 [+14.83, +23.72]	same	0.0006 / 0.0006	higher error	below 0.1%
XG-2c R1(0.25) – P1@cci5y	10	four main regions	–8.37 [–10.97, –7.39] / –12.87 [–14.69, –11.10]	same	0.0006 / 0.0006	lower error (HK region-level CI [–16.02, –0.50])	below 0.1%
XG-3c R1(0.25) – P1@cci5y	all	four main regions	–7.51 [–10.31, –6.27] / –11.75 [–13.63, –9.88]	same	0.0006 / 0.0006	lower error (HK CI [–17.05, +2.43])	below 0.1%

Flags: registration deviation (WRAPUP 10); designed after earlier results were viewed; XG-1w to XG-3w blind; XG-1c partly unblinded (raw CCI v4 minus P0 viewed in LGX L29); XG-2c and XG-3c partly unblinded (CCI v4 scale recalibration viewed); all confirmatory rows carry a few-blocks flag after masking (fewer than eight scoring blocks per split). The two masks are co-primary and gave the same branch, so no mask-dependence flag applies. CCI v5 has no training sites, so its values are the same under both masks. B:product, the product map without labels; P1@product, the product recalibrated with the same target labels. ALT definitions differ: CCI is the annual maximum thaw depth of a model pixel, Wei v2 is a machine-learning map trained on CALM labels (including CALM sites inside the target regions), and our labels mix GPR and probe measurements. Source-coefficient RMSE before and after masking (cell-weighted): Wei pool 24.93 cm, 25.28 cm (5 km) and 28.34 cm (25 km); Lena Delta 21.69, 21.74 and 27.86 cm; Canada 28.16 and 28.82 cm (both masks); CCI pool 30.64 cm (unchanged). Regional XG-1c rows (5 km): Lena Delta +1.19 [+0.13, +5.38] / +9.34 [+6.57, +12.22] (depends on split independence), Canada +82.38 [+49.01, +99.54] / +53.00 [+41.74, +64.55], E Russia +5.84 [+0.08, +18.20] / +10.44 [+4.25, +17.24] (depends on split independence), all higher error; W Russia +5.78 [–5.86, +17.69] / +3.89 [–6.71, +15.83], undetermined. XG-2c lower error in all four regions; XG-3c undetermined in Canada (–1.47 [–7.55, +0.96] / –10.94 [–13.71, –8.20]) and lower error elsewhere. Wei rows: XG-3w lower error in the Lena Delta only (5 km –1.08 [–1.76, –0.59] / –1.49 [–2.24, –0.77]); the Lena comparison depends strongly on imputed product values (excluding blocks with imputed values removes all Lena blocks, so the Wei pool row cannot be decided; CCI v5 verdicts are unchanged). Cell-level 5 km sensitivity (refitted): XG-1w +2.09 [+0.41, +5.23] / +4.45 [+2.50, +6.46], higher error (P 0.016); XG-2w and XG-3w undetermined; sensitivities do not change the confirmatory verdicts. Registered sentences (rendered): XG-1c, ‘Without labels, the existing CCI v5 map had 23.79 cm higher error than the source-coefficient Stefan model (4 of 4 regions, after 5 and 25 km block masks of training sites); error was higher in the Lena Delta (Δ +1.19 cm, CI [0.13, 5.38]), Canada (+82.38 cm, [49.01, 99.54]) and E Russia (+5.84 cm, [0.08, 18.20]).’ XG-2c, ‘With the same ten target labels, the recalibrated-anchor residual had 8.37 cm lower error than CCI v5 recalibrated with those labels’ (the HK region-level CI excludes zero, so a region-general sentence is allowed). XG-3c, ‘... with all target labels, 7.51 cm lower error’ (the HK CI includes zero; no region-general sentence). XG-1w, ‘Without labels, no difference was established between the existing Wei v2 map and the source-coefficient Stefan model (2 of 2 regions, after 5 and 25 km block masks; product values imputed for 47.4%). Wei v2 also used CALM sites inside the target regions for training, whereas the source-coefficient Stefan model used only labels outside the target region and its 100 km buffer.’ XG-2w and XG-3w, ‘No difference was established between the recalibrated-anchor residual with the same 10 (all) target labels and Wei v2 recalibrated with those labels (product values imputed for 47.4%).’ No sentence states that physics-anchored ML is worse than a product that shares training data (WRAPUP 10). Source: data/processed/xbatch/XG_product_comparison/sealed/xg_tests.csv, xg_confirm_rows.csv, xg_tests_cell.csv, xg_fill_sensitivity.csv; gate passed 5 October 2026, 03:42:02.

e. XH validation ladder: RMSE by scoring scheme (cm, cell-weighted [95% CI] / block-equal; descriptive)

Region	Item	W1R random cell	W1S 0.05° site	W1B 0.5° block	W1K kNNDM	V-G region holdout
Alaska	median distance to nearest training cell (km)	0.00	0.88	26.06	58.73	
	D0	11.54 [9.61, 14.99] / 11.94	13.28 / 13.65	16.21 / 14.02	17.30 [14.15, 19.68] / 14.12	35.56 / 31.91
	P1 (V-G: P0)	14.24 [12.08, 18.60] / 15.37	14.30 / 15.38	14.41 / 15.43	14.53 [12.43, 18.84] / 15.70	14.68 / 14.31
	R1 (V-G: R0)	13.27 / 14.39	13.82 / 14.88	13.97 / 14.97	14.07 / 15.41	15.00 / 16.76
Lena Delta	median distance (km)	0.01	1.02	9.55	26.42	
	D0	13.94 [10.86, 21.12] / 21.44	18.28 / 23.25	19.77 / 24.51	22.10 [17.26, 37.01] / 26.42	25.31 / 24.60
	P1 (V-G: P0)	20.89 [13.42, 37.22] / 25.92	21.11 / 26.07	21.18 / 26.23	21.40 [13.67, 38.01] / 26.37	21.64 / 24.95
	R1 (V-G: R0)	18.35 / 24.08	19.54 / 24.55	20.17 / 24.94	21.03 / 25.48	21.93 / 24.67
Canada	median distance (km)	0.01	3.86	25.33	236.53	
	D0	18.96 [14.01, 24.14] / 20.90	23.90 / 20.08	24.48 / 21.46	33.16 [22.52, 39.09] / 28.48	25.24 / 20.22

Region	Item	W1R random cell	W1S 0.05° site	W1B 0.5° block	W1K kNNNDM	V-G region holdout
	P1 (V-G: P0)	26.50 [18.43, 33.13] / 20.99	27.98 / 21.31	29.53 / 21.22	30.78 [20.19, 39.06] / 23.40	26.48 / 20.96
	R1 (V-G: R0)	23.59 / 19.80	26.27 / 19.83	27.83 / 20.07	30.94 / 22.57	25.60 / 20.05

f. XH optimism difference $\Delta\Delta$ from random-cell to kNNNDM splits (cm; descriptive)

Item	Learner	Alaska	Lena Delta	Canada	Three-region stratified mean
XH-1 ($M = P1$)	CatBoost (main)	+5.46 [+2.13, +8.53] / +1.85 [+0.75, +2.97]	+7.66 [+4.20, +15.86] / +4.52 [+0.59, +8.68]	+9.92 [+3.35, +17.28] / +5.17 [+0.94, +9.26]	+7.68 [+4.51, +11.25] / +3.85 [+1.84, +5.83]
	CatBoost (second setting)	+5.27 / +2.70	+7.61 / +4.76	+8.77 / +5.09	+7.22 [+4.84, +10.42] / +4.18 [+2.28, +6.11]
	random forest	+6.95 / +3.26	+9.99 / +5.78	+6.41 / +1.30	+7.78 [+5.37, +10.26] / +3.44 [+1.97, +5.06]
	CatBoost (main), Canada $\pm 2^\circ$			+7.79 [+1.82, +14.86] / +5.03 [+1.08, +8.92]	+6.97 [+3.99, +10.49] / +3.80 [+1.90, +5.76]
XH-2 ($M = R1, \lambda 0.25$)	CatBoost (main)	+4.96 [+1.51, +8.44] / +1.15 [+0.29, +2.03]	+5.49 [+2.99, +11.04] / +3.58 [+0.62, +6.58]	+6.85 [+2.42, +11.97] / +4.81 [+1.04, +8.58]	+5.77 [+3.25, +8.43] / +3.18 [+1.54, +4.79]
	CatBoost (second setting)	+4.15 / +1.70	+5.20 / +3.32	+5.83 / +4.78	+5.06 [+3.35, +7.32] / +3.27 [+1.74, +4.80]
	random forest	+5.59 / +2.04	+7.22 / +4.21	+3.99 / +0.82	+5.60 [+3.67, +7.45] / +2.36 [+1.24, +3.55]

Flags: designed after earlier results were viewed; descriptive (no verdict words, no multiplicity); Alaska rows are an unblinded replication, Lena Delta and Canada rows blind. Values: RMSE in cm, cell-weighted [95% CI] / block-equal. D0 direct CatBoost; P1 Stefan coefficient fitted on the training fold; R1 residual ML ($\lambda 0.25$). V-G rows reuse the region-holdout results of an earlier run (source-coefficient Stefan P0 and residual R0, no target labels); the sealed reuse table was empty, so these values were read from data/processed/lgx/ladder/lgv_metrics.csv without new fits. $\Delta\Delta = [\text{RMSE}_K(\text{D0}) - \text{RMSE}_R(\text{D0})] - [\text{RMSE}_K(M) - \text{RMSE}_R(M)]$, in cm, cell-weighted [95% CI] / block-equal [95% CI] from 10,000 block resamples of cell-level squared errors. Canada prediction-domain sensitivity ($\pm 2^\circ$, W1K): D0 29.99 / 26.97, P1 29.74 / 22.03, R1 29.31 / 21.06 cm. Alaska contest-continuity rows (ridge, W1R to W1K, cell-weighted): direct ridge 12.65 to 14.18 cm, residual ($\lambda 0.75$) 12.77 to 13.72 cm. Registered descriptive sentence (rendered): ‘From random-cell to kNNNDM splits, the RMSE of direct ML increased by 5.76 cm (Alaska), 8.17 cm (Lena Delta) and 14.20 cm (Canada), and that of the recalibrated Stefan model by 0.29, 0.51 and 4.28 cm (cell-weighted; block-equal, direct ML 2.18, 4.97 and 7.58 cm, recalibrated Stefan 0.32, 0.45 and 2.41 cm; the difference of the two increases, $\Delta\Delta$, was +5.46 [+2.13, +8.53], +7.66 [+4.20, +15.86] and +9.92 [+3.35, +17.28] cm, three-region stratified mean +7.68 [+4.51, +11.25] cm, block-equal three-region +3.85 [+1.84, +5.83] cm.’ With random-cell splits, RMSE was 11.54, 13.94 and 18.96 cm for direct ML and 14.24, 20.89 and 26.50 cm for the recalibrated Stefan model (cell-weighted). Accompanying sentence: random validation has been argued to estimate map accuracy validly for probability samples (Wadoux et al., 2021); the labels of this study are not a probability sample. Source: data/processed/xbatch/XH_validation_ladder/sealed/ (xh_ladder.csv, xh_dd.csv, xh_stage_contrasts.csv); gates passed (xh_gate.csv).

g. XI warm-block spatial proxy for climate extrapolation, Canada licence-verified expanded edition (not a true extrapolation test)

Hypothesis and contrast	n	warm_trim (main): cell / block-equal	Verdict, main	warm (basic): cell / block-equal	Verdict, basic	Reported verdict (weaker)	Holm P
XI-a EP(D0 – P1), higher-error criterion	100	–3.12 [–3.95, –1.02] / +0.33 [–1.52, +2.18]	undetermined	–4.68 [–5.90, –2.50] / –2.29 [–4.55, +0.02]	undetermined	undetermined	1.00
	all	–6.17 [–7.00, –3.02] / –1.12 [–3.36, +1.22]	undetermined	–6.49 [–7.66, –3.77] / –3.07 [–5.59, –0.51]	lower error (depends on split independence)	undetermined (both reported); hypothesis rejected	1.00
XI-b EP(R1 – D0), lower-error criterion	100	+0.79 [–0.23, +1.41] / –0.86 [–2.27, +0.54]	undetermined	+2.37 [+1.20, +3.34] / +1.30 [–0.43, +3.04]	undetermined	undetermined	0.90
	all	+3.99 [+1.90, +4.61] / +0.49 [–1.47, +2.38]	undetermined	+3.58 [+1.95, +4.57] / +2.30 [+0.31, +4.31]	higher error (depends on split independence)	undetermined (both reported); hypothesis rejected	1.00
XI-c R1 (CV λ) – P1 in W, non-inferiority	100	–2.44 [–2.83, –1.78] / –2.35 [–2.92, –1.79]	lower error, non-inferior	–2.91 [–3.31, –2.34] / –2.94 [–3.59, –2.23]	lower error, non-inferior	lower error	<0.0001
	all	–2.41 [–3.09, –2.03] / –2.88 [–3.40, –2.36]	lower error, non-inferior	–2.88 [–3.47, –2.27] / –3.20 [–3.98, –2.38]	lower error, non-inferior	lower error; hypothesis supported	<0.0001

Flags: designed after earlier results were viewed; Canada (licence-verified expanded edition) rows blind; Alaska rows an unblinded replication; Holm $m = 6$ (auxiliary column). This is a spatial proxy with warm blocks of the same period, not a true extrapolation test. $EP(A - B) = \Delta_W(A - B) - \Delta_I(A - B)$, the contrast in the extrapolation blocks W minus that in the interpolation blocks I (cm). Extrapolation width (median $\sqrt{TDD}$ of W scoring cells minus the maximum of A, split mean): warm_trim $0.97 (^{\circ}C\ d)^{1/2}$ (all W cells outside A); basic warm $-0.002 (^{\circ}C\ d)^{1/2}$ (58.4% outside A); cold 1.74; Alaska warm 0.63. RMSE levels (cell-weighted, n 100 and all): in W, P1 44.38 and 45.10 cm, R1 41.95 and 42.66 cm, D0 41.99 and 40.32 cm; in I, P1 21.93 and 21.98 cm, R1 21.94 and 21.81 cm, D0 23.00 and 23.59 cm. Alaska warm and Canada cold rows were undetermined for all three contrasts. Registered sentences (rendered): XI-a, ‘In a spatial proxy using warm blocks of the same period (extrapolation width $0.97 (^{\circ}C\ d)^{1/2}$ in $\sqrt{TDD}$), no difference was established in the extrapolation loss of direct ML relative to the recalibrated Stefan model (100 labels, all labels)’; XI-b, ‘In the same spatial proxy, no difference was established between the extrapolation losses of the recalibrated-anchor residual and of direct ML (100 labels, all labels)’; with all labels the two editions disagree (basic edition lower error for XI-a and higher error for XI-b), both are reported and the weaker verdict is used; XI-c, ‘In the extrapolation blocks of the same proxy, the recalibrated-anchor residual (cross-validated λ) had 2.44 cm (100 labels) and 2.41 cm (all labels) lower error than the recalibrated Stefan model (cell-weighted) and met the non-inferiority criterion (upper bounds of both CIs below 0.5 cm).’ Source: data/processed/xbatch/XI_climate_extrapolation_retest/sealed/ (xi_hyp.csv, xi_tests.csv, xi_holm.csv, xi_curve.csv); gate passed (xi_gate.csv).

h. XE public inputs at 10 to 500 m within regions, stage 1 (xh0; auxiliary contrast XE-c)

Contrast (within-grid RMSE, R1 with CV λ)	n	Pool	Three-target mean: cell / block-equal	Four-way verdict	Target rows
XE-c R1(xh0) – R1(x25)	200	3/3	–0.09 [–0.10, +0.06] / +0.06 [+0.01, +0.12]	equivalent	Alaska, Canada equivalent; Lena Delta equivalent (margin-dependent, depends on split independence, few blocks)
	500	partial (2/3)	–0.16 [–0.18, +0.05] / +0.05 [–0.05, +0.15]	equivalent	Alaska equivalent; Lena Delta equivalent (same flags)
	1,000	partial (2/3)	–0.17 [–0.22, +0.05] / –0.01 [–0.11, +0.11]	equivalent	Alaska equivalent; Lena Delta equivalent (same flags)
	all	3/3	–0.12 [–0.16, +0.02] / –0.08 [–0.18, +0.02]	equivalent	Alaska, Canada equivalent; Lena Delta undetermined (–0.34 [–0.45, +0.05] / –0.22 [–0.50, +0.06])
XE-c R1(xh0) – P1	200	3/3	–0.02 [–0.06, +0.21] / +0.23 [+0.15, +0.30]	equivalent (margin-dependent)	Alaska equivalent; Lena Delta undetermined; Canada higher error (+0.33 [+0.25, +0.45] / +0.30 [+0.22, +0.39], size below 0.5 cm)
	500	partial (2/3)	–0.24 [–0.30, +0.15] / +0.27 [+0.12, +0.41]	equivalent (margin-dependent; depends on split independence)	Alaska equivalent; Lena Delta undetermined
	1,000	partial (2/3)	–0.19 [–0.27, +0.33] / +0.36 [+0.18, +0.54]	undetermined (margin-dependent)	Alaska equivalent; Lena Delta undetermined
	all	3/3	–0.03 [–0.08, +0.35] / +0.29 [+0.16, +0.43]	equivalent (margin-dependent; depends on split independence)	Alaska equivalent; Lena Delta undetermined; Canada higher error (+0.25 [+0.16, +0.38] / +0.21 [+0.12, +0.31], size below 0.5 cm)
XE-a, XE-b, XE-a0			Pending: registered data deadline 8 October 2026 (XE stage 2: XE-a, XE-b, XE-a0 and the xt2 table).		

Flags: designed after earlier results were viewed; xh0 partly unblinded (the ten H0 inputs were inputs of H19, and the within-grid R1 – P1 contrast of WF9-a had been viewed). All rows are the auxiliary contrast XE-c (four-way verdict, not counted as confirmatory, outside the XE Holm family). xh0 = the 25 covariates plus ten H0 inputs (topographic wetness index and its standard deviation, flow accumulation, two curvatures, relief, surface-water occurrence and distance, tree cover). Within-grid share explained, $1 - SSE_w(M)/SSE_w(P1)$, all labels (descriptive, no CI): Alaska 0.07% (x25) and 0.69% (xh0), Lena Delta –0.06% and 3.88%, Canada –2.18% and –2.15%. Total RMSE, x25 to xh0: Alaska 14.06 to 14.00 cm, Lena Delta 20.89 to 20.31 cm, Canada 29.72 to 26.75 cm (between-grid component 23.62 to 19.75 cm), P1 14.53, 20.62 and 29.10 cm; these total and between-grid contrasts are not registered and not judged. No pre-fixed sentence applies to XE-c; only facts are reported. The reproduction gate was rescored at the registered local-to-cloud tolerance (gate level corrected, not a design change). Source: data/processed/xbatch/XE_hires_covariates/sealed/ (xe_r1b_xh0_tests.csv, xe_r1b_xh0_decomp.csv).

h (continued). XE-e diagnostic: share of the within-grid residual of P1 explained by on-site soil moisture (block cross-validation)

Region	Model	Non-GPR VWC, registered subset (main)	Non-GPR VWC, finite-predictor cells	GPR VWC, registered subset	GPR VWC, finite cells
Alaska	linear regression	−3.54%	−0.79%	−0.54%	−5.56%
	CatBoost (mean of seeds 0 and 1)	−4.89%	−2.40%	+0.82%	−4.16%
Canada	linear regression	−4.58%	+2.55%	n.a.	n.a.
	CatBoost	−4.48%	−6.59%	n.a.	n.a.

XE-e, diagnostic (exploratory): share of the within-grid residual of the recalibrated Stefan model P1 explained by on-site volumetric water content (shallow and deep layers) under five-fold block cross-validation. Flags: designed after earlier results were viewed; blind (the relation of VWC with P1 residuals had not been computed); the registered rule says XE-e changes no verdict. Site data: ABoVE soil thaw depth and moisture validation, version 2 (15 m radius). Registered subsets: Alaska 2,468 cells in 31 blocks, Canada 602 cells in 15 blocks; shares computed on cells in grid groups of at least two cells (2,466 and 599); non-GPR VWC finite in 1,329 and 595 cells. GPR VWC shares error with ALT and is a sensitivity only; Canada has no GPR VWC (n.a.). Reference: a 0.5 cm reduction of within-grid RMSE corresponds to explained shares of 7.9% (Alaska) and 5.0% (Canada). Description: under block cross-validation on-site VWC did not explain the within-grid residual of P1 (main shares −4.89% to −3.54%, sensitivities −6.59% to +2.55%). Run locally on 5 October 2026 (05:41, 8 s); sealed table first opened 05:42:35 KST. Source: data/processed/xbatch/XE_hires_covariates/sealed/x_e_xe_e.csv and x_e_xe_e_meta.json; plan 8.6.

i. XD placement algorithm (XD-1 to XD-3; $R1 \lambda 0.25$, cm; exploratory)

Hypothesis and contrast	n	Pool or target	Δ cell [95% CI]	Δ block-equal [95% CI]	Four-way verdict	Non-inferiority ($P \times 2$)	Holm P
XD-1 S8a – S1	10	PE1 (5/5)	−0.33 [−1.20, +0.66]	−0.16 [−0.91, +0.57]	undetermined; hypothesis rejected		1.00
	40	PE1, partial (2/5)	−0.70 [−1.61, +0.47]	−0.96 [−1.76, −0.22]	undetermined		1.00
XD-2 S8w – S8a (post hoc design, Lena target)	10	Lena Delta (transfer)	−1.86 [−3.30, +0.10]	+0.49 [−0.96, +1.97]	undetermined; hypothesis rejected (higher error in eight Alaska and Canada rows)		1.00
	40	Lena Delta (transfer)	−0.18 [−1.32, +0.67]	−0.51 [−1.49, +0.41]	undetermined		1.00
XD-3 S8a – S2	10	PE1 (5/5)	+0.32 [+0.01, +0.61]	+0.28 [+0.00, +0.56]	higher error (depends on split independence; size below 0.5 cm)	not met (0.20)	1.00
	40	PE1, partial (2/5)	−0.63 [−1.06, −0.21]	−0.62 [−0.98, −0.27]	lower error	met (<0.0001)	<0.0001
XD-3 S8a – S4	10	PE1 (5/5)	+0.09 [−0.36, +0.43]	−0.07 [−0.40, +0.27]	equivalent (depends on split independence)	met (0.020; before correction only)	0.27
	40	PE1, partial (2/5)	+0.22 [−0.20, +0.62]	+0.06 [−0.27, +0.39]	undetermined (margin-dependent; depends on split independence)	not met (0.16)	1.00
XD-3 S8b – S2	10	PE1 (5/5)	+0.45 [+0.12, +0.79]	+0.42 [+0.13, +0.72]	higher error (size below 0.5 cm)	not met (0.75)	1.00
	40	PE1, partial (2/5)	−0.48 [−0.85, −0.08]	−0.47 [−0.80, −0.17]	lower error	met (<0.0001)	<0.0001
XD-3 S8b – S4	10	PE1 (5/5)	+0.22 [−0.15, +0.53]	+0.07 [−0.22, +0.38]	undetermined (margin-dependent; draw-dependent)	not met (0.074)	0.89
	40	PE1, partial (2/5)	+0.38 [−0.01, +0.79]	+0.21 [−0.13, +0.57]	undetermined (margin-dependent)	not met (0.61)	1.00
XD-3 S8c – S2	10	PE1 (5/5)	−0.36 [−0.87, +0.01]	−0.34 [−0.71, +0.03]	undetermined (margin-dependent; draw-dependent)	met (<0.0001)	<0.0001
	40	PE1, partial (2/5)	−1.13 [−1.67, −0.60]	−0.90 [−1.37, −0.42]	lower error	met (<0.0001)	<0.0001
XD-3 S8c – S4	10	PE1 (5/5)	−0.59 [−1.21, −0.19]	−0.68 [−1.09, −0.29]	lower error (depends on split independence)	met (<0.0001)	<0.0001
	40	PE1, partial (2/5)	−0.28 [−0.57, −0.00]	−0.22 [−0.47, +0.04]	undetermined (margin-dependent; draw-dependent)	met (<0.0001)	<0.0001
XD-3 S8p – S2	10	PE1 (5/5)	+0.52 [−0.37, +1.28]	+0.31 [−0.44, +1.06]	undetermined	not met (0.93)	1.00
	40	PE1, partial (2/5)	−0.75 [−1.92, +0.12]	−0.50 [−1.33, +0.34]	undetermined	met (0.019; before correction only)	0.27
XD-3 S8p – S4	10	PE1 (5/5)	+0.29 [−0.49, +0.88]	−0.04 [−0.63, +0.57]	undetermined (margin-dependent)	not met (0.39)	1.00
	40	PE1, partial (2/5)	+0.10 [−0.92, +0.77]	+0.18 [−0.42, +0.78]	undetermined (margin-dependent)	not met (0.31)	1.00

j. XD-4 learned placement policy S9* (S9-DS) against Algorithm P (= S1) in test tasks outside Alaska (descriptive, no test)

Test task or family	v cell-weighted	v block-equal	Direction
LE-1	+0.408	+0.149	higher error in both weightings
LE-2	-0.027	-0.006	lower error in both weightings
CA-2	-0.008	+0.005	weightings disagree
CA-3	-0.036	-0.004	lower error in both weightings
Tibetan Plateau	+0.132	+0.086	higher error in both weightings
Central Russia (not counted; no n 40)	+0.031	+0.018	descriptive
Family Lena Delta (LE-1, LE-2)	+0.190	+0.071	higher error in both weightings
Family Canada (CA-2, CA-3)	-0.022	+0.0002	not consistent
Family Tibetan Plateau	+0.132	+0.086	higher error in both weightings

k. XD-6 region traits and placement effect (descriptive)

Target (transfer)	Blocks	Effective blocks	Covariate between-block share	$\ln(\text{cell-weighted } E / \text{block-mean } E)$	S8a - S1, n 10	S8a - S1, n 40
Alaska	36.2	3.49	0.78	+0.095	-0.96 / -1.97 (lower error)	-1.14 / -2.23 (lower error)
Canada	17.8	3.89	0.83	-0.044	-0.81 / +0.52 (undetermined)	-2.39 / -1.71 (lower error)
Lena Delta	7.6	1.72	0.39	-0.069	+1.70 / +1.02 (undetermined)	+0.98 / -0.22 (undetermined)
W Russia	10.6	7.73	0.92	+0.114	-0.06 / -0.05 (undetermined)	
E Russia	10.8	7.81	0.90	-0.301	+0.56 / -0.72 (undetermined)	
Central Russia	5.4	2.87	0.76	+0.075	-3.03 / -1.55 (undetermined)	
Tibetan Plateau	16.8	10.25	0.78	+0.009	+2.30 / +2.37 (undetermined)	+3.01 / +1.15 (undetermined)
AL-1	3.75	1.26	0.35	+0.006	+0.35 / -1.65 (undetermined)	-1.11 / -1.82 (lower error)
AL-2	8.6	1.17	0.32	-0.102	-0.31 / -0.71 (lower error)	-0.34 / -0.81 (lower error)
AL-3	10.2	2.89	0.71	-0.009	+1.43 / +1.30 (higher error)	+0.35 / +0.15 (undetermined)
AL-4	1.33	1.00	0.18	-0.001	not decidable	not decidable
AL-5	6.8	1.95	0.72	+0.204	-2.10 / -2.90 (lower error)	-2.39 / -4.30 (undetermined)
AL-6	3.25	1.00	0.14	-0.000	not decidable	not decidable

Flags: exploratory (Supplementary Information); designed after earlier results were viewed; XD-1 and XD-3 partly unblinded (S2 and S4 of WF8 viewed); XD-2 partly unblinded and a post hoc design targeting the Lena Delta loss; XD-4 blind (placements of the test tasks never computed before), descriptive, no verdict words and no P values (three families; minimum family-level sign-test P 0.125). Contrasts use R1 (λ 0.25); contrasts between different label sets use two-stage CIs, S8w - S8a uses block resampling with the same labels; Holm m = 20 (auxiliary column). S8a, γ 0 with farthest-first block order and the cell nearest the block centre first; S8b, farthest-first within blocks; S8c, γ 0.5; S8p, γ 1; S8w, S8a labels with design-weighted coefficients. Algorithm P (appendix XC-0): rule 2 excluded S2, S4, S8a, S8b, S8c and S8p on Alaskan targets, so Algorithm P = S1 (cell-random sampling); XC-5b, XC-5c and S-XC3 were therefore not tested and remain in the XC Holm family with P = 1. XD-2 higher-error rows: Alaska transfer n 10 +2.02 [+1.36, +2.80] / +1.94 [+1.02, +2.81], n 40 +1.44 / +3.06; Canada transfer n 10 +4.89 [+1.07, +6.65] / +2.82, n 40 +1.99 / +1.21 (both depend on split independence); Alaska within region n 20 +1.21 / +2.35; Canada within region n 50 +1.78 / +1.07, n 100 +1.30 / +0.87 (both depend on split independence), n 200 +1.02 / +0.86. Post hoc rows outside the registered hypotheses (not counted): S8c - S1 lower error in the four-way verdict at n 10 and 40 but undetermined with the two-stage common CI; these rows do not change Algorithm P. XD-4: v = mean over splits, draws, $n \in \{20, 40\}$ and policy seeds of $(\text{RMSE}(\text{S9}^*) - \text{RMSE}(\text{Algorithm P})) / \text{RMSE}(\text{Algorithm P})$; task RMSE differences (cm, cell-weighted [two-stage CI] (block-equal)): LE-1 n 20 +7.60 [+0.61, +9.12] (+2.43), n 40 +8.82 [+1.30, +10.43] (+3.49); LE-2 n 20 -0.62, n 40 -0.29; CA-2 n 20 +0.19 [-2.01, +2.14], n 40 -0.34 [-1.31, +1.23]; CA-3 n 20 -1.11 [-1.89, +0.30], n 40 -1.18 [-2.06, +0.20]; Tibetan Plateau n 20 +21.72 [+16.47, +26.48] (+13.65), n 40 +7.45 [+4.03, +10.32] (+2.65). Registered sentences (rendered): XD-1, 'No difference between placement combining block allocation and covariate coverage and random placement was established with 10 labels' and '... with 40 labels' (transfer, PE1); XD-2, 'Design-weighted coefficients increased error in Alaska (transfer n 10 and 40, within region n 20) and Canada (transfer n 10 and 40, within region n 50, 100 and 200)' (post hoc design, Lena target); XD-3, 'The error increase of S8 variant V relative to S2 (or S4) was within 0.5 cm' for S8a - S2 (n 40), S8a - S4 (n 10; significant before correction only), S8b - S2 (n 40), S8c - S2 (n 10, 40), S8c - S4 (n 10, 40) and S8p - S2 (n 40; significant before correction only), '... was larger' for S8a - S2 and S8b - S2 at n 10 (statistically distinguishable but below 0.5 cm in size), and 'non-inferiority was not established' for the six other rows; XD-4, 'A placement policy learned and frozen in Alaskan sub-regions had lower error than Algorithm P in 2 of 5 test tasks outside Alaska (0 of 3 families; both weightings, no test)' and '... higher error in 2 of 5 test tasks (2 of 3 families; both weightings, no test).' Common limitation: the candidate

pool consists of already measured cells, so gains are not guaranteed to transfer to field candidates (all land grid cells). XD-6 is descriptive (trait means over splits; effects as cell-weighted / block-equal cm); correlations of traits with effects were not computed. XD-5: part k (continued). Source: data/processed/xbatch/XD_placement_policy/sealed/ (xd_alg_tests.csv, xd_alg_contrasts.csv, xd_alg_holm.csv, xd6_traits.csv; xd4_summary.json, xd4_tasks.csv, xd4_families.csv, xd4_ci.csv); gate passed (gate/xd_gate.csv).

k (continued). XD-5 regret against the best of 200 candidate sets (descriptive)

Test task	U(S9*)	U(Algorithm P = S1)	U(oracle)	Median of 200 candidate sets	Regret, S9*	Regret, Algorithm P
CA-2	−0.005	0	−0.089	+0.002	0.084	0.089
CA-3	−0.024	0	−0.085	−0.028	0.061	0.085
LE-1	+0.275	0	−0.069	+0.001	0.344	0.069
LE-2	−0.017	0	−0.051	−0.024	0.034	0.051
Tibetan Plateau	+0.106	0	−0.090	−0.009	0.195	0.090
Central Russia (not counted)	+0.023	0	n.a.	n.a.	n.a.	n.a.

k (continued). XD-5 cross-validation utility of the learned policies (post hoc)

Task	Family	S9-DS, leave family out	S9-DS, leave task out	S9-GBM, leave family out	S9-GBM, leave task out
LE-1	Lena Delta	+0.074	+0.034	−0.024	−0.013
LE-2	Lena Delta	−0.015	−0.008	−0.029	−0.020
CA-2	Canada	+0.066	+0.081	−0.004	+0.082
CA-3	Canada	−0.031	−0.037	−0.044	−0.025
Tibetan Plateau	Tibetan Plateau	+0.053	+0.072	−0.009	−0.009

XD-5, descriptive (Supplementary Information; no verdict words and no P values), built after the XD-4 tables were opened; local run 5 October 2026, 03:30 to 05:37 KST; sealed tables first opened 05:38:08 KST. U is the relative change of RMSE against the mean of S1 draws, averaged over the two weightings (negative, lower error); regret = $U - U(\text{oracle})$, with the oracle the best of 200 candidate label sets (mean over splits and n). Because Algorithm P = S1, the S9* − S1 description equals XD-4. S9* had lower regret than Algorithm P in 3 of 5 test tasks (CA-2, CA-3, LE-2) and higher regret in 2 (LE-1, Tibetan Plateau); the LE-1 mean is set by four rows of splits 2 and 4 (U(S9*) +0.38, +0.73, +0.71, +0.64; the other four rows −0.11 to −0.03). Cross-validation utility (post hoc comparison after S9* = S9-DS had been frozen): leaving a family out, S9-DS was negative in 2 of 5 tasks (LE-2, CA-3) and S9-GBM in 5 of 5 (−0.044 to −0.004). For the Tibetan Plateau both schemes use the same training set; S9-GBM gave the same value (−0.009) and S9-DS gave +0.053 and +0.072, because the minibatch-order seed includes the fold name. The S9-DS utility is therefore sensitive to the minibatch-order seed; this is not a determinism failure (the frozen model gave identical selections on two GPUs, Jaccard 1.000). Alaska-family values (leave the Alaska family out) are in plan 8.11. Common limitation: the candidate pool consists of already measured cells, so gains are not guaranteed to transfer to field candidates. Source: data/processed/xbatch/XD_placement_policy/sealed/ (xd5_regret.csv, xd5_cv_ds.csv, xd5_cv_gbm.csv, xd5_meta.json); plan 8.11.

l. XJ temperature-derived auxiliary labels

Hypothesis	w	n	Δ cell [95% CI]	Δ block-equal [95% CI]	Four-way verdict	Holm P (m 15)
XJ-1 R1@ w − R1, four main regions, n 0 and 10, w 0.1, 0.3, 1 (six rows)					not tested (licence basis not recorded)	1.00
XJ-2 D1@ w − D1, n 0, w 0.1, 0.3, 1 (three rows)					not tested (licence basis not recorded)	1.00
XJ-3 P1_aux(w) − P1, Tibetan Plateau	0.1	3	−20.85 [−21.80, −19.89]	−21.81 [−23.20, −20.11]	lower error	0.0015
	0.1	10	−8.14 [−8.88, −7.10]	−5.94 [−7.67, −4.02]	lower error	0.0015
	0.3	3	−47.54 [−50.12, −44.50]	−45.18 [−50.11, −39.44]	lower error	0.0015
	0.3	10	−19.63 [−21.74, −16.53]	−12.66 [−17.30, −7.67]	lower error	0.0015
	1	3	−83.17 [−90.21, −73.41]	−66.86 [−79.47, −53.35]	lower error	0.0015
	1	10	−37.50 [−43.30, −28.88]	−20.57 [−31.04, −9.55]	lower error	0.004

Flags: designed after earlier results were viewed; XJ-3 partly unblinded (L39). Target-side auxiliary rows: 39 temperature-derived rows (F3, CC BY 4.0), of which rows outside the B blocks of a split entered the A pool (15, 16, 8, 24 and 28 rows in splits 1 to 5); only direct-label cells are scored. Baseline RMSE (cell-weighted): P0 244.51 cm; P1 201.05 cm (n 3) and 148.14 cm (n 10); P1_aux(w 1) 117.89 and 110.63 cm. Descriptive n 0 rows: −34.85 (w 0.1), −75.81 (0.3) and −124.60 cm (1). Registered sentences (rendered): XJ-3, ‘Temperature-derived auxiliary rows (weight w) reduced the transfer error with k labels by a cm. Temperature-derived labels differ in definition from direct measurements (L39), so this result is not generalized as evidence for label augmentation’, with (w , k , a) =

(0.1, 3, 20.85), (0.1, 10, 8.14), (0.3, 3, 47.54), (0.3, 10, 19.63), (1, 3, 83.17), (1, 10, 37.50); XJ-1 and XJ-2, ‘The source-side design of temperature-derived auxiliary rows was not tested (licence basis not recorded).’ All rows concern one target (Tibetan Plateau). Source: data/processed/xbatch/XJ_tempderived_aux_labels/sealed/ (xj_hyp.csv, xj_tests.csv, xj_holm.csv); gate passed (xj_gate.csv).

m. XK error by grid size within regions (added registration; descriptive)

Region	Grid	Grid cells	Blocks	P1 cell / block	R1 cell / block	P1 – R1 cell [95% CI] / block-equal [95% CI]	CCI v4 cell / block
Alaska	1 km	13,606	74	14.30 / 18.16	13.80 / 17.31	0.50 [0.18, 1.10] / 0.86 [0.29, 1.54]	19.90 / 28.22
	0.05°	105	36	12.94 / 15.29	11.27 / 13.60	1.67 [0.60, 2.69] / 1.70 [0.67, 2.68]	20.96 / 26.78
	0.1°	83	35	12.76 / 15.13	11.22 / 13.40	1.54 [0.50, 2.54] / 1.72 [0.68, 2.69]	21.30 / 26.66
	0.25°	56	35	13.16 / 14.78	11.55 / 13.05	1.61 [0.54, 2.72] / 1.72 [0.64, 2.81]	23.13 / 26.35
Lena Delta	1 km	2,958	20	21.02 / 34.20	21.01 / 32.70	0.01 [–1.39, 1.41] / 1.50 [–0.38, 3.20]	22.45 / 40.23
	0.05°	56	13	24.66 / 26.74	24.97 / 25.90	–0.30 [–3.32, 1.32] / 0.84 [–2.19, 3.48]	28.65 / 31.17
	0.1°	37	12	26.38 / 31.93	26.75 / 31.20	–0.38 [–3.77, 1.72] / 0.72 [–2.23, 3.70]	30.62 / 36.25
	0.25°	22	12	15.85 / 18.64	15.07 / 17.39	0.78 [–1.25, 3.04] / 1.25 [–1.60, 4.26]	20.14 / 23.33
Canada	1 km	747	36	28.84 / 27.00	26.22 / 26.64	2.62 [0.00, 4.42] / 0.36 [–3.38, 3.34]	30.05 / 40.77
	0.05°	37	16	20.29 / 18.75	19.18 / 16.40	1.11 [–1.15, 4.49] / 2.34 [0.34, 4.64]	32.50 / 29.61
	0.1°	26	15	20.16 / 19.27	18.80 / 16.84	1.35 [–0.50, 4.22] / 2.43 [0.53, 4.70]	31.78 / 29.51
	0.25°	19	14	15.85 / 16.27	14.89 / 13.53	0.96 [–1.93, 4.45] / 2.74 [–0.04, 4.87]	33.87 / 28.93

m (continued). Products against R1 at the same grid size, RMSE(product) – RMSE(R1)

Region (cells)	Grid	CCI v4	CCI v5	Wei 2026	Wei 2026 (5 km mask)	Yi-Kimball
Alaska (13,605)	1 km	6.10 [2.54, 9.11] / 10.87	54.22 [12.08, 90.76] / 53.21	2.23 [0.69, 5.01] / –0.94	3.02 [0.83, 5.73] / 3.32 (9,131 cells)	46.09 [10.97, 99.91] / 51.90 (10,605 cells)
	0.1°	10.08 [6.42, 14.12] / 13.26	53.20 [34.47, 71.94] / 63.02	3.88 [0.77, 6.78] / 3.14	4.03 [0.74, 7.71] / 3.08	39.10 [26.43, 52.86] / 48.49
	0.25°	0.78 [–0.73, 5.49] / 7.69	–0.31 [–1.87, 1.44] / 1.79	9.91 [–4.04, 14.05] / –2.08	10.00 [–4.81, 14.13] / –2.03 (1,330 cells)	
Lena Delta (1,342)	1 km	3.14 [–0.04, 6.40] / 4.08	–0.61 [–4.72, 2.64] / 0.09	–1.14 [–6.72, 8.57] / –1.33	–1.26 [–7.10, 9.24] / –1.32	
	0.1°	3.76 [–2.75, 16.12] / 13.75	54.25 [23.70, 78.39] / 61.55	2.99 [–3.66, 10.78] / –2.61	3.32 [–4.23, 11.33] / 7.36 (726 cells)	
	0.25°	12.98 [1.58, 23.37] / 12.67	60.09 [31.09, 87.72] / 67.23	5.42 [–0.14, 13.07] / 7.27	5.42 [0.36, 13.70] / 7.27	

Flags: added registration of 5 October 2026 (commit d721d24, before computation); designed after earlier results were viewed; descriptive, no verdict words. Out-of-block predictions from 0.5° block five-fold cross-validation; grid cells with fewer than three label cells dropped; 1,000 block resamples. First table: models set (cells where P1, R1 and CCI v4 are finite), no mask; RMSE in cm, cell-weighted / block-equal. Second table: RMSE(product) – RMSE(R1) in cm, cell-weighted [95% CI] / block-equal, common set (all product values finite; Alaska Yi-Kimball on its own set), no mask, plus the 5 km mask edition for Wei 2026. Bias at label cells (product – label, 1 km, cell-weighted, common set), reported as fact without a cause: CCI v5 +35.63 cm (Alaska), +58.60 cm (Canada) and +0.27 cm (Lena Delta); CCI v4 –5.09, +10.35 and –6.04 cm. Pre-fixed interpretation rules: rule 1 (all RMSE decrease with grid size) does not hold, so its sentence is not used; rule 2 does not fix the weighting and the two weightings pick different grid sizes in all three regions, so only the fact is reported: ‘In Alaska the RMSE difference between P1 and R1 was 0.50 cm at 1 km and 1.54 to 1.67 cm at 0.05°, 0.1° and 0.25° (cell-weighted).’ Products are compared only as RMSE differences at the same grid size, for example ‘At the 0.1° grid in Alaska, the RMSE of CCI v4, CCI v5, Wei 2026 and Yi-Kimball was larger than that of R1 by 10.08, 53.20, 3.88 and 39.10 cm’ and ‘At the 0.1° grid in the Lena Delta, the RMSE of CCI v5 and Wei 2026 was smaller than that of R1 by 0.61 and 1.14 cm (CIs include zero)’ (cell-weighted); product rankings are not generalized, and Wei 2026 used CALM sites for training (WRAPUP 10). Supplementary Fig. S6. Source: data/processed/xbatch/XK_support_scale/xk_support_scale_v1.csv and xk_meta.json.

n. XL 1 km maps compared with ALT products (added registration; descriptive)

Region	Map	Mean	Quartiles (25–75%)	Mean difference	Mean absolute difference	Pearson r	Spearman ρ	Share of mapped cells with values
Alaska	this study (recalibrated anchor plus residual ML)	56.09	48.41–63.61					
	recalibrated Stefan	57.46	48.92–66.06	–1.38	2.57	0.956	0.961	1.000
	this map, 0.1° cell means	56.09	48.49–63.53	0.00	1.37	0.983	0.983	1.000
	CCI v5 (1997–2023 mean)	105.46	61.30–133.89	–49.37	50.45	0.464	0.642	0.999
	Wei 2026 (2000–2024 mean)	70.79	57.64–71.53	–14.66	18.14	–0.154	0.090	0.994
	Aalto 2018 (2000–2014)	78.82	60.40–102.61	–22.63	37.16	–0.104	0.106	0.991
	Yi-Kimball (2001–2015 mean)	105.15	54.33–137.00	–49.33	53.37	0.338	0.461	0.862
Lena Delta	this study	42.18	40.90–43.19					

Region	Map	Mean	Quartiles (25–75%)	Mean difference	Mean absolute difference	Pearson r	Spearman ρ	Share of mapped cells with values
	recalibrated Stefan	41.09	40.12–42.10	1.10	1.61	0.361	0.387	1.000
	this map, 0.1° cell means	42.18	41.14–43.16	0.00	0.81	0.783	0.816	1.000
	CCI v5	41.24	32.96–48.15	0.94	8.99	−0.226	−0.262	1.000
	Wei 2026	58.64	53.32–63.07	−16.50	16.51	0.045	0.068	0.940
	Aalto 2018	55.16	48.81–60.46	−12.98	13.09	−0.015	−0.023	0.986

Flags: added registration of 5 October 2026 (commit d721d24, before computation); designed after earlier results were viewed; descriptive, no verdict. Mapped cells (cm; difference = this map minus the compared map): Alaska 858,517 of 899,633 domain cells (permafrost fraction at least 50%), Lena Delta 38,086 of 53,011. The 0.1° cell means explain 97% (Alaska, r 0.983) and 61% (Lena Delta, r 0.783) of the variance of the 1 km map; the standard deviation of 1 km values within a 0.1° cell is 1.8 cm in Alaska (total 9.5 cm) and 1.1 cm in the Lena Delta (total 1.8 cm). Registered map sentence (rendered): ‘The 1 km map is a display resolution; its information resolution is bounded by the climate (about 10 km) and soil (about 5 km) inputs.’ The maps carry no accuracy claim; accuracy is compared only at label cells (XG and XK). Areas of large difference are named without a cause: Interior Alaska (CCI v5 and Yi-Kimball, more than 50 cm), about 61–63° N, 141–150° W (Wei 2026), and west of about 148° W at 62–64° N (this map deeper than Aalto 2018); in the Lena Delta, the northern and eastern delta (this map deeper than CCI v5) and most of the grid (this map shallower than Wei 2026 and Aalto 2018). In Alaska, 73.2% of mapped cells lie outside the training range of at least one of the 25 covariates (mainly terrain roughness, slope and topographic position); the constant-width split-conformal interval is ± 20.15 cm (Alaska) and ± 25.39 cm (Lena Delta), and its coverage is not guaranteed in extrapolated cells. Supplementary Figs S1 to S5. Source: data/processed/xbatch/XL_map_products/xl_summary_v1.csv and xl_meta.json.

o. XC workflow applied in sequence, in new random splits of the same regions (PE1 pool; Algorithm P = S1)

Hypothesis and contrast	n	Pool	Δ cell [95% CI]	Δ block-equal [95% CI]	Four-way verdict	Holm P (rank, multiplier)	Branch
XC-1b A1 – A2 (workflow – random placement with fixed recipe)	40	partial (2/5: Lena Delta, Canada)	−0.86 [−1.03, −0.39]	−0.36 [−0.72, +0.003]	undetermined	0.21 (5, 4)	undetermined (minimum detectable effect about 0.5 cm; two-stage 0.46 cm)
XC-1c A1 – A2	160	partial (2/5)	−0.93 [−1.25, −0.40]	−0.54 [−0.94, −0.13]	lower error	0.040 (4, 5)	lower error
XC-2a A1 – A3 (workflow – recalibrated Stefan), non-inferiority 0.5 cm	10	5/5	+0.08 [−0.07, +0.20]	−0.005 [−0.15, +0.14]	equivalent	<0.0001 (1, 8)	non-inferior; XC-2s not superior
XC-2b A1 – A3	40	partial (2/5)	−1.35 [−1.64, −0.80]	−1.00 [−1.42, −0.59]	lower error	<0.0001 (2, 7)	non-inferior and superior (XC-2s adjusted P 0.0008)
XC-2c A1 – A3	160	partial (2/5)	−1.51 [−1.99, −0.88]	−1.29 [−1.78, −0.80]	lower error	<0.0001 (3, 6)	non-inferior and superior (XC-2s adjusted P 0.0008)
XC-5b A1 – A5 (placement under the selection rule)	40				not tested	1.00 (6, 3)	not decidable (Algorithm P = S1, rule 5)
XC-5c A1 – A5	160				not tested	1.00 (7, 2)	not decidable (Algorithm P = S1, rule 5)
XC-F3 A1 – A2, external family F3	10	F3			not tested	1.00 (8, 1)	not decidable (A1 = A2 at n 10); appendix pending: registered data deadline 11 October 2026 (XC-F3, after XF)

o (continued). Regional rows

Contrast	n	Region	Cell-weighted [95% CI]	Block-equal [95% CI]	Verdict
A1 – A2	40	Lena Delta	+0.21 [+0.02, +0.50]	+0.11 [−0.10, +0.32]	undetermined (margin-dependent)
	40	Canada	−1.94 [−2.27, −1.07]	−0.83 [−1.52, −0.15]	lower error
	160	Lena Delta	+0.54 [−0.05, +0.82]	−0.10 [−0.49, +0.27]	undetermined (margin-dependent)
A1 – A3	160	Canada	−2.41 [−2.77, −1.31]	−0.98 [−1.69, −0.28]	lower error
	10	Lena Delta	+0.08 [−0.26, +0.14]	−0.35 [−0.54, −0.18]	undetermined
	10	Canada	−0.84	−0.85	lower error
	10	W Russia	−0.28	−0.27	undetermined

Contrast	<i>n</i>	Region	Cell-weighted [95% CI]	Block-equal [95% CI]	Verdict
A1 – A3, selection family (S-XC4 b, S-XC8)	10	E Russia	+1.34 [+0.95, +1.84]	+1.30 [+0.82, +1.78]	higher error
	10	Central Russia	+0.10	+0.14	equivalent (depends on split independence)
	40	Lena Delta	+0.25 [–0.13, +0.41]	–0.24 [–0.49, –0.01]	equivalent (depends on split independence)
	40	Canada	–2.96	–1.75	lower error
	160	Lena Delta	+0.53 [–0.35, +0.75]	–0.61 [–1.15, –0.10]	undetermined
	160	Canada	–3.55	–1.96	lower error
	40	Alaska (parent excluded)	+1.51 [+0.73, +1.79]	+0.48 [+0.28, +0.67]	higher error
	160	Alaska (parent excluded)	+2.34 [+0.73, +2.86]	+0.68 [+0.36, +1.04]	higher error

Flags: designed after earlier results were viewed; re-test in reused regions (new random splits of the same cells, seeds 201 to 210); XC-1, XC-2 and XC-5 partly unblinded (WF2, WF4, WF8, LGX-N4), XC-F3 blind (registered before running). Holm $m = 8$; the Holm-adjusted P sets the wording; no row needed the 'significant before correction only' flag (XC-1c 0.040). Run on the local server, 5 October 2026, 03:16 to 05:19 KST (208 units, no failure); sealed tables first opened 05:20:39 KST; reproduction gate passed (128 keys, local-to-cloud tolerance). Algorithm P = S1 (appendix XC-0, rule 5): the placement step of the workflow A1 was random nested labels, so A1 = A2 at n 10 (both R1 with λ 0.25) and A1 = A5 at n 40 and 160 (both the selection rule W); XC-1b and XC-1c therefore equal the auxiliary contrast S-XC1 at n 40 and 160 (rule W against the fixed recipe on the same labels), and the placement step contributed nothing. XC-5b, XC-5c, XC-F3, S-XC0 and S-XC3 were not tested and stay in the Holm family with $P = 1$. Because the two arms of the main contrasts share label sets, the main CI is the same-label block-resampling CI; common-resampling and two-stage upper bounds gave the same four-way and non-inferiority verdicts. HK region-level intervals include zero for all five tested rows (XC-2a [–0.92, +1.04] cm), so no region-general sentence is written. Leave one region out (auxiliary): XC-1b and XC-1c are undetermined with the Lena Delta only (+0.21, +0.54) and lower error with Canada only (–1.94, –2.41); without Canada, XC-2a has higher error (+0.31 [+0.12, +0.44] / +0.21 [+0.04, +0.37], both upper bounds below 0.5 cm); without E Russia, XC-2a has lower error (–0.24 / –0.33). S-XC4 (a), targets more than 2 cm worse than P0 (A1 / A2 / A3), independent-region family: n 10 1 (E Russia) / 1 (E Russia) / 0; n 40 0 / 0 / 1 (Canada); n 160 1 (Alaska) / 1 (Canada) / 1 (Canada); all labels 1 (Alaska) / 2 (Canada, E Russia) / 1 (Canada). S-XC4 (b), regions more than 0.5 cm worse than the recalibrated Stefan model (cell-weighted point estimate): n 10, 1 of 7 (E Russia +1.34); n 40, 1 of 4 (Alaska, selection family, +1.51); n 160, 2 of 3 (Lena Delta +0.53, Alaska +2.34); Alaska at n 10 was +0.05 cell-weighted but +1.75 [+1.34, +2.16] block-equal; Tibetan Plateau n 10 –21.00 / –17.54, n 40 –29.68 / –23.46. Auxiliary contrasts (verdict words, not counted): S-XC1 (rule W – fixed R1, same labels) n 10 –0.87 [–1.38, –0.38] / –0.89 [–1.43, –0.35], lower error (depends on split independence); S-XC2 (stacking candidates added to W) n 40 –0.12 / +0.30 equivalent, n 160 –0.41 / +0.03 undetermined; S-XC5 regret A1 – A6 n 10 +1.73 / +1.41 (higher error), n 40 +0.24 / +0.21, n 160 +0.42 / +0.20 (undetermined); S-XC6 (post hoc threshold) median RMSE(P0) – RMSE(A1 at the largest n) 0.37 / 0.21 cm for 16 targets with $|b_{10}| \geq 8$ cm and –1.81 / –0.57 cm for 5 targets below; S-XC8, PE1 all labels A1 – A2 –1.29 / –1.33 and A1 – A3 –1.43 / –1.56 (lower error); S-XC9, equal and $|b_{10}|$ -proportional budgets gave the same allocation; S-XC10 (A1 – P0, margin 0.5 cm) n 10 –1.87 / –1.58 (lower error, non-inferior), n 40 –0.22 / –0.14 and n 160 –0.06 / –0.20 (undetermined, non-inferior, margin-dependent), E Russia n 10 +2.30 [+1.35, +4.54] / +4.00 (higher error); S-XC11 (descriptive) share of reducible error removed, Canada 0.20 to 0.25, Lena Delta –0.04 to –0.09, Alaska –0.42 to –1.64; S-XC12 ρ 0.17 [–0.30, 0.86] / –0.06 [–0.49, 0.65] (21 targets, 7 families). Within-region auxiliary arm XC-r (descriptive verdicts, outside the Holm family): rule W – R1 (cross-validated λ) three targets n 200 –0.35 [–0.49, +0.06] / +0.15 [–0.14, +0.46] equivalent (depends on split independence, margin-dependent), n 500 +0.03 / +0.14 and n 1,000 +0.36 / +0.05 undetermined (Alaska rows equivalent at all three). Registered sentences (rendered; all with 'in new random splits of the same regions'): common sentence, 'Algorithm P was chosen in the Alaska family, and the cells of the evaluation regions were used to design the components (placement, rule W); the evaluation splits are new partitions of the same cells', with the fact that the placement step was random placement; XC-1b, 'the workflow that applies the placement algorithm, the ten-label bias diagnostic and the label-count method rule in sequence established no difference from random placement with the fixed residual recipe at 40 labels (minimum detectable effect about 0.5 cm)', Lena Delta row +0.21 cm [0.02, 0.50], undetermined; XC-1c, '... had 0.93 cm (cell-weighted) lower error than random placement with the fixed residual recipe at 160 labels (Lena Delta and Canada, two regions)', Lena Delta row +0.54 cm [–0.05, 0.82], undetermined; XC-2a, 'the error increase of the workflow relative to the recalibrated Stefan model was within 0.5 cm (non-inferior, 10 labels)', 'in E Russia the error was larger (Δ +1.34 cm, CI [0.95, 1.84])', 'of seven independent regions (five PE1 regions, Alaska, Tibetan Plateau) the workflow was more than 0.5 cm worse than the recalibrated Stefan model in one (E Russia, +1.34 cm)' and 'non-inferiority refers to the pooled mean and did not hold in E Russia', Lena Delta row +0.08, undetermined; XC-2b and XC-2c, non-inferior at 40 and 160 labels and '1.35 cm' and '1.51 cm lower error than the recalibrated Stefan model', with one of four evaluated regions (Alaska, selection family, +1.51 cm) and two of three (Lena Delta +0.53 cm, Alaska +2.34 cm) more than 0.5 cm worse, and 'non-inferiority refers to the pooled mean and did not hold' in those regions; Lena Delta rows +0.25 (equivalent) and +0.53 (undetermined); XC-5b, XC-5c, not tested (Algorithm P = S1); XC-F3, not tested (A1 = A2 at n 10); the registered 'no eligible region' sentence is written after the XF data deadline. Source: data/processed/xbatch/XC_workflow_end_to_end/sealed/ (xc_hyp.csv, xc_contrasts_main.csv, xc_holm.csv, xc_sentences.json, xc_harm_a.csv, xc_harm_b.csv, xc_loo.csv, xc_contrasts_aux.csv, xc_xcr_contrasts.csv); docs/EXPERIMENT_PLAN_FINAL_BATCH_2026-10-04.md 8.10.

p. XF

Analysis	Status
XF new independent regions	Pending: registered data deadline 11 October 2026 (XF).

q. XM open 10 to 20 m land-cover and vegetation inputs of the 1 km label cell, within regions (added registration; exploratory; R1 with CV λ , $xw - x25$, cm)

Hypothesis	Component	n	Pool or target	Δ cell [95% CI]	Δ block-equal [95% CI]	Four-way verdict
XM-a (main)	within-grid	all	three-target mean (3/3)	−0.15 [−0.25, −0.05]	−0.25 [−0.35, −0.14]	lower error (size below 0.5 cm); Holm P 0.019
	within-grid	all	Alaska / Lena Delta / Canada	−0.03 / −0.52 / +0.09	−0.05 / −0.72 / +0.03	lower error / lower error / equivalent
	within-grid	1,000	Alaska and Lena Delta mean (partial, 2/3)	−0.21 [−0.28, −0.03]	−0.25 [−0.39, −0.11]	lower error (size below 0.5 cm); Holm P 0.04
	within-grid	500	Alaska and Lena Delta mean (partial, 2/3)	−0.17 [−0.20, +0.03]	−0.13 [−0.25, −0.01]	equivalent; Holm P 0.135
XM-b (auxiliary)	total	all	three-target mean (3/3)	−0.70 [−0.93, −0.44]	−0.74 [−1.03, −0.44]	lower error
	total	all	Alaska / Lena Delta / Canada	−0.12 / −1.50 / −0.49	+0.03 / −2.08 / −0.16	equivalent / lower error / undetermined
XM-c (auxiliary)	between-grid	all	Lena Delta	−1.80 [−2.35, −1.10]	−2.11 [−2.96, −1.25]	lower error

XM, added registration (docs/EXPERIMENT_PLAN_FINAL_BATCH_ADDENDUM_XM_2026-10-05.md; commit ff7c97f, 5 October 2026, 13:23 KST, before feature extraction). Flags: designed after earlier results were viewed (XE xh0 and XE-e tables); exploratory (Supplementary Information); partly unblinded (the tree-cover and water-occurrence inputs of xh0 overlap with the WorldCover tree and water fractions, and the xh0 within-grid contrast had been viewed); outside the XE Holm family. Inputs: xw = the 25 covariates plus the fractions of 11 ESA WorldCover 2021 v200 classes (10 m) in the 1 km label cell and three Sentinel-2 L2A (20 m) summer indices (July and August 2019 to 2023; cell median NDVI, median NDMI and NDVI standard deviation); the WorldCover-only variant xw_lc is auxiliary and not shown. Contrast: R1 with λ chosen by block cross-validation, xw minus $x25$, on the same labels, splits and seeds; components as in Methods. Holm family: XM-a at n 500, 1,000 and all ($m = 3$). Verdict: XM-a partially supported (lower error at 1,000 and all labels, equivalent at 500). Registered sentence (rendered): ‘Adding the WorldCover 10 m land-cover fractions and the Sentinel-2 20 m summer vegetation indices of the 1 km cell reduced the ERA5-Land within-grid error of R1 by 0.15 cm (within-grid share explained 1.2%) (statistically distinguishable, size below 0.5 cm). As an exploratory result it does not change the within-grid statement or the map wording of the main text and is reported as a candidate for a confirmatory test. XM does not distinguish the causes of the remaining variation.’ Facts, without interpretation: the total RMSE gain was concentrated in the Lena Delta (−1.50 cm), mostly in the between-grid component (−1.80 cm); in Alaska and Canada the total RMSE change was within 0.5 cm or undetermined. Coverage of label rows: WorldCover 100% in all three targets; Sentinel-2 100% (Alaska), 99.28% (Lena Delta) and 99.87% (Canada); no group was dropped by the 90% rule, so xw is the main variant. Sealed tables first opened 5 October 2026, 14:20:47 KST, after the reproduction gate of the re-used $x25$ shards passed (35,442 keys, no failure). Source: data/processed/xbatch/XM_landcover_vegetation/sealed/ (xm_tests.csv, xm_hypotheses.csv, xm_holm.csv); addendum XM, results.

Supplementary Table S3. Zero-label risk table (WF0, post hoc description, no verdict words).

Number of the 30 targets (25 LG targets and five licence-verified LGD targets; seven independent regions and 20 sub-regions) in which the method minus P0 exceeded +2 cm, with median and maximum (cm). Cell-random draws, $\alpha = 1$, CatBoost. D0_1.0 direct ML; R1_1.0 residual weight 1.0; R1_0.25 low-weight residual (equal to R0 at zero labels); D1_1.0 physics augmentation; R2_1.0 augmentation plus anchor plus residual with λ 1.0. The count uses the point estimate only. Counting only the seven independent regions, targets above +2 cm at zero labels were D0 5/7, R1_1.0 2/7, D1 1/7, R1_0.25 0/7 and R2_1.0 0/7. Counting targets whose two CIs both exceeded zero (higher error) instead of the 2 cm threshold gives D0_1.0 14, R1_1.0 10, D1_1.0 8, R1_0.25 6 and R2_1.0 5 of 30 (post hoc calculation, C1 README 2.8). Source: paper/claims/C1_label0_safety/tables/wf0_risk.csv.

Risk table (C1 README 2.5)

<i>n</i>	Method	Targets	Targets above +2 cm	Median (cm)	Maximum (cm)
0	P1(= P0)	30	0	+0.00	+0.00
0	D0_1.0(direct ML)	30	18	+2.69	+38.46
0	R1_1.0(residual weight 1.0)	30	13	+1.55	+38.54
0	R1_0.25(low-weight residual)	30	5	+0.15	+6.29
0	D1_1.0(physics augmentation)	30	2	+0.13	+3.89
0	R2_1.0(augmentation + anchor + residual)	30	1	−0.02	+6.03
10	P1	30	2	+0.06	+2.27
10	D0_1.0	30	12	+1.49	+12.89
10	R1_1.0	30	11	+1.34	+15.19
10	R1_0.25	30	2	+0.49	+2.71
10	D1_1.0	30	5	+0.43	+5.00
10	R2_1.0	30	2	+0.31	+7.50
all	P1	30	4	+0.13	+4.64
all	D0_1.0	30	7	+0.27	+7.35
all	R1_1.0	30	5	+0.16	+9.14
all	R1_0.25	30	4	−0.13	+3.38
all	D1_1.0	30	5	−0.17	+12.72
all	R2_1.0	30	7	−0.36	+10.70

Supplementary Table S4. Selection rule W and bias diagnosis (WF4-a, WF4-b, WF4-c, selection counts).

W chooses among seven candidates by block cross-validation within the target labels (Methods); fixed recipe R1 with λ 0.25. Non-inferiority margin 0.5 cm. Designed after the label-grid results had been viewed and registered before running. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: |x parent region excluded from the source, |i rest of the parent kept, |r within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. WF4-a pooled contrast W – R1(0.25) (C5 README 2.1)

ID	n (pool)	Δ	CI cell / block-equal	Verdict	Common CI, verdict	Holm P	P_{ni} , non-inferior	Δ (% of P0)
E1	10(regions 4/4)	-0.25	[-1.11, 0.73] / [-1.09, 0.69]	undetermined	[-1.56, 1.30] / [-1.63, 1.20], undetermined	0.66	0.07, False	-0.82
E2	40(regions 2/4, Lena Delta and Canada)	-0.87	[-1.20, -0.27] / [-1.28, -0.46]	lower error	[-1.31, 0.07] / [-1.47, -0.29], undetermined (ci_dependence = depends on split independence)	0.01(0.0054)	0.00, True	-2.85
E3	160(regions 2/4, Lena Delta and Canada)	-1.32	[-1.70, -0.68] / [-1.77, -0.83]	lower error	[-1.86, -0.34] / [-2.02, -0.63], lower error	0.00(0.0004)	0.00, True	-4.31
E4	all(regions 4/4)	-0.74	[-1.65, 0.19] / [-1.75, -0.07]	undetermined	[-2.02, 0.62] / [-2.18, 0.33], undetermined (ci_dependence empty; the cell-weighted upper limit 0.62 exceeds 0.5 cm, so non-inferiority is not met under the common CI; this check)	0.25	0.00(0.0047), True	-2.41

b. WF4-a regional rows (C5 README 2.2)

ID	Target	n	Δ	CI cell / block-equal	Verdict	Common-CI verdict
E6	Canada	40	-2.10	[-2.71, -0.95] / [-2.56, -0.98]	lower error	lower error
E7	Canada	160	-2.72	[-3.41, -1.51] / [-3.21, -1.40]	lower error	lower error
E8	Canada	10	-0.86	[-1.35, 0.11] / [-1.53, -0.15]	undetermined	undetermined
E9	Canada	all	-3.06	[-3.81, -1.74] / [-3.50, -1.49]	lower error	lower error
E10	Lena Delta	40	0.35	[0.03, 0.67] / [-0.21, 0.26]	undetermined	undetermined
E11	Lena Delta	160	0.07	[-0.36, 0.48] / [-0.61, -0.02]	undetermined	undetermined
E12	Lena Delta	10	0.70	[0.56, 1.03] / [0.80, 1.31]	higher error	higher error
E13	Lena Delta	all	0.46	[-1.15, 1.81] / [-0.69, 1.02]	undetermined	undetermined
E14	W Russia	10, all	-0.42, 0.49	[-3.47, 3.20] / [-3.20, 3.47]; [-1.90, 3.43] / [-1.71, 3.93]	undetermined, undetermined	undetermined, undetermined
E15	E Russia	10, all	-0.43, -0.84	[-2.08, 0.69] / [-2.12, -0.19]; [-3.23, 0.46] / [-3.60, -1.14]	undetermined, undetermined	undetermined, undetermined

c. WF4-b, targets with lower error than P1 (C5 README 2.3)

ID	n	W lower error	R1(0.25) lower error	W higher error	R1(0.25) higher error
E18	160(17/30)	3(AL-5 i, CA-3 i, Canada)	4(AL-5 i, CA-2 x, Canada, Lena Delta)	3(AL-1 x, AL-2 i, AL-2 x)	3(AL-1 i, AL-1 x, AL-2 i)
E19	all(21/30)	2(Canada, Tibetan Plateau)	8(AL-2 x, CA-2 i, CA-2 x, Canada, LE-1 x, Lena Delta, Central Russia(LGD edition, Russia_C-lgd), Tibetan Plateau)	3(AL-1 i, AL-2 x, AL-3 x)	1(AL-1 x)

d. W against the physics baselines (curve table, auxiliary) (C5 README 2.4)

ID	Target	<i>n</i>	W – P0 cell / block-equal	Verdict	W – P1 cell / block-equal	Verdict
E20	Lena	10	0.89 [0.33, 1.72] / [0.87, 1.94]	higher error	0.67 [0.47, 0.89] / [0.47, 0.94]	higher error
E21	Delta					
E21	Lena	40	0.20 [-0.37, 1.04] / [0.34, 1.39]	undetermined	0.22 [-0.20, 0.48] / [-0.70, -0.08]	undetermined
E22	Delta					
E22	Lena	160	0.02 [-0.51, 0.81] / [0.07, 1.13]	undetermined	-0.07 [-0.64, 0.29] / [-1.19, -0.39]	undetermined
E23	Delta					
E23	Canada	10	0.51 [-0.97, 1.52] / [-1.54, 0.62]	undetermined	-1.75 [-2.50, -0.63] / [-2.84, -0.95]	lower error
E24	Canada	40	0.90 [-0.53, 1.76] / [-1.24, 1.28]	undetermined	-3.08 [-3.93, -1.66] / [-3.82, -1.71]	lower error
E25	Canada	160	0.35 [-1.20, 1.11] / [-1.84, 0.83]	undetermined	-3.84 [-4.80, -2.29] / [-4.55, -2.22]	lower error
E26	Canada	all	0.32 [-1.34, 1.14] / [-1.91, 0.99]	undetermined	-4.32 [-5.35, -2.67] / [-4.96, -2.41]	lower error
E27a	Alaska(x)	10	0.91 [0.58, 1.42] / [1.14, 2.13]	higher error	0.93 [0.67, 1.19] / [0.55, 0.98]	higher error
E27b	Alaska(x)	40	0.98 [0.44, 1.57] / [0.93, 1.91]	higher error	0.75 [0.28, 1.09] / [-0.17, 0.29]	undetermined
E27c	Alaska(x)	160	2.14 [0.28, 3.18] / [0.71, 1.94]	higher error	2.19 [0.23, 3.11] / [-0.45, 0.64]	undetermined
E27d	Alaska(x)	all	0.26 [-0.41, 1.12] / [-0.04, 1.80]	undetermined	0.19 [-0.54, 0.82] / [-1.30, 0.30]	undetermined
E28a	W Russia	10	-12.94 [-17.44, -8.02] / [-18.63, -7.89]	lower error	-0.97 [-4.09, 2.72] / [-3.64, 3.42]	undetermined
E28b	W Russia	all	-13.49 [-17.64, -8.93] / [-19.24, -8.62]	lower error	0.01 [-2.48, 2.99] / [-2.09, 3.85]	undetermined
E28c	E Russia	10	0.00 [0.00, 0.00] / [0.00, 0.00](file value absolute value 1e-14 below)	equivalent	0.34 [-1.30, 0.92] / [-1.47, 0.10]	undetermined
E28d	E Russia	all	0.00 [0.00, 0.00] / [0.00, 0.00](same)	equivalent	0.09 [-2.46, 1.09] / [-2.92, -0.58]	undetermined

e. Methods chosen by W (counts over splits and draws; this check) (C5 README 2.5)

ID	Target	<i>n</i> 10	<i>n</i> 40	<i>n</i> 160	All
E29	Canada	R1@1.0 19, P0 4, P2 2	R1@1.0 22, P0 2, P2 1	R1@1.0 24, P0 1	R1@1.0 5
E30	Lena Delta	P0 8, P2 7, R1@1.0 4, R1@0.25 2, D1 2, R2@0.25 2	P0 11, R1@1.0 5, P2 4, R1@0.25 3, R2@0.25 2	P0 12, R1@1.0 8, R1@0.25 3, P2 2	P0 2, R1@0.25 2, R1@1.0 1
E31	W Russia	P2 23, R1@1.0 2	none	none	P2 4, R1@1.0 1
E32	E Russia	P0 25	none	none	P0 5

f. Point estimates against fixed recipes (this check) (C5 README 2.5)

ID	Target	<i>n</i>	W – R1(0.25)	W – R1(1.0)
E33	Canada	10, 40, 160, all	-0.86, -2.10, -2.72, -3.06	0.98, 0.12, 0.00, 0.00
E34	Lena Delta	10, 40, 160, all	0.70, 0.35, 0.07, 0.46	0.30, 0.35, -0.12, 1.11
E35	Alaska(x)	10, 40, 160, all	0.36, 0.29, 1.89, 0.54	-8.78, -5.88, -3.91, -0.15
E36	E Russia	10, all	-0.43, -0.84	-4.63, -5.29
E37	W Russia	10, all	-0.42, 0.49	0.31, 1.04

g. Bias diagnosis records (WF4-c, WF0) (C2 README 2.1)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
A1	recalibration gain(P0 – P1) best ML gain(P0 – best ML) rank correlation, all labels	0.66, p 8.15e-05, 30 targets	none(label: 'post hoc descriptive(verdict word none)'). note: best ML five recipe among post hoc selection optimistic
A2	P0 RMSE best ML gain rank correlation	0.51, p 0.00362	none(post hoc descriptive)
A3	labels 10 diagnostic rule W gain(P0 – W, all labels) rank correlation	0.38 [0.17, 0.55], 28 targets, resamples 10000 times, one-sided p 0.0002	Supported. Column verdict of the rows with scope == 'verdict': 'Supported: positive rank correlation (ρ 0.38, CI [0.17, 0.55])'. Same rows: role main, confirmatory False, blind True, design_note 'designed after viewing results'
A4	signed bias gain rank correlation	-0.01 [-0.17, 0.18]	auxiliary(verdict word none)
A5	diagnostic fixed R1(λ 0.25) gain(P0 – R1) rank correlation	0.47(CI column none)	auxiliary
A6	P1 – P0, labels 10, main 4 regions mean(AB4)	-2.45 [-3.04, -1.88], block-equal -2.62, Holm p 0.001	lower error. abstract rule (a). S-a calculation same quantity(draws seed differs) analytic P1 – P0 splits distribution viewed(2026-09-30 03:25)
A7	R1 – P1, labels 10, main 4 regions mean(AB5)	-0.18 [-0.53, -0.01], block-equal -0.29, Holm p 0.136	lower error(before correction). abstract rule (b) 'no difference established'
A8	R1 – P0, all labels, main 4 regions mean(AB8)	-2.64 [-3.48, -1.85], block-equal -2.64, Holm p 0.001	lower error. abstract rule (a)
A9	R1 – P1, all labels, main 4 regions mean(AB9)	-0.40 [-0.76, -0.22], block-equal -0.51, Holm p 0.001	lower error. size below 0.5 cm
A10	A8 – A9 = P1 – P0(all labels, main 4 regions mean)	-2.23(A8 85 %)	none(arithmetic)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
A11	AB8 regions rows(R1 – P0, all labels)	Lena Delta -0.78 $[-1.38, 0.08]$, Canada $+3.38$ $[1.10, 4.40]$, W Russia -13.98 $[-16.05, -12.06]$, E Russia $+0.84$ $[-0.43, 3.19]$	W Russia lower error, Canada higher error, Lena Delta·E Russia undetermined(improvement significant regions 1/4)
A12	AB9 regions rows(R1 – P1, all labels)	Lena Delta -0.80 $[-1.51, -0.44]$, Canada -1.26 $[-1.57, -0.90]$, W Russia -0.48 $[-1.05, 0.01]$, E Russia $+0.93$ $[0.00(\text{fourth decimal } 0.0002), 1.28]$	Lena Delta and Canada lower error, W Russia undetermined, E Russia higher error. In the LGX auxiliary column of the same contrast (data/processed/paper_figs/fig3_c.csv, L4 rows), E Russia is undetermined, $[-0.02, 1.28]$ (depends on the resampling)
A13	R1 – P0, all labels, Lena Delta and Canada·Alaska(x) 3 regions mean	$+0.77$ $[0.02, 1.28]$, block-equal $+1.15$ $[0.66, 1.63]$	higher error. ci_dependence 'depends on split independence'
A14	R1 – P*, all labels, main 4 regions mean(P* = P0@tdm)	-1.64 $[-2.95, -0.74]$, block-equal -2.23 $[-3.17, -1.24]$	lower error

h. Source cross-validation tuning of neural networks, tuned minus default (LGF-N2; cm, four main regions) (C5 README 2.6)

ID	Learner	R1 n 0 λ 0.25	R1 n 10 λ 0.25	R1 all λ 0.25	D0 n 0	D0 n 10	D0 all	R1 n 0 λ 1.0	R1 n 10 λ 1.0	R1 all λ 1.0
E38	MLP	0.06 equivalent	0.43 unde- terminated	0.25 equivalent‡	1.51 higher error‡	2.96 higher error	1.10 higher error‡	-0.27 unde- terminated	0.70 unde- terminated	-0.01 unde- terminated
E39	multi-head MLP	0.24 higher error	0.05 equivalent	-0.02 equivalent	1.33 higher error	2.56 higher error	1.67 higher error	0.75 higher error‡	-0.22 unde- terminated	0.12 unde- terminated
E40	reduced FT-T	0.01 equivalent	0.02 equivalent	-0.19 lower error‡	-0.82 unde- terminated	-0.36 unde- terminated	-0.16 unde- terminated	0.20 unde- terminated	0.10 unde- terminated	-0.76 lower error‡
E41	RealMLP	-0.06 equivalent‡	0.19 equivalent‡	-0.08 equivalent	0.51 unde- terminated	-0.23 unde- terminated	-0.13 unde- terminated	0.23 unde- terminated	0.74 unde- terminated	-0.34 unde- terminated

i. LGF-N2 verdicts by learner (C5 README 2.6)

ID	Learner	Verdict wording (summary of the file values)	Default kept (k_default)
E42	MLP	source-CV tuning increased the transfer error(D0 n 0·10·all higher error)(contrasts with insufficient precision 6/9)	R 0/4, D 1/4
E43	multi-head MLP	source-CV tuning increased the transfer error(R1 n 0 λ 0.25· λ 1.0, D0 n 0·10·all higher error)(contrasts with insufficient precision 4/9)	R 2/4, D 1/4
E44	reduced FT-T	the tuned version had lower error(R1 all λ 0.25· λ 1.0 lower error)(contrasts with insufficient precision 6/9)	R 2/4, D 2/4
E45	RealMLP	precision insufficient(equivalence not decidable, contrast 6/9)	R 2/4, D 2/4
E46	all	condition for a learner-general sentence not met(learners meeting the robust-sentence rule: none)	

Supplementary Table S5. Combination structures and augmentation (L10, L2, L12, L17, L4, recalibration method).

Values: cell-weighted Δ [95% CI] / block-equal [95% CI] in cm. Pools at 40 and 160 labels contain the Lena Delta and Canada only. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: | x parent region excluded from the source, | i rest of the parent kept, | r within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. Residual structure versus physics-input structure, 0 to 10 labels (C3 README 2.1)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E01	R0 – F1k, n 0	–2.72 [–4.34, –1.73] / [–3.93, –1.97]	lower error(pool 4/4. region rows lower error 2/4, E08)
E02	R0 – F1a, n 0	–2.49 [–3.35, –1.56] / [–2.97, –1.11]	lower error(pool 4/4. region rows lower error 2/4, E08b)
E03	R1 – F1k, n 10	–3.74 [–5.18, –2.70] / [–5.12, –2.96]	lower error(pool 4/4. region rows lower error 2/4, E07)
E04	R1 – F1n, n 10	–2.52 [–3.26, –1.56] / [–2.90, –1.16]	lower error(pool 4/4. region rows lower error 2/4, higher error 1/4 Canada, E08a)
E05	L10 hypothesis verdict	'Supported: the residual structure has lower error than the physics-input structure'; holm_p 0.0023 (auxiliary column within the item)	supported(blind)
E06	AB7 = R1 – F1k, n 10	–3.74 [–5.23, –2.75] / [–5.14, –2.98]; holm_p 0.001; '(a) direction stated'. note: F1k inputs have recalibration coefficient E_n none(LG 6A.3). same n P1 – P0 –2.45(E36)	lower error
E07	AB7 regions rows	Lena Delta –1.09 [–2.37, 0.19] / [–0.24, 2.27]; Canada +3.44 [–1.57, 5.38] / [–1.67, 1.90]; W Russia –13.98 [–16.60, –11.22] / [–18.68, –12.13]; E Russia –3.33 [–4.78, –0.76] / [–3.92, –0.12] ‡	undetermined, undetermined, lower error, lower error
E08	R0 – F1k, n 0 regions rows	Lena Delta –3.00 [–4.28, –1.60] / [–1.63, 1.22]; Canada –0.31 [–6.00, 2.11] / [–6.30, –1.90]; W Russia –3.96 [–5.81, –1.73] / [–6.54, –2.57] ‡; E Russia –3.61 [–5.53, –1.20] / [–5.12, –0.75] ‡	undetermined, undetermined, lower error, lower error
E08a	R1 – F1n, n 10 regions rows	Lena Delta –1.86 [–3.26, –0.22] / [–0.60, 2.11]; Canada +3.12 [1.42, 3.97] / [0.07, 2.74] ‡; W Russia –7.50 [–9.08, –5.83] / [–10.45, –6.14]; E Russia –3.83 [–5.27, –0.72] / [–3.97, –0.03] ‡	undetermined, higher error, lower error, lower error
E08b	R0 – F1a, n 0 regions rows	Lena Delta –2.83 [–4.12, –1.61] / [–1.59, 1.18]; Canada +1.53 [–0.04, 2.29] / [–1.09, 1.58]; W Russia –4.12 [–6.16, –1.57] / [–6.84, –2.24] ‡; E Russia –4.55 [–6.27, –1.90] / [–5.86, –1.40]	undetermined, undetermined, lower error, lower error
E09	R1 – F1k, R1 – F1n, n 3(auxiliary)	–3.02 [–4.65, –1.93] / [–4.14, –2.16]; –3.17 [–4.00, –2.10] / [–3.39, –1.51]	lower error, lower error(both pool 4/4)
E09a	same contrast n 3 regions rows(auxiliary)	R1 – F1k: Lena Delta –2.05 [–3.25, –0.70] / [–0.88, 1.73]; Canada +1.40 [–4.62, 3.83] / [–4.73, –0.52]; W Russia –7.96 [–10.07, –5.50] / [–10.88, –6.18]; E Russia –3.44 [–5.01, –0.78] / [–3.91, 0.11]. R1 – F1n: Lena Delta –2.61 [–3.92, –1.07] / [–1.11, 1.68]; Canada +2.02 [0.55, 2.81] / [–0.57, 1.98]; W Russia –7.95 [–10.14, –5.35] / [–10.96, –5.98]; E Russia –4.13 [–5.79, –1.05] / [–4.41, –0.20] ‡	F1k: undetermined, undetermined, lower error, undetermined. F1n: undetermined, undetermined, lower error, lower error
E10	R1 – F1k, R1 – F1n, all(auxiliary)	–2.93 [–3.90, –1.73] / [–3.88, –1.80]; –0.77 [–1.60, 0.37] / [–1.21, 0.48]	lower error, undetermined
E11	missing substitute sensitivity R1 – F1k@tr, n 10	–3.75 [–5.21, –2.71] / [–5.16, –2.99]; 'missing substitute statistics training side median changed four-way verdict same'	lower error

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E11a	L29 P* companion: R0@tddm – F1k(n 0), R1@tddm – F1k(n 10)	–3.45 [–4.73, –2.37] / [–3.98, –2.09]; –4.37 [–5.63, –3.29] / [–5.16, –3.01]. Region rows: R0@tddm – F1k Lena Delta –3.92, Canada –1.28, W Russia –4.61, E Russia –3.98; R1@tddm – F1k Lena Delta –2.29, Canada +1.88, W Russia –13.44, E Russia –3.61. Verdict: 'P* = P0@tddm. The contrasts of L4, L8 and L10 are also reported against P*'	lower error, lower error(auxiliary rows, pool 4/4. region rows lower error 2/4, W Russia:E)

b. 40 to 160 labels: reversal and regional rows (C3 README 2.2)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E12	R1 – F1k, n 40	+2.38 [0.52, 3.25] / [1.17, 2.95]; 'depends on split independence'	higher error(pool 2/4: Lena Delta and Canada. region rows higher error 1/2, Canada)
E13	R1 – F1n, n 40	+1.85 [0.73, 2.61] / [1.34, 3.09]	higher error(pool 2/4. region rows higher error 1/2, Canada)
E14	R1 – F1k, n 160	+2.23 [0.75, 3.04] / [1.30, 3.09]	higher error(pool 2/4. region rows higher error 1/2, Canada)
E15	R1 – F1n, n 160	+1.98 [0.80, 2.73] / [1.35, 3.12]	higher error(pool 2/4. region rows higher error 1/2, Canada)
E16	of the four above region rows, n 40	F1k: Lena Delta –0.38 [–1.38, 0.73] / [0.64, 2.71], Canada +5.14 [1.58, 6.60] / [0.97, 3.91] ‡, Alaska(x) –7.46 [–9.21, –3.03] / [–3.66, –1.83]. F1n: Lena Delta –0.76, Canada +4.46 [2.54, 5.41] / [1.50, 4.09], Alaska(x) –7.12 [–8.73, –3.31] / [–4.38, –2.62]	Lena Delta undetermined, Canada higher error, Alaska(x) lower error
E17	same region rows, n 160	F1k: Lena Delta –0.39, Canada +4.86 [2.04, 6.10] / [1.28, 4.17], Alaska(x) –7.20 [–9.12, –2.07] / [–2.37, –0.54]. F1n: Lena Delta –0.71, Canada +4.67 [2.52, 5.72] / [1.52, 4.16], Alaska(x) –6.06 [–7.64, –1.95] / [–2.59, –0.91]	Lena Delta undetermined, Canada higher error, Alaska(x) lower error
E18	3 regions mean(Alaska included), n 40-160	F1k n 40 –0.90 [–2.19, 0.63] / [–0.22, 1.12]; F1n n 40 –1.14; F1k n 160 –0.91; F1n n 160 –0.70	all undetermined
E19	P1* contrast, n 40-160	P1@ed – P1: –1.87 [–2.64, –0.34] / [–1.32, –0.13] ‡, –1.93 [–2.63, –0.56] / [–1.20, –0.07] ‡. R1 – P1@ed: +1.31 [–0.13, 1.96] / [–0.54, 0.57], +1.30 [0.01, 1.86] / [–0.68, 0.41]	P1@ed lower error(pool 2/4); R1 – P1@ed undetermined(pool 2/4)
E20	TabPFN direct – TabPFN residual(L36)	n 0 +1.33 [0.63, 2.39] / [0.63, 2.23]; n 10 +0.97; n 40 –1.79 [–2.49, –0.86] / [–2.98, –1.19]; n 160 –2.45 [–3.48, –0.97] / [–3.41, –1.42]; all –0.60. regions n 40: Lena Delta –0.10, Canada –3.47, Alaska(x) +8.57; n 160: Lena Delta –0.57, Canada –4.33, Alaska(x) +6.52	higher error, higher error, lower error(pool 2/4), lower error(pool 2/4), undetermined. regions: undetermined, lower error, higher error

c. Augmented combination and stacking (C3 README 2.3)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E21	R2 – R1, n 10-40-160	n 10 –0.13 [–0.31, 0.24] / [–0.32, 0.20]; n 40 +0.34 [0.25, 0.49] / [0.21, 0.50]; n 160 +0.08 [–0.09, 0.42] / [0.02, 0.31]	ns, worse, ns. hypothesis verdict rows(scope=verdict): 'rejected(augmentation zero labels only)'
E22	four-way auxiliary column of the same contrast	P4: n 10 –0.13 equivalent, n 40 +0.34 higher error('size below 0.5 cm'), n 160 +0.08 equivalent ‡. MEAN3: +0.01, +0.04, –0.09 all equivalent	as above same
E23	AB6 = R2 – R1, n 10	–0.13 [–0.31, 0.22] / [–0.31, 0.20]; holm_p_eq 0.002; '(c) limit stated('0.5 cm same within')'	equivalent(blind)
E24	AB6 regions rows	Lena Delta +0.02 [–0.08, 0.19] / [0.04, 0.35]; Canada +0.76 [0.56, 1.00] / [0.48, 0.98]; W Russia –0.25; E Russia –1.05 [–1.53, 0.15] / [–1.31, –0.06]; Alaska(x) –0.76 [–1.30, –0.34] / [–1.68, –0.76]	equivalent, higher error, undetermined, undetermined, lower error
E25	R3 – R1, n 10-40-160	n 10 +0.49 [–0.17, 1.41] / [–0.39, 1.38]; n 40 –1.23 [–1.63, –0.50] / [–1.90, –0.79]; n 160 –1.83 [–2.59, –0.40] / [–2.10, –0.61]	ns, improve(2 regions), improve(2 regions). hypothesis verdict rows: 'partial'
E26	four-way auxiliary column of the same contrast	n 10 +0.49 / n 40 –1.23 / n 160 –1.83	undetermined, lower error, lower error

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E27	R3 – R1, 3 regions mean	n 40 +0.30 [–0.67, 0.84] / [–1.23, –0.37]; n 160 –0.02 [–1.17, 0.92] / [–1.29, –0.24]	undetermined, undetermined
E28	R3 – R1 regions rows	n 40: Lena Delta –0.00, Canada –2.46 [–3.19, –1.03] / [–2.82, –1.00], Alaska(x) +3.35 [0.56, 4.25] / [–0.42, 0.91]. n 160: Lena Delta +0.20, Canada –3.86 [–5.22, –1.02] / [–3.03, –0.66], Alaska(x) +3.59 [0.53, 4.57] / [–0.29, 1.02]	Lena Delta undetermined, Canada lower error, Alaska(x) undetermined

d. Multiplicative residual and label shuffle (C3 README 2.4)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E29	RM – R1, n 10, λ 0.25	+0.35 [0.08, 0.46] / [0.12, 0.44]; small_effect True; 'depends on split independence'; holm_p 0.0768 'significant before correction'	higher error
E30	RM – R1, n 10, λ 1.0	+1.09 [0.19, 1.38] / [0.43, 1.62]; 'depends on split independence'; holm_p 0.1904 'significant before correction'	higher error
E31	RM – R1, all, λ 0.25	+0.25 [0.07, 0.37] / [0.18, 0.55]; small_effect True; 'depends on split independence'; holm_p 0.0900 'significant before correction'	higher error
E32	RM – R1, all, λ 1.0	+0.70 [–0.12, 1.13] / [0.45, 1.78]	undetermined
E33	RM – R1, 3 regions mean	n 10 λ 0.25 +0.04 equivalent; n 10 λ 1.0 –0.94 [–2.19, –0.07] / [–1.06, –0.18] lower error ‡; all λ 0.25 +0.13 higher error ‡('size below 0.5 cm'); all λ 1.0 +0.16 undetermined	rowsby
E34	R1 – R1s(label shuffle)	n 10: –0.12, –0.26; all labels: –0.60 [–0.80, –0.36] / [–0.64, –0.32], –1.69 [–2.48, –0.84] / [–2.01, –0.94]. 'The relation between the covariates and the residuals of the target labels contributes: n=–1 λ =0.25'	equivalent, equivalent, lower error, lower error

e. Net value of the residual over the recalibrated model (C3 README 2.5)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E35	R1 – P1(CatBoost, Rescale)	n 3 –0.08 [–0.43, 0.11] / [–0.47, –0.00]; n 10 –0.18 [–0.52, –0.02] / [–0.52, –0.09]; n 40 –0.56; n 160 –0.64; n 320 –0.83; all –0.40 [–0.76, –0.23] / [–0.76, –0.30]. 'net value of ML with sparse labels'	ns, improve(pool 4/4), improve(pool 2/4) $\times$ 3, improve(pool 4/4). P1* flag: n 3:320 'P1* confirmednot n', n 40:160 E19 R1 – P1@ed(undetermined) companion
E36	AB4 = P1 – P0, AB5 = R1 – P1(n 10)	AB4 –2.45 [–3.04, –1.88] / [–3.30, –1.87], holm_p 0.001, 'S-a viewing after cited'. AB5 –0.18 [–0.53, –0.01] / [–0.51, –0.08] ‡, 'distinguishable, size below 0.5 cm', holm_p 0.136, '(b) abstract 'no difference established', main text 'significant before correction'	AB4 lower error(pool 4/4. region rows lower error 1/4, E36a), AB5 lower error
E36a	AB4 regions rows	Lena Delta +0.22 [–0.32, 0.92] / [0.25, 1.14]; Canada +2.26 [0.71, 2.94] / [0.67, 2.15]; W Russia –11.96 [–13.62, –10.53] / [–15.71, –10.54]; E Russia –0.34 [–0.90, 1.30] / [–0.08, 1.48]	undetermined, higher error, lower error, undetermined(lower error regions 1/4)
E37	extended pool R1 – P1, n 10	P4 –0.18 [–0.52, –0.02] / [–0.51, –0.08] ‡; PE1 –0.13 [–0.42, 0.01] / [–0.41, 0.01] ‡, l28 'weakened', cross-environment flag; Central Russia(PE1-new) +0.06 [–0.40, 0.51] / [–0.39, 0.81], cross-environment flag	lower error, equivalent, undetermined
E38	TabPFN residual R1[T] – P1	n 3 –0.32 [–0.46, –0.02] / [–0.38, –0.00] ‡, holm_p 0.2366 'significant before correction'; n 10 –0.42 [–0.54, –0.18] / [–0.47, –0.14], holm_p 0.0036; n 40 –0.62; n 160 –0.93; n 320 –1.10; all –0.80. MEAN3: n 3 –0.18, n 10 –0.18. companion R1[C] – P1 n 10 –0.16 ‡	Lower error (n 3 and 10 'size below 0.5 cm', blind False, replication (unblinded); n 3 'significant before correction' in the auxiliary Holm column within the item). P1* flag: n 3 and 320 'n at which P1* was not checked', n 40 and 160 'P1* contrast not computed (P1 contrast only, auxiliary)'. MEAN3 equivalent and equivalent n 3 undetermined, n $\geq$ 10 lower error. P1* flag: n 3 and 320 'n at which P1* was not checked', n 40 and 160 'P1* contrast not computed (P1 contrast only, auxiliary)'. MEAN3 n 10 equivalent, n 40 lower error ('size below 0.5 cm')
E39	TabICL residual R1[I] – P1	n 3 –0.42 [–0.59, 0.01] / [–0.40, 0.11]; n 10 –0.55 [–0.71, –0.21] / [–0.63, –0.10] ‡, holm_p 0.0180(LGF auxiliary bundle); n 40 –0.79; n 160 –1.18; n 320 –1.43; all –1.00 [–1.23, –0.63] / [–1.15, –0.60]. MEAN3: n 10 –0.02, n 40 –0.32 ‡. companion R1[C] – P1 n 10 –0.16 ‡. platform local-3090-torch2.6.0-numpy1.26.4	

f. Shrinkage recalibration versus local least squares (C3 README 2.6)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E40	P2 – P1, n 3·10, 4 regions mean	n 3 +1.76 [0.90, 3.05] / [1.62, 3.55]; n 10 +0.84 [0.03, 1.91] / [0.42, 2.33]; role 'descriptive(verdict not used)'	higher error, higher error
E41	P2 – P1, n 3, region rows	Lena Delta +2.91 [2.17, 3.87] / [3.03, 4.47], RMSE 24.76 vs 21.84; Canada +5.40 [3.91, 6.23] / [4.04, 5.76], 34.54 vs 29.14; W Russia –4.97 [–7.99, –1.62] / [–8.82, –1.78], 33.36 vs 38.33; E Russia +3.69 [2.58, 7.18] / [5.83, 7.97], 33.32 vs 29.64	higher error, higher error, lower error, higher error
E42	P2 – P1, n 10, region rows	Lena Delta +1.01; Canada +3.25; W Russia –1.44 [–4.62, 2.35] / [–4.41, 2.93]; E Russia +0.55	higher error, higher error, undetermined, undetermined
E43	Tibetan Plateau P2 – P1	n 3 –81.77 [–96.22, –64.75] / [–81.01, –42.14]; n 10 –40.02 [–53.74, –24.18] / [–35.99, –0.51] ‡. two rows all cross-environment flag. RMSE n 3: P2 119.28, P1 201.05, P0 244.51. n 10: P2 108.12, P1 148.14	improve, improve
E44	Central Russia P2 – P1	n 3 +2.32 [–0.43, 4.72] / [0.82, 5.95]; n 10 –0.57 [–2.14, 1.00] / [–1.14, 1.95]. two rows all cross-environment flag	ns, ns

Supplementary Table S6. Within-region label-count tests (WF1, WF6, SAR).

Designed after the transfer results had been viewed and registered before running. WF6-a: 25 splits (24 valid in the Lena Delta), comparator P1. WF1: first test, five splits, comparator the best cross-validated physics calibration (Pbest); WF1-b adds nine SAR covariates (x34) in Alaska. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: |x parent region excluded from the source, |i rest of the parent kept, |r within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. WF6-a, Alaska (C4 README 2.1)

ID	Contrast	Δ cell [CI]	Δ block-equal [CI]	Verdict	Common-CI verdict	RMSE A / B
E1.R1.n200	R1(λ cv)-P1 n200	-0.18 [-0.34, 0.01]	-0.25 [-0.42, -0.07]	equivalent	undetermined	14.22 / 14.40
E1.R1.n500	R1(λ cv)-P1 n500	-0.39 [-0.54, -0.21]	-0.39 [-0.57, -0.21]	lower error(flag: size below 0.5 cm)	undetermined	13.99 / 14.38
E1.R1.n1000	R1(λ cv)-P1 n1000	-0.54 [-0.69, -0.32]	-0.60 [-0.80, -0.40]	lower error	lower error	13.87 / 14.41
E1.R1.nall	R1(λ cv)-P1 nall	-0.52 [-0.70, -0.31]	-0.68 [-0.91, -0.44]	lower error	undetermined	13.88 / 14.40
E1.R2.n200	R2(λ cv)-P1 n200	-0.20 [-0.24, -0.11]	-0.07 [-0.14, -0.01]	lower error(below 0.5 cm)	equivalent	14.20 / 14.40
E1.R2.n500	R2(λ cv)-P1 n500	-0.31 [-0.38, -0.18]	-0.17 [-0.26, -0.07]	lower error(below 0.5 cm)	equivalent	14.07 / 14.38
E1.R2.n1000	R2(λ cv)-P1 n1000	-0.42 [-0.51, -0.26]	-0.33 [-0.44, -0.21]	lower error(below 0.5 cm)	lower error	13.99 / 14.41
E1.R2.nall	R2(λ cv)-P1 nall	0.70 [0.06, 0.90]	-0.48 [-0.88, -0.08]	undetermined	undetermined	15.10 / 14.40
E1.Re.n200	Re(λ cv)-P1 n200	1.11 [0.78, 1.44]	1.30 [0.88, 1.75]	higher error	higher error	15.51 / 14.40
E1.Re.n500	Re(λ cv)-P1 n500	0.80 [0.53, 1.17]	1.05 [0.63, 1.48]	higher error	undetermined	15.19 / 14.38
E1.Re.n1000	Re(λ cv)-P1 n1000	0.48 [0.20, 0.87]	0.52 [0.12, 0.94]	higher error(below 0.5 cm)	undetermined	14.89 / 14.41
E1.Re.nall	Re(λ cv)-P1 nall	0.50 [0.22, 0.85]	0.35 [-0.07, 0.77]	undetermined	undetermined	14.90 / 14.40
E1.D0.n200	D0(catboost)-P1 n200	3.51 [1.71, 3.56]	-0.21 [-0.71, 0.30]	undetermined	undetermined	17.91 / 14.40
E1.D0.n500	D0(catboost)-P1 n500	3.16 [1.46, 3.20]	-0.56 [-1.08, -0.04]	undetermined	undetermined	17.55 / 14.38
E1.D0.n1000	D0(catboost)-P1 n1000	2.76 [1.16, 2.78]	-0.65 [-1.15, -0.14]	undetermined	undetermined	17.17 / 14.41
E1.D0.nall	D0(catboost)-P1 nall	1.27 [0.30, 1.37]	-1.06 [-1.56, -0.58]	undetermined	undetermined	15.66 / 14.40

b. WF6-a, Lena Delta and Canada (C4 README 2.2)

ID	Target	Contrast	Δ cell [CI]	Δ block-equal [CI]	Verdict	Splits
E2.Lena.R1.n200	Lena r	R1(λ cv)-P1 n200	0.22 [-0.79, 0.64]	-0.20 [-0.75, 0.34]	undetermined	24
E2.Lena.R1.n500	Lena r	R1(λ cv)-P1 n500	0.34 [-0.76, 0.78]	-0.36 [-1.16, 0.38]	undetermined	24
E2.Lena.R1.n1000	Lena r	R1(λ cv)-P1 n1000	0.61 [-0.57, 1.38]	-0.08 [-1.04, 0.81]	undetermined	23
E2.Lena.R1.nall	Lena r	R1(λ cv)-P1 nall	0.47 [-0.56, 1.28]	-0.14 [-1.07, 0.72]	undetermined	24
E2.Lena.Re.n200	Lena r	Re(λ cv)-P1 n200	-1.03 [-1.80, -0.43]	0.37 [-0.12, 0.86]	undetermined	24
E2.Lena.D0.n1000	Lena r	D0(catboost)-P1 n1000	-0.78 [-1.94, 0.17]	-1.08 [-2.00, -0.21]	undetermined	23
E2.Canada.R1.n200	Canada r	R1(λ cv)-P1 n200	-0.40 [-1.05, 0.19]	-0.84 [-1.36, -0.28]	undetermined	25
E2.Canada.R1.nall	Canada r	R1(λ cv)-P1 nall	0.45 [-0.73, 1.24]	-0.23 [-1.09, 0.65]	undetermined	25
E2.Canada.Re.n200	Canada r	Re(λ cv)-P1 n200	-2.13 [-2.72, -0.23]	2.17 [1.29, 3.06]	undetermined	25

ID	Target	Contrast	Δ cell [CI]	Δ block-equal [CI]	Verdict	Splits
E2.Canada.Re.nall	Canada r	Re(λ cv)-P1 nall	−1.73 [−2.45, 0.09]	1.86 [0.92, 2.83]	undetermined	25
E2.Canada.D0.n200	Canada r	D0(catboost)-P1 n200	−1.50 [−1.99, −0.24]	−0.03 [−1.03, 1.01]	undetermined	25
E2.Canada.D0.nall	Canada r	D0(catboost)-P1 nall	−1.50 [−2.19, −0.12]	−0.30 [−1.44, 0.86]	undetermined	25

c. WF6-b, stratified mean of three targets (C4 README 2.3)

ID	n	Δ cell [CI]	Δ block-equal [CI]	Verdict
E3.WF6-b.n500	500	0.02 [−0.30, 0.23]	−0.25 [−0.55, 0.03]	undetermined
E3.WF6-b.n1000	1,000	−0.02 [−0.53, 0.54]	−0.03 [−0.53, 0.43]	undetermined
E3.WF6-b.nall	all	0.31 [−0.28, 1.16]	0.05 [−0.57, 0.65]	undetermined

d. WF1-a, first test (C4 README 2.4)

ID	Target	Contrast	Δ cell [CI]	Δ block-equal [CI]	Verdict	RMSE A / B
E6.Alaska.R1.n1000	Alaska r	R1(λ cv)-Pbest n1000	−0.49 [−0.82, −0.01]	0.24 [−0.23, 0.72]	undetermined	14.26 / 14.75
E6.Alaska.R1.n2000	Alaska r	R1(λ cv)-Pbest n2000	−0.58 [−0.89, −0.22]	−0.10 [−0.57, 0.41]	undetermined	14.13 / 14.71
E6.Alaska.R1.n5000	Alaska r	R1(λ cv)-Pbest n5000	−0.37 [−0.66, 0.03]	0.26 [−0.22, 0.74]	undetermined	14.14 / 14.51
E6.Alaska.R1.nall	Alaska r	R1(λ cv)-Pbest nall	−0.77 [−2.18, 0.08]	0.28 [−0.52, 1.12]	undetermined	14.17 / 14.93
E6.Alaska.R2.nall	Alaska r	R2(λ cv)-Pbest nall	1.05 [−0.66, 2.05]	1.00 [0.18, 1.89]	undetermined	15.98 / 14.93
E6.AL-2.R1.n1000	AL-2 r	R1(λ cv)-Pbest n1000	−1.35 [−1.64, −0.47]	−0.85 [−1.53, −0.19]	lower error	12.38 / 13.73
E6.AL-2.R2.n1000	AL-2 r	R2(λ cv)-Pbest n1000	−1.42 [−1.70, −0.54]	−0.99 [−1.73, −0.29]	lower error	12.31 / 13.73
E6.AL-2.R1.nall	AL-2 r	R1(λ cv)-Pbest nall	−0.83 [−1.31, −0.18]	−0.99 [−1.70, −0.29]	lower error	12.32 / 13.15
E6.AL-2.R2.nall	AL-2 r	R2(λ cv)-Pbest nall	−0.74 [−1.20, 0.04]	−0.75 [−1.47, −0.04]	undetermined	12.41 / 13.15

e. WF1-b (SAR) and WF1-c (risk counts) (C4 README 2.5)

ID	Contrast	Role	Δ cell [CI]	Δ block-equal [CI]	Verdict
E7.R1x34.n500	R1@x34-R1 n500 λ cv	decision rule	0.07 [−0.04, 0.18]	0.20 [0.04, 0.36]	equivalent
E7.R1x34.n1000	R1@x34-R1 n1000 λ cv	decision rule	−0.11 [−0.21, 0.05]	0.06 [−0.15, 0.28]	equivalent
E7.R1x34.n2000	R1@x34-R1 n2000 λ cv	decision rule	−0.09 [−0.21, 0.15]	0.08 [−0.23, 0.37]	equivalent
E7.R1x34.n5000	R1@x34-R1 n5000 λ cv	decision rule	−0.08 [−0.18, 0.08]	0.15 [−0.09, 0.41]	equivalent
E7.R1x34.nall	R1@x34-R1 nall λ cv	decision rule	−0.04 [−0.20, 0.21]	0.23 [−0.08, 0.56]	undetermined
E7.D1x34.n500	D1@x34-D1 n500	auxiliary	0.24 [0.01, 0.35]	0.11 [0.01, 0.22]	higher error
E7.D1x34.n1000	D1@x34-D1 n1000	auxiliary	0.38 [0.03, 0.53]	0.16 [0.01, 0.32]	higher error
E7.D1x34.n2000	D1@x34-D1 n2000	auxiliary	0.41 [0.10, 0.55]	0.34 [0.07, 0.62]	higher error
E7.D1x34.n5000	D1@x34-D1 n5000	auxiliary	0.81 [0.05, 1.08]	0.61 [0.04, 1.25]	higher error
E7.D1x34.nall	D1@x34-D1 nall	auxiliary	0.53 [−0.21, 0.82]	0.57 [−0.12, 1.28]	undetermined

f. Error floors (covariate-conditional, scoring cells) and contest values (cm) (C4 README 2.6)

ID	Column	Value
E8.floor	floor_rmse_cm, n_cells_used	11.31, 13,333
E8.floor.Lena / Canada	floor_rmse_cm	14.83 / 17.90
E8.contest.stefan	rmse_cm	14.46(file 14.457)
E8.contest.resid	rmse_cm	13.33
E8.contest.min	rmse_cm	13.33
E8.contest.nrows_lam_pos	row count	56
E8.cv.block_canonical.*	rmse_cm, nnd_median_km	stefan 14.46, stefan_ridge_1075 13.33, ridge 13.62, catboost_lo 16.12, median nearest distance 28.37 km

g. Derived values: relative RMSE change and reducible squared error (F = error floor) (C4 README 2.6)

ID	Calculation	Value
D1.wf6.rmse_rel_P1_pct	$(\text{rmse_B} - \text{rmse_A})/\text{rmse_B} \times 100$, E1.R1.n1000 rows	3.75 %
D1.wf6.reducible_P1 / R1	$\text{rmse_B}^2 - F^2$, $\text{rmse_A}^2 - F^2$	79.71 / 64.42 cm ²
D1.wf6.reducible_cut_pct	$(79.71 - 64.42)/79.71 \times 100$	19.17 %
D2.contest.rmse_rel_pct	$(14.457 - 13.330)/14.457 \times 100$	7.80 %
D2.contest.reducible_stefan / resid	$14.457^2 - F^2$, $13.330^2 - F^2$	81.13 / 49.82 cm ²
D2.contest.reducible_cut_pct	$(81.13 - 49.82)/81.13 \times 100$	38.60 %
D3.c2.rho_recal_vs_mlvsp1	Spearman(recal_gain, ml_vs_p1), tables/wf0_misspec.csv 30 rows. signs as in the source column	ρ 0.08, p 0.67
D3.c2.rho_recal_vs_mlbest	Spearman(recal_gain, ml_best_delta). signs as in the source column	ρ -0.66
(revision check) gain basis	ml_vs_p1 = ml_best_delta - P1(best ML - P1, maximum difference 3.6e-15)so Spearman(recal_gain, P1 - best ML) = -0.08, Spearman(recal_gain, P0 - best ML) = 0.66	ρ -0.08(p 0.67), ρ 0.66

Supplementary Table S7. Gain decomposition and climate extrapolation (WF9, WF10).

Scoring cells grouped by 0.5° block and air $\sqrt{\text{TDD}}$ (registered: soil $\sqrt{\text{TDD}}$, sensitivity rows b and e). Shares are computed from the squared-error sums (Methods, equation for SSE). Explained within-grid share = $1 - (\text{within-grid RMSE of the method} / \text{within-grid RMSE of P1})^2$. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: $|x$ parent region excluded from the source, $|i$ rest of the parent kept, $|r$ within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. Decomposition of the within-region gain in Alaska (all labels, λ by CV) (C8 README 2.1)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
A1	Alaska within region all, R1(cross-validation λ) – P1, total	-0.52 [-0.70, -0.31] / -0.68 [-0.91, -0.44]	lower error
A2	same contrast, between grids store	-1.03 [-1.30, -0.59] / -0.89 [-1.16, -0.61]	lower error
A3	same contrast, within grid store(WF9-a targets rows)	-0.01 [-0.03, 0.04] / 0.04 [0.01, 0.06]	equivalent
A4	soil definition: total	-0.52 [-0.70, -0.31] / -0.68 [-0.91, -0.44]	lower error
A5	soil definition: between grids	-1.03 [-1.30, -0.59] / -0.89 [-1.16, -0.61]	lower error
A6	soil definition: within grid	-0.01 [-0.03, 0.04] / 0.04 [0.01, 0.06]	equivalent

b. Shares of the squared error (descriptive, no CI) (C8 README 2.2)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
B1	P1 total SSE of within grid share, Alaska within region all	0.72 (71.9 %)	descriptive
B2	same, Lena Delta	0.62 (61.7 %)	descriptive
B3	same, Canada	0.37 (37.4 %)	descriptive
B4	R1(cross-validation λ) total SSE gain of within grid component share, Alaska within region	0.01 (0.9 %)	descriptive
B5	R1(λ 0.25), Alaska within region	0.07 (6.7 %)	descriptive
B6	R1(λ 0.25), Lena Delta within region	0.53 (53.5 %)	descriptive
B7	R1(λ 0.25), Canada within region	-0.00 (-0.1 %)	descriptive
B8	R1(λ 0.25), Lena Delta transfer all	0.25 (25.2 %)	descriptive
B9	R1(λ 0.25), Alaska transfer all	0.36 (35.6 %)	descriptive
B10	soil definition: R1(cross-validation λ), Alaska within region	0.01 (1.0 %)	descriptive
B11	soil definition: R1(λ 0.25), Lena Delta within region	0.24 (23.7 %)	descriptive
B12	P1 decomp RMSE column(total; between grids; within grid), Alaska within region	14.53; 7.68; 12.37	descriptive
B13	R1(cross-validation λ) decomp RMSE column, Alaska within region	14.06; 6.76; 12.36	descriptive
B14	R1(cross-validation λ) total SSE gain within grid share, Canada within region	-12,133.57; 0.21	descriptive

c. Explained within-grid share (descriptive, no CI) (C8 README 2.3)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
C1	R1(cross-validation λ), Alaska within region	0.00 (0.1 %)	descriptive
C2	R1(λ 0.25), Alaska within region	0.01 (0.7 %)	descriptive
C3	D0(catboost), Alaska within region	-0.02 (-2.0 %)	descriptive
C4	R1(λ 0.25), Lena Delta within region	0.06 (5.6 %)	descriptive
C5	D0(catboost), Lena Delta within region	0.06 (6.2 %)	descriptive
C6	R1(cross-validation λ), Canada within region	-0.02 (-2.2 %)	descriptive
C7	D0(catboost), Canada within region	-0.08 (-8.3 %)	descriptive
C8	R1(λ 0.25), Lena Delta transfer	0.03 (3.2 %)	descriptive
C9	R1(λ 0.25), Alaska transfer	0.02 (2.0 %)	descriptive
C10	R1(λ 0.25), Canada transfer	-0.00 (-0.4 %)	descriptive
C11	R1(λ 0.25), E Russia transfer	-0.08 (-8.5 %)	descriptive
C12	within region ML over all keys of maximum(air definition)	0.08 (7.8 %)	descriptive
C13	within region ML over all keys of maximum(soil definition)	0.04 (4.0 %)	descriptive
C14	transfer ML over all keys of maximum(air definition, soil definition same value)	0.07 (7.2 %)	descriptive

d. WF9-a within regions, air $\sqrt{\text{TDD}}$ grouping (C8 README 2.4)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
D1	stratified mean n 500(Alaska-Lena Delta)	-0.08 [-0.16, 0.15] / 0.22 [0.09, 0.35]	equivalent
D2	stratified mean n 1,000(Alaska-Lena Delta)	-0.02 [-0.12, 0.35] / 0.36 [0.19, 0.54]	undetermined
D3	stratified mean all(3 regions)	0.09 [0.02, 0.40] / 0.37 [0.25, 0.50]	higher error
D4	Lena Delta all	0.05 [-0.18, 0.92] / 0.87 [0.51, 1.24]	undetermined
D5	Canada all	0.23 [0.16, 0.34] / 0.21 [0.12, 0.31]	higher error
D6	verdict wording	partial(regions 2/3, splits 23/24): rejected: higher error contrast exists(n=all) [distinguishable, size below 0.5 cm: n=all]	

e. WF9-a, soil $\sqrt{\text{TDD}}$ grouping (sensitivity) (C8 README 2.5)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
E1	stratified mean n 500(Alaska-Lena Delta)	0.27 [0.07, 0.41] / 0.28 [0.15, 0.41]	higher error
E2	stratified mean n 1,000(Alaska-Lena Delta)	0.42 [0.20, 0.64] / 0.44 [0.26, 0.61]	higher error
E3	stratified mean all(3 regions)	0.40 [0.24, 0.59] / 0.42 [0.30, 0.55]	higher error
E4	Lena Delta all	0.99 [0.49, 1.52] / 1.02 [0.67, 1.39]	higher error
E5	Canada all	0.23 [0.16, 0.33] / 0.21 [0.11, 0.31]	higher error
E6	verdict wording	partial(regions 2/3, splits 23/24): rejected: higher error contrast exists(n=500, n=1000, n=all) [distinguishable, size below 0.5 cm: n=500, n=1000, n=all]	

f. WF9-c in transfer, air grouping (C8 README 2.6)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
F1	stratified mean n 0(Lena Delta and Canada)	-0.01 [-0.04, 0.01] / -0.02 [-0.06, 0.00]	equivalent
F2	stratified mean n 10	-0.01 [-0.04, 0.01] / -0.04 [-0.07, -0.00]	equivalent
F3	stratified mean all	-0.11 [-0.25, -0.01] / -0.12 [-0.22, -0.03]	lower error
F4	Lena Delta all	-0.27 [-0.52, -0.05] / -0.22 [-0.42, -0.04]	lower error
F5	E Russia all(scoring blocks few CI none)	1.08 [NA, NA] / 1.08 [NA, NA]	not decidable
F6	verdict wording	partial(regions 2/4): partially supported: lower error contrast = n=all [distinguishable, size below 0.5 cm: n=all]	

g. WF9-c, soil grouping (sensitivity) (C8 README 2.7)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
G1	stratified mean n 0(Lena Delta and Canada)	0.00 [-0.04, 0.02] / -0.02 [-0.05, 0.01]	equivalent
G2	stratified mean n 10	0.01 [-0.03, 0.02] / -0.03 [-0.07, 0.00]	equivalent
G3	stratified mean all	-0.04 [-0.20, 0.04] / -0.10 [-0.21, -0.01]	equivalent
G4	Lena Delta all	-0.13 [-0.42, 0.05] / -0.19 [-0.39, -0.01]	equivalent
G5	E Russia all(CI none)	1.08 [NA, NA] / 1.08 [NA, NA]	not decidable
G6	verdict wording	partial(regions 2/4): rejected: lower error contrast none	

h. Descriptive contrasts outside the registered hypotheses (C8 README 2.8)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
H1	R1(λ 0.25) – P1 within grid, 3 regions stratified mean(air)	-0.16 [-0.17, -0.07] / -0.09 [-0.13, -0.04]	lower error
H2	same, Lena Delta(air)	-0.44 [-0.50, -0.19] / -0.26 [-0.40, -0.14]	lower error
H3	same, 3 regions stratified mean(soil)	-0.08 [-0.11, -0.02] / -0.07 [-0.12, -0.03]	lower error
H4	same, Lena Delta(soil)	-0.19 [-0.30, -0.04] / -0.22 [-0.35, -0.10]	lower error
H5	R1(λ 0.25) – P1 between grids, Alaska	-0.99 [-1.11, -0.69] / -0.58 [-0.71, -0.45]	lower error
H6	same, Lena Delta	-0.51 [-1.14, -0.35] / -1.27 [-1.65, -0.90]	lower error
H7	same, Canada	-0.93 [-1.39, -0.48] / -1.39 [-1.69, -1.08]	lower error
H8	D0(catboost) – P1 within grid, 3 regions stratified mean	0.16 [0.10, 0.42] / 0.46 [0.33, 0.59]	higher error
H9	Re(Stefan-CCI anchor, cross-validation λ) – P1 within grid, 3 regions stratified mean	0.31 [0.21, 0.59] / 0.42 [0.28, 0.56]	higher error

i. WF10 climate extrapolation (warm variant) (C8 README 2.9)

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
I1	WF10-a verdict wording	Not decidable (contrasts 0/2; missing contrasts: n=100, n=all)	
I2	WF10-b verdict wording	Not decidable (contrasts 0/2; missing contrasts: n=100, n=all)	
I3	WF10-c verdict wording	Not decidable (contrasts 0/2; missing contrasts: n=100, n=all)	

ID	Contrast or quantity	Value (cm unless stated): cell-weighted [CI] / block-equal [CI]	Verdict
I4	EP(D0 – P1), Alaska warm, n 100	7.02 [3.91, 9.51] / 0.75 [-1.55, 3.04]	undetermined
I5	same, n 500(descriptive rows)	4.55 [1.58, 7.09] / -1.31 [-3.54, 0.92]	undetermined
I6	same, all	3.39 [0.61, 5.95] / -1.33 [-3.41, 0.77]	undetermined
I7	EP(R1 – D0), Alaska warm, n 100	-5.56 [-7.87, -3.02] / -1.38 [-3.29, 0.52]	undetermined
I8	same, all	-2.45 [-4.73, 0.09] / 1.59 [-0.25, 3.38]	undetermined
I9	W R1 – P1, Alaska warm, n 100	0.87 [-0.12, 1.16] / -1.32 [-1.83, -0.78]	undetermined
I10	same, all	0.13 [-0.54, 0.41] / -0.51 [-0.93, -0.06]	undetermined
I11	Canada warm extrapolation scoring(W) number of blocks	3–3	
I12	Alaska warm W number of blocks	17–17	
I13	incomplete(partial) work units	Alaska warm splits 10 partial	

Supplementary Table S8. Label placement strategies (WF2, WF8, L43).

Strategy minus random cells (S1) at the same budget; within-region designs (target suffix |r) and transfer (|x). Values in cm; RMSE A / B are the RMSE of the strategy and of random cells. WF2-a rows marked descriptive are reported without verdict words in the main text. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: |x parent region excluded from the source, |i rest of the parent kept, |r within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. Block stratification (S2) and covariate max-min distance (S4), WF2-a (C6 README 2.1)

ID	Target; contrast (role)	Δ cell [CI]	Δ block-equal [CI]	RMSE A / B	Verdict
A1	Canada r; S2-S1 R1(0.25) n20 (descriptive)	-3.46 [-4.80, -0.69]	-2.66 [-3.92, -1.40]	27.76 / 31.22	lower error
A2	Canada r; S2-S1 R1(0.25) n50 (descriptive)	-2.74 [-3.52, -0.87]	-1.73 [-2.69, -0.79]	28.27 / 31.01	lower error
A3	Canada r; S2-S1 R1(0.25) n100 (descriptive)	-2.54 [-3.22, -1.12]	-1.98 [-2.69, -1.30]	28.75 / 31.29	lower error
A4	Canada r; S2-S1 R1(0.25) n200 (descriptive)	-1.44 [-2.00, -0.42]	-1.28 [-1.75, -0.80]	29.16 / 30.59	lower error
A5	Canada r; S4-S1 R1(0.25) n20 (descriptive)	-2.24 [-3.41, 0.03]	-2.10 [-3.28, -0.95]	28.98 / 31.22	undetermined
A6	Canada r; S4-S1 R1(0.25) n50 (descriptive)	-2.63 [-3.49, -0.70]	-1.64 [-2.55, -0.76]	28.38 / 31.01	lower error
A7	Canada r; S4-S1 R1(0.25) n100 (descriptive)	-3.21 [-4.27, -1.10]	-2.24 [-3.09, -1.40]	28.09 / 31.29	lower error
A8	Canada r; S4-S1 R1(0.25) n200 (descriptive)	-1.25 [-1.75, -0.37]	-1.20 [-1.63, -0.77]	29.34 / 30.59	lower error
A9	AL-1 r; S4-S1 R1(0.25) n50 (descriptive)	-0.77 [-2.64, -0.36]	-1.82 [-2.87, -0.77]	16.36 / 17.13	lower error
A10	AL-1 r; S4-S1 R1(0.25) n100 (descriptive)	-0.50 [-2.29, -0.20]	-1.99 [-2.86, -1.12]	16.13 / 16.64	lower error
A11	AL-1 r; S4-S1 R1(0.25) n200 (descriptive)	-0.32 [-1.09, -0.16]	-1.04 [-1.46, -0.62]	16.23 / 16.55	lower error
A12	AL-2 r; S2-S1 R1(0.25) n50 (descriptive)	-0.70 [-1.29, -0.53]	-0.67 [-1.34, -0.01]	12.34 / 13.04	lower error
A13	AL-2 r; S2-S1 R1(0.25) n100 (descriptive)	-0.65 [-1.31, -0.45]	-0.73 [-1.45, -0.03]	12.34 / 12.98	lower error
A14	AL-2 r; S2-S1 R1(0.25) n200 (descriptive)	-0.36 [-1.02, -0.21]	-0.55 [-1.14, 0.02]	12.31 / 12.67	undetermined
A15	AL-2 r; S4-S1 R1(0.25) n50 (descriptive)	-0.39 [-0.97, -0.25]	-0.51 [-0.94, -0.09]	12.65 / 13.04	lower error
A16	Lena r; S2-S1 R1(0.25) n50 (descriptive)	1.47 [0.37, 2.29]	-0.05 [-1.14, 0.98]	23.12 / 21.65	undetermined
A17	Lena r; S2-S1 R1(0.25) n100 (descriptive)	1.65 [0.90, 2.40]	0.88 [0.15, 1.57]	22.83 / 21.17	higher error
A18	Lena r; S2-S1 R1(0.25) n200 (descriptive)	1.79 [0.84, 2.46]	0.73 [-0.12, 1.54]	22.52 / 20.73	undetermined
A19	Lena r; S2-S1 R1(0.25) n500 (descriptive)	1.48 [0.70, 2.12]	0.76 [-0.04, 1.52]	22.23 / 20.75	undetermined
A20	Lena r; S4-S1 R1(0.25) n50 (descriptive)	0.11 [-1.12, 1.01]	-0.62 [-1.35, 0.09]	21.75 / 21.65	undetermined
A21	Lena r; S4-S1 R1(0.25) n100 (descriptive)	0.13 [-0.57, 0.72]	-0.12 [-0.61, 0.36]	21.30 / 21.17	undetermined
A22	Lena r; S4-S1 R1(0.25) n200 (descriptive)	0.44 [-0.14, 0.78]	-0.20 [-0.65, 0.24]	21.17 / 20.73	undetermined
A23	Alaska r; S2-S1 R1(0.25) n50 (main)	0.45 [-0.23, 0.77]	-1.30 [-1.97, -0.62]	14.73 / 14.28	undetermined
A24	Alaska r; S2-S1 R1(0.25) n100 (main)	0.34 [-0.45, 0.66]	-1.73 [-2.47, -0.97]	14.68 / 14.33	undetermined
A25	Alaska r; S2-S1 R1(0.25) n200 (main)	0.61 [-0.05, 0.89]	-1.08 [-1.64, -0.51]	14.66 / 14.05	undetermined
A26	Alaska r; S3-S1 R1(0.25) n50 (main)	-0.04 [-0.23, 0.08]	-0.35 [-0.58, -0.14]	14.23 / 14.28	undetermined
A27	Alaska r; S3-S1 R1(0.25) n100 (main)	-0.11 [-0.33, -0.03]	-0.62 [-0.88, -0.37]	14.23 / 14.33	lower error
A28	Alaska r; S3-S1 R1(0.25) n200 (main)	0.08 [-0.01, 0.17]	0.16 [0.02, 0.29]	14.13 / 14.05	equivalent
A29	Alaska r; S4-S1 R1(0.25) n50 (main)	0.28 [-0.34, 0.62]	-1.15 [-1.79, -0.49]	14.56 / 14.28	undetermined
A30	Alaska r; S4-S1 R1(0.25) n100 (main)	0.44 [-0.40, 0.81]	-1.71 [-2.48, -0.94]	14.77 / 14.33	undetermined
A31	Alaska r; S4-S1 R1(0.25) n200 (main)	0.74 [0.01, 1.09]	-1.09 [-1.68, -0.49]	14.79 / 14.05	undetermined

b. Sequential active selection (S6), WF2-a (C6 README 2.2)

ID	Target; contrast (role)	Δ cell [CI]	Δ block-equal [CI]	RMSE A / B	Verdict
B1	Alaska r; S6-S1 R1(0.25) n50 (main)	1.88 [1.58, 2.29]	2.40 [1.97, 2.81]	16.15 / 14.28	higher error
B2	Alaska r; S6-S1 R1(0.25) n100 (main)	1.25 [1.02, 1.50]	1.15 [0.92, 1.39]	15.58 / 14.33	higher error
B3	Alaska r; S6-S1 R1(0.25) n200 (main)	0.69 [0.50, 0.91]	0.60 [0.29, 0.89]	14.74 / 14.05	higher error
B4	Lena r; S6-S1 R1(0.25) n50 (descriptive)	2.12 [1.07, 2.66]	1.19 [0.38, 1.93]	23.77 / 21.65	higher error
B5	Lena r; S6-S1 R1(0.25) n100 (descriptive)	2.56 [1.46, 3.18]	1.81 [0.86, 2.62]	23.73 / 21.17	higher error
B6	Lena r; S6-S1 R1(0.25) n200 (descriptive)	2.79 [1.48, 3.45]	1.29 [0.14, 2.37]	23.52 / 20.73	higher error
B7	AL-1 r; S6-S1 R1(0.25) n50 (descriptive)	0.44 [0.39, 1.86]	1.34 [0.67, 2.01]	17.57 / 17.13	higher error
B8	AL-1 r; S6-S1 R1(0.25) n100 (descriptive)	0.89 [0.85, 1.88]	1.61 [1.15, 2.07]	17.52 / 16.64	higher error
B9	AL-1 r; S6-S1 R1(0.25) n200 (descriptive)	0.12 [0.07, 0.82]	0.52 [0.19, 0.86]	16.63 / 16.50	higher error
B10	AL-2 r; S6-S1 R1(0.25) n50 (descriptive)	0.21 [-0.65, 0.38]	-0.46 [-1.13, 0.20]	13.25 / 13.04	undetermined
B11	AL-2 r; S6-S1 R1(0.25) n100 (descriptive)	-0.06 [-0.62, 0.04]	-0.42 [-0.89, 0.05]	12.93 / 12.98	undetermined
B12	AL-2 r; S6-S1 R1(0.25) n200 (descriptive)	0.13 [-0.27, 0.19]	-0.15 [-0.44, 0.13]	12.80 / 12.67	equivalent
B13	Canada r; S6-S1 R1(0.25) n50 (descriptive)	-0.98 [-1.43, -0.10]	-0.14 [-0.60, 0.32]	30.03 / 31.01	undetermined
B14	Canada r; S6-S1 R1(0.25) n100 (descriptive)	-1.50 [-2.00, -0.54]	-0.95 [-1.38, -0.52]	29.79 / 31.29	lower error
B15	Canada r; S6-S1 R1(0.25) n200 (descriptive)	-0.68 [-1.03, -0.09]	-0.71 [-1.06, -0.36]	29.91 / 30.59	lower error

c. Source-extrapolation first (S5) and $\sqrt{\text{TDD}}$ first (S7), WF2-a (C6 README 2.3)

ID	Target; contrast (role)	Δ cell [CI]	Δ block-equal [CI]	RMSE A / B	Verdict
F1	Alaska r; S5-S1 R1(0.25) n100 (main)	0.23 [0.08, 0.61]	0.61 [0.31, 0.92]	14.58 / 14.35	higher error
F2	Alaska r; S5-S1 R1(0.25) n200 (main)	1.17 [0.70, 1.81]	1.57 [0.89, 2.25]	15.19 / 14.02	higher error
F3	AL-2 r; S5-S1 R1(0.25) n50 (descriptive)	1.55 [1.16, 1.73]	1.05 [0.53, 1.52]	14.59 / 13.04	higher error
F4	AL-2 r; S5-S1 R1(0.25) n200 (descriptive)	1.04 [0.31, 1.21]	0.50 [0.01, 1.00]	13.69 / 12.65	higher error
F5	Lena r; S5-S1 R1(0.25) n100 (descriptive)	-0.20 [-0.55, 0.45]	-0.09 [-0.63, 0.48]	20.97 / 21.17	undetermined
F6	Canada r; S5-S1 R1(0.25) n100 (descriptive)	-2.63 [-3.70, -0.37]	-0.89 [-1.79, 0.02]	28.67 / 31.30	undetermined
F7	AL-1 r; S5-S1 R1(0.25) n100 (descriptive)	-0.29 [-4.10, 0.43]	-3.72 [-5.67, -1.77]	16.37 / 16.66	undetermined
F8	Lena r; S7-S1 R1(0.25) n50 (descriptive)	3.61 [2.63, 5.20]	4.54 [3.13, 5.76]	25.26 / 21.65	higher error
F9	Lena r; S7-S1 R1(0.25) n100 (descriptive)	5.06 [3.92, 7.16]	6.72 [5.11, 8.20]	26.24 / 21.17	higher error
F10	Lena r; S7-S1 R1(0.25) n200 (descriptive)	2.91 [1.85, 4.51]	3.89 [2.75, 4.90]	23.64 / 20.73	higher error
F11	Alaska r; S7-S1 R1(0.25) n100 (main)	0.47 [-0.03, 0.90]	-0.69 [-1.34, -0.01]	14.80 / 14.33	undetermined
F12	Canada r; S7-S1 R1(0.25) n100 (descriptive)	-2.04 [-3.22, -0.00]	-0.93 [-1.64, -0.22]	29.25 / 31.29	lower error
F13	AL-1 r; S7-S1 R1(0.25) n100 (descriptive)	-0.66 [-1.87, 1.23]	0.75 [-0.23, 1.70]	15.98 / 16.64	undetermined
F14	AL-2 r; S7-S1 R1(0.25) n100 (descriptive)	-0.47 [-1.02, 0.00]	-0.44 [-0.99, 0.11]	12.52 / 12.98	undetermined

d. Transfer of the best Alaskan strategy, WF2-b (C6 README 2.4)

ID	Target; contrast (role)	Δ cell [CI]	Δ block-equal [CI]	RMSE A / B	Verdict
C1	Lena r; S3-S1 R1(0.25) n100 (main)	0.33 [0.04, 0.77]	0.68 [0.36, 1.00]	21.50 / 21.17	higher error
C2	Canada r; S3-S1 R1(0.25) n100 (main)	-0.48 [-0.65, -0.25]	-0.24 [-0.44, -0.02]	30.81 / 31.29	lower error

e. Transfer conditions, WF8 (C6 README 2.5)

ID	Target; contrast (role)	Δ cell [CI]	Δ block-equal [CI]	RMSE A / B	Verdict
D1	Canada x; S4-S1 R1(0.25) n40 (main)	-2.39 [-3.40, -0.49]	-1.64 [-2.49, -0.76]	28.77 / 31.16	lower error
D2	Lena x; S4-S1 R1(0.25) n40 (main)	0.55 [-0.44, 1.27]	-0.41 [-1.09, 0.24]	22.10 / 21.55	undetermined
D3	Lena x; S2-S1 R1(0.25) n10 (main)	1.89 [0.30, 3.08]	0.86 [-0.27, 1.93]	23.78 / 21.89	undetermined
D4	Lena x; S4-S1 R1(0.25) n10 (main)	1.90 [-0.02, 3.37]	1.09 [-0.15, 2.28]	23.79 / 21.89	undetermined
D5	Canada x; S2-S1 R1(0.25) n10 (main)	-1.35 [-3.08, 2.20]	0.18 [-1.32, 1.67]	28.18 / 29.53	undetermined
D6	Canada x; S4-S1 R1(0.25) n10 (main)	-1.08 [-2.70, 2.02]	0.06 [-1.39, 1.48]	28.46 / 29.53	undetermined
D7	Russia_W x; S2-S1 R1(0.25) n10 (main)	-0.72 [-1.13, -0.32]	-0.80 [-1.41, -0.18]	29.73 / 30.46	lower error
D8	Russia_W x; S4-S1 R1(0.25) n10 (main)	-0.38 [-0.89, 0.12]	-0.42 [-1.11, 0.29]	30.08 / 30.46	undetermined
D9	Russia_E x; S2-S1 R1(0.25) n10 (main)	0.36 [-0.94, 0.82]	-0.63 [-1.29, 0.07]	30.52 / 30.15	undetermined
D10	Russia_E x; S4-S1 R1(0.25) n10 (main)	0.06 [-0.33, 0.54]	0.36 [0.08, 0.62]	30.21 / 30.15	undetermined
D11	Alaska x; S2-S1 R1(0.25) n10 (main)	-0.79 [-1.33, -0.26]	-1.37 [-1.82, -0.91]	14.30 / 15.09	lower error
D12	Alaska x; S4-S1 R1(0.25) n10 (main)	-0.74 [-1.34, -0.12]	-1.02 [-1.50, -0.54]	14.35 / 15.09	lower error
D13	Canada x; S2-S1 R1(0.25) n40 (descriptive)	-2.32 [-3.43, -0.18]	-1.43 [-2.39, -0.45]	28.85 / 31.16	lower error
D14	Lena x; S2-S1 R1(0.25) n40 (descriptive)	2.18 [0.84, 2.88]	0.74 [-0.38, 1.75]	23.72 / 21.55	undetermined
D15	Alaska x; S2-S1 R1(0.25) n40 (descriptive)	-1.13 [-2.06, -0.48]	-2.17 [-2.94, -1.41]	14.09 / 15.22	lower error
D16	Alaska x; S4-S1 R1(0.25) n40 (descriptive)	-1.19 [-2.08, -0.52]	-2.11 [-2.87, -1.36]	14.04 / 15.22	lower error

f. Block-dispersed versus cell-random placement, L43 (C6 README 2.6)

ID	Target; contrast (role)	Δ cell [CI]	Δ block-equal [CI]	RMSE A / B	Verdict
E1	MEAN[Lena x,Canada x,Russia_W x,Russia_E x]; P1, n 10 (main)	-0.07 [-0.86, 0.89]	-0.21 [-0.81, 0.41]		undetermined
E2	MEAN[Lena x,Canada x]; P1, n 40 (main)	-0.24 [-1.19, 0.99]	-0.52 [-1.36, 0.30]		undetermined
E3	MEAN[Lena x,Canada x]; R1, n 40 (auxiliary)	-0.07 [-1.02, 1.15]	-0.34 [-1.19, 0.48]		undetermined
E4	MEAN[Lena x,Canada x]; R1, n 160 (auxiliary)	0.26 [-0.32, 0.80]	-0.20 [-0.78, 0.35]		undetermined
E5	MEAN[Lena x,Canada x]; P1, n 160 (auxiliary)	0.32 [-0.33, 0.90]	-0.18 [-0.81, 0.40]		undetermined
E6	Lena x; P1, n 10 (main)	1.55 [-0.18, 2.97]	0.58 [-0.60, 1.85]		undetermined
E7	Canada x; P1, n 10 (main)	-1.62 [-3.57, 2.06]	-0.28 [-1.98, 1.43]		undetermined
E8	Russia_W x; P1, n 10 (main)	-0.77 [-1.58, -0.05]	-0.85 [-1.82, 0.00]		undetermined
E9	Russia_E x; P1, n 10 (main)	0.56 [-0.82, 1.13]	-0.27 [-1.10, 0.56]		undetermined
E10	Lena x; P1, n 40 (main)	2.10 [0.73, 2.97]	0.69 [-0.41, 1.78]		undetermined
E11	Canada x; P1, n 40 (main)	-2.59 [-4.01, -0.24]	-1.72 [-2.96, -0.55]		lower error
E12	Canada x; R1, n 40 (auxiliary)	-2.32 [-3.74, -0.04]	-1.43 [-2.64, -0.27]		lower error
E13	Alaska x; R1, n 40 (auxiliary)	-1.13 [-2.36, -0.36]	-2.17 [-3.15, -1.27]		lower error

g. Verdict counts by strategy (post hoc count) (C6 README 2.9)

Strategy	WF2 lower	WF2 equiva- lent	WF2 unde- terminated	WF2 higher	WF8 lower	WF8 unde- terminated	WF8 higher
S2 block stratification	8	1	14	1	4	4	0
S3 covariate-cluster stratification	3	12	8	1			
S4 covariate max-min distance	8	2	14	0	3	5	0
S5 source extrapolation first	2	0	15	7			
S6 sequential active selection	3	8	4	9			
S7 $\sqrt{\text{TDD}}$ first	2	2	15	5			

Supplementary Table S9. Additional independent regions (L1e, L4e, L8e, L38, L39).

Licence-verified edition is the main edition; the North Atlantic rows are from the full-data edition and carry a few-blocks flag. All Tibetan scored cells lie above the source elevation range, so Tibetan contrasts are predicted outcomes and are not used as evidence for independent regions. New-region shards ran on the local server and P4 shards on the cloud (cross-environment flag). Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: |*x* parent region excluded from the source, |*i* rest of the parent kept, |*r* within-region design. *n* = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. Pool verdicts (L1e, L4e, L8e) (SI README 2.5)

ID	Filter	Value	Verdict
D-L1e-V-P4, -PE1, -PE2	test_id=L1e, pool $\in$ {P4, PE1, PE2}, scope=verdict, column verdict, stat	significant improvement regions 0/4, 0/5, 1/6(Tibet_LGD: n 0, 3, 10, 40)	supported $\times 3$
D-L1e-compare	test_id=L1e, pool=P4·PE1·PE2, scope=compare	'4 regions verdict extended pool(regions 6) is maintained'	
D-L4e-V-P4	test_id=L4e, pool=P4, scope=verdict	'net value of ML with sparse labels'(minimum n 10)	
D-L4e-V-PE1	pool=PE1	'partial(regions 2/5): net value with sparse labels'(whole pool minimum n none, n $\leq$ 10 not met)	
D-L4e-V-PE2	pool=PE2	'net value of ML with sparse labels'(minimum n 3), flag 'shrinkage depends on prior misspecification'	
D-L4e-PE1-n10	test_id=L4e, pool=PE1, item=R1-P1, scope=MEAN, n=10	-0.13 [-0.43, 0.01]; block-equal -0.19 [-0.42, 0.00]	ns
D-L4e-PE2-n10	pool=PE2, item=R1-P1, scope=MEAN, n=10	-3.59 [-4.12, -3.13]; block-equal -2.80 [-3.43, -2.19]	improve(flag 'shrinkage depends on prior misspecification')
D-L4e-PE2-R1P2-n10	pool=PE2, item=R1-P2, scope=MEAN, n=10	2.61 [0.33, 4.44]; block-equal -0.74 [-3.14, 1.73]	ns
D-L4e-pool-P4-10	test_id=L4e, pool=P4, item='(a)1 pool contrast R1-P1 n10', scope=compare_pool	-0.18 [-0.52, -0.02]; block-equal -0.29 [-0.51, -0.08]	lower error
D-L4e-pool-PE1-10	pool=PE1, same item	-0.13 [-0.42, 0.01]; block-equal -0.19 [-0.41, 0.01]	equivalent
D-L4e-pool-PE1-new-10	pool=PE1-new(Russia_C), same item	0.06 [-0.40, 0.51]; block-equal 0.20 [-0.39, 0.81]	undetermined
D-L4e-pool-PE1-new-1	pool=PE1-new, item='(a)1 pool contrast R1-P1 nall'	-1.40 [-2.95, -0.19]; block-equal -1.54 [-3.12, -0.10]	lower error
D-L4e-compare	test_id=L4e, pool=P4·PE1·PE2, scope=compare	'4 regions verdict extended pool is not maintained. PE1: L28 weakened'	
D-L8e-P4	test_id=L8e, pool=P4, item=R1-P0, scope=MEAN	-2.64 [-3.47, -1.90]; block-equal -2.64 [-3.53, -1.67]	improve
D-L8e-PE1	pool=PE1	-3.19 [-4.19, -2.22]; block-equal -2.62 [-3.71, -1.53]	improve
D-L8e-PE2	pool=PE2	-26.78 [-29.09, -24.21]; block-equal -23.13 [-26.03, -20.20]	improve(flag 'unblinded(label statistics viewed): Tibetan Plateau included')
D-L8e-PE2-rel	pool=PE2, scope=MEAN_rel	-0.16 [-0.19, -0.13](relative units)	improve
D-L8e-compare	test_id=L8e, pool=P4·PE1·PE2, scope=compare	'4 regions verdict extended pool(regions 6) is maintained'	

b. Region-level contrasts (L38) (SI README 2.5)

ID	Item, target, <i>n</i>	Value (cm)	Verdict	Note
D-L38-Tibet-D0-P0-n0	(a) D0-P0, Tibet_LGD x, 0	-59.98 [-62.71, -57.10]; block-equal -62.91 [-65.29, -60.32]	lower error	P0 RMSE 244.51(rmse_B). unblinded flag
D-L38-Tibet-D0-P0-n3, -n10, -n40	(a) D0-P0, Tibet_LGD x, 3·10·40	-130.18, -150.92, -157.59	lower error $\times 3$	L1 form significant improvement(deep regime)
D-L38-Tibet-R1-P0-n-1	(b) R1-P0, Tibet_LGD x, all	-144.73 [-157.42, -129.90]; block-equal -125.70 [-142.62, -108.64]	lower error	

ID	Item, target, n	Value (cm)	Verdict	Note
D-L38-Tibet-P1-P0-n10	(c) P1-P0, Tibet_LGD x, 10	−96.37 [−100.70, −91.92]; block-equal −97.75 [−102.91, −92.22]	lower error	P1 RMSE 148.14
D-L38-Tibet-R1-P1-n10	(c) R1-P1, Tibet_LGD x, 10	−20.90 [−23.40, −17.89]; block-equal −15.82 [−19.40, −12.33]	lower error	flag 'shrinkage depends on prior misspecification'. R1 127.23, P1 148.14
D-L38-Tibet-R1-P2-n10	(c) R1-P2, Tibet_LGD x, 10	19.11 [5.74, 30.14]; block-equal 2.37 [−11.81, 16.77]	undetermined	(a)10 reported alongside. P2 RMSE 108.12. All cell-weighted CIs lie above 0 (P2 more accurate) and only the block-equal CIs include 0. Not to be written as 'indistinguishable'
D-L38-Tibet-R1-P3-n10	(c) R1-P3, Tibet_LGD x, 10	16.65 [9.53, 22.48]; block-equal 5.20 [−3.29, 13.58]	undetermined	(a)10 companion. R1 − P2 same form(cell-weighted CI 0 above, block-equal CI 0 included)
D-L38-Tibet-P2-P1-n10	(c) P2-P1, Tibet_LGD x, 10	−40.02 [−53.21, −24.07]; block-equal −18.19 [−35.84, −0.69]	lower error	flag 'depends on split independence'
D-L38-RussiaC-D0-P0- n0, -n3, -n10, -n-1	(a) D0-P0, Russia_C x, 0·3· 10·all	2.75, 2.44, 1.81, −4.58	higher error, higher error, undeter- mined, undeter- mined	P0 RMSE 32.12
D-L38-RussiaC-R1-P0- n-1	(b) R1-P0, Russia_C x, all	−5.42 [−8.96, −1.64]; block-equal −2.55 [−6.83, 1.48]	undetermined	Region where replication failed (L38 (b)). All cell-weighted CIs lie below 0 and the block-equal CIs include 0 (undetermined, the two weightings disagree). Cell-weighted point estimates and block-equal CIs are not combined in one bracket
D-L38-RussiaC-P1-P0- n10	(c) P1-P0, Russia_C x, 10	−2.11 [−3.37, −0.61]; block-equal −0.70 [−2.24, 0.76]	undetermined	
D-L38-RussiaC-R1-P1- n10	(c) R1-P1, Russia_C x, 10	0.06 [−0.40, 0.51]; block-equal 0.20 [−0.39, 0.81]	undetermined	
D-L38-RussiaW-ref-*	pool=P4 reference, Russia_ W x: (b) R1-P0 all; (c) P1-P0 10; (c) R1-P1 10	−13.98; −11.96; −0.56 [−1.21, −0.02], block-equal −0.24 [−0.84, 0.31]	lower error, lower error, undeter- mined	LG main run CatBoost shards. C2 reduced basis
D-L38-V	test_id=L38, pool=new regions, scope=verdict	'diverging regions: Russia_ C(shallow; (a) met, (b) undetermined); Tibet_ LGD(deep; (a) not met, (b) lower error)'		stat '(a) met 1/2, (b) lower error 1/2. symmetric reference(P4): (a) 4/4, (b) 1/ 4'
D-L38-V-note	same rows, column note	'Tibetan Plateau scoring cells outside source support share: elevation 1.000, $\sqrt{\text{TDD}}$ 0.333 ...; Tibetan Plateau result NOVELTY-N7 sentence basis not used'		

c. Label definition in the Tibetan Plateau (L39) (SI README 2.5)

ID	Filter	Value	Verdict
D-L39-V	test_id=L39, scope=verdict	'robust'(Tibetan Plateau labels definition: GPR vs temperature-derived)	robust
D-L39-tempderived-R1P1- n10	test_id=L39, item=R1-P1 n10, target=Tibet_LGD~L39 x	−20.92 [−22.59, −18.23]; block-equal −21.64 [−23.90, −19.20]	lower error(label set at other locations)
D-L40-V-*	test_id=L40, scope=verdict, target ∈ {Russia_W~exp~lic, Russia_E~exp, Canada~exp~lic, Russia_E~expnokyt}, column verdict, stat	four expanded editions all 'weakened'. four edition all R1-P0 nall P1-P0 n10 'robust', weakened D0-P0 contrast came from	weakened ×4

ID	Filter	Value	Verdict
D-L41-V-*	test_id=L41, scope=verdict, columns verdict, stat, note, same_as; target cell counts from the columns n_cells and eligible of the source tables/lgd_eligibility_v1.csv	Tibetan Plateau: (a), (c) and (h) have no target cells and are 'variant not possible (ineligible in the eligibility table)'. (b), (d), (e), (f) and (g) are 'robust' but all use the run table of the main setting (same_as, 132 cells), and the note records that every row equals the main setting because the TMx of the main setting is used unchanged; the Tibetan Plateau therefore has no label-set sensitivity. Central Russia: (a) variant not possible (12 cells, ineligible), (h) point estimate only, (b) 'weakened' (D0-P0 n3 higher error to undetermined), (e) 'weakened' (D0-P0 n10 undetermined to higher error), (d) and (f) 'robust' (refitted variants); (c) and (g) 'robust' are same_as runs and not sensitivity results	(b)(e)(d)(f) substantive variants
D-L42-V-*	test_id=L42, scope=verdict, column verdict, stat	Tibet 'robust', Russia_C 'weakened'(D0-P0 n0·n3 higher error → undetermined, R1-P1 n10 undetermined → lower error), NAtlantic 'rows none(ineligible)'	
D-elig-*	Source Delig, spec ∈ {NAtlantic_lic, Tibet_L42_lic, Russia_C_L42_lic}, columns n_cells, nb_union, min_nb_eval_used, eligible, regime	NAtlantic 22cells, blocks union 6, ineligible; Tibet 132cells, 34, eligible, deep; Russia_C 57cells, 13, eligible, shallow	

d. North Atlantic, full-data edition (L38) (SI README 2.5)

ID	Item, n	Value (cm)	Verdict	Note
Dfull-L38-NAtl-D0-P0-n0	(a) D0-P0, 0	−9.91 [−15.95, −2.85]; block-equal [−14.53, 2.40]	undetermined	flag 'few blocks'. P0 RMSE 29.80
Dfull-L38-NAtl-D0-P0-n10	(a) D0-P0, 10	−9.15 [−14.87, −1.62]; block-equal [−12.73, 4.15]	undetermined	
Dfull-L38-NAtl-R1-P0-n-1	(b) R1-P0, all	−4.65 [−8.50, 1.42]; block-equal [−6.40, 5.86]	undetermined	
Dfull-L38-NAtl-P1-P0-n10, R1-P1-n10	(c) P1-P0, R1-P1, 10	−4.22; −0.89	undetermined ×2	
Dfull-L1e-V-PE1	test_id=L1e, pool=PE1, scope=verdict	significant improvement regions 0/6(NAtlantic included)	supported	
Dfull-L8e-V-PE1	test_id=L8e, pool=PE1, scope=verdict	Δ −3.44 [−4.45, −2.14], block-equal [−3.54, −0.87], regions 6/6	supported	
Dfull-L4e-compare	test_id=L4e, pool=P4·PE1·PE2, scope=compare	'4 regions verdict extended pool is not maintained'(R1-P1 n10: P4 lower error → PE1 undetermined)		

Supplementary Table S10. Coverage of 90% prediction intervals by step: within-region CQR, transfer CQR, pooled generative intervals and hierarchical conformal intervals.

Nominal coverage 0.90. Coverage [95% CI] and mean width (cm); cell-weighted values. The hierarchical conformal coverage of 0.86 uses a different protocol from the zero-label interval comparison of Fig. 4d and is therefore reported here. Values are copied from paper/claims/SI_uncertainty/README.md section 2 (no new calculation; section 4 recommendation). Detailed rows follow.

a. Steps

Step	Coverage [CI]	Width (cm)	Verdict	Source
Within-region CQR (S11, contest; Alaska, 6-fold block out-of-fold)	0.93 [0.89, 0.97]	53.64	descriptive	U1, SI_uncertainty README 2.2
Transfer CQR trained in Alaska (H17; four main regions, two conditions)	0.32–0.73		H17 confirmed (upper CI < 0.90 in 4/4)	U2, README 2.3
Pooled generative quantile intervals, nflow (LGU-B1, λ 1.0)	0.76 [0.67, 0.83]		supported (replication, unblinded)	U4, README 2.5
Pooled generative quantile intervals, cfm (LGU-B1, λ 1.0)	0.65 [0.56, 0.76]		supported (replication, unblinded)	U4, README 2.5
Hierarchical conformal, hier2_cdf (h37, four main regions)	0.86 [0.80, 0.92]	80.54	partially supported (F10)	U3, README 2.4
Hierarchical conformal, constant normalizer (LGU-B, zero labels)	0.86 [0.83, 0.90]	81.29		U4, README 2.5

b. U1 within-region intervals in Alaska (SI README 2.2)

Item	Columns	Value	Verdict
uncalibrated quantile interval(3 seed ensemble interval, quantile bounds mean)	coverage / cov_ci_lo / cov_ci_hi / width_cm	0.45 / 0.32 / 0.62 / 14.65	descriptive(contest 'adopted')
CQR(3 seed ensemble interval, quantile bounds mean)	coverage / cov_ci_lo / cov_ci_hi / width_cm	0.93 / 0.89 / 0.97 / 53.64	descriptive(contest 'adopted')
before correction interval score	interval_score / is_ci_lo / is_ci_hi	123.91 / 88.04 / 153.78	descriptive
CQR interval score	interval_score / is_ci_lo / is_ci_hi	68.85 / 58.64 / 94.92	descriptive(manuscript action: descriptive relaxed)
CQR decomposition	is_width_part / is_under_part / is_over_part	53.64 / 2.66 / 12.55	descriptive
CQR coverage(E4 re-scored, S11 same ensemble interval)	coverage / cov_ci_lo / cov_ci_hi	0.93 / 0.89 / 0.96	descriptive
constant width reference interval score	interval_score / is_ci_lo / is_ci_hi	65.39 / 55.61 / 92.19	descriptive
constant width reference decomposition	is_width_part / is_under_part / is_over_part	43.46 / 3.84 / 18.09	descriptive
constant width reference coverage	coverage / cov_ci_lo / cov_ci_hi	0.90 / 0.85 / 0.92	descriptive
constant width half-width(post hoc reference)	half_width_cm	21.73	not applicable
CQR — constant width(arithmetic difference)	interval_score	+3.46	Descriptive. No paired CI in the source table. The interval [0.6, 6.5] of the source document is not reproduced exactly (it depends on the draws; reading notes of the source README)
CQR conditional coverage(observed ALT 5quantile)	coverage	0.85 / 1.00 / 1.00 / 1.00 / 0.83	descriptive
CQR width 5quantile both ends	coverage / interval_score	W1 0.86 / 75.52, W5 0.97 / 76.59	descriptive
CQR blocksby coverage(cells 20 or more 33blocks)	coverage(median) / lo(10 percentile) / hi(90 percentile) / interval_score(coverage 0.8 below blocks share)	0.98 / 0.70 / 1.00 / 0.15	descriptive
reference: M1 edition Alaska within region CQR(x25)	coverage / cov_ci_lo / cov_ci_hi / width_mean / interval_score	0.91 / 0.83 / 0.94 / 48.94 / 70.35	descriptive

c. U2 transfer CQR (coverage) (SI README 2.3)

Method	Condition	Lena Delta	Canada	W Russia	E Russia	Verdict
cqr_ak	noinfo	0.57 [0.48, 0.66] (0.5700)	0.60 [0.43, 0.74] (0.5953)	0.32 [0.20, 0.43] (0.3214)	0.73 [0.56, 0.89] (0.7284)	H17 confirmed(4/4 CI upper bound < 0.90)

Method	Condition	Lena Delta	Canada	W Russia	E Russia	Verdict
cqr_ak	covonly	0.57 [0.46, 0.64] (0.5748)	0.53 [0.35, 0.74] (0.5292)	0.34 [0.17, 0.45] (0.3415)	0.66 [0.48, 0.90] (0.6597, upper bound 0.8964)	H17 confirmed(4/4)
cqr_iw	noinfo	0.35 [0.26, 0.41]	0.52 [0.38, 0.68]	0.30 [0.17, 0.43]	0.69 [0.53, 0.87]	descriptive(recovery none)
cqr_iw	covonly	0.33 [0.21, 0.40]	0.47 [0.30, 0.66]	0.34 [0.15, 0.46]	0.62 [0.44, 0.87]	descriptive
anchor_ak	noinfo	0.92 [0.79, 0.95]	0.77 [0.68, 0.87]	0.57 [0.44, 0.71]	0.84 [0.71, 1.00]	descriptive
anchor_iw	noinfo	0.93 [0.79, 0.96]	0.88 [0.79, 0.94]	0.56 [0.41, 0.70]	0.84 [0.71, 1.00]	descriptive

d. *U3 zero-label hierarchical conformal intervals (SI README 2.4)*

Item	Columns	Value	Verdict
hier2_cdf(main method) main 4 regions	coverage / coverage_lo / coverage_hi / coverage_beq / width_cm / interval_score	0.86 / 0.80 / 0.92 / 0.85 / 80.54 / 131.98	partially supported(F10)
cell pooling(pooled)	coverage / coverage_lo / coverage_hi / coverage_beq / width_cm / interval_score	0.73 / 0.65 / 0.81 / 0.74 / 56.34 / 152.86	descriptive
hier2_sub	coverage / coverage_lo / coverage_hi / width_cm	0.83 / 0.77 / 0.89 / 72.86	descriptive
hier2_blk	coverage / coverage_lo / coverage_hi / width_cm	0.87 / 0.81 / 0.93 / 83.30	descriptive
hier2_cdf Lena Delta	coverage / coverage_lo / coverage_hi / coverage_beq / width_cm / K_groups	0.89 / 0.80 / 0.96 / 0.81 / 73.03 / 4	descriptive
hier2_cdf Canada	same	0.89 / 0.80 / 0.97 / 0.88 / 95.35 / 4	descriptive
hier2_cdf W Russia	same	0.75 / 0.59 / 0.88 / 0.74 / 71.13 / 4	descriptive(outside the target band)
hier2_cdf E Russia	same	0.93 / 0.80 / 1.00 / 0.97 / 82.64 / 4	descriptive
hier2_cdf Alaska(reference)	same	0.98 / 0.96 / 0.99 / 0.97 / 78.54 / 4	descriptive
F10 verdict rows	in_band / coverage(MAIN6)	4 / 0.88	partially supported. ci_flag: 'hier2_cdf in [0.85,0.95]: 4/6; ref_cqr_ak<0.85: 5/6; regions_in_band=Russia_C,Russia_ E,Canada,Lena'

e. *U4 normalizers and generative intervals (LGU-B) (SI README 2.5)*

Hypothesis	Cell-weighted	Block-equal	Verdict (blinding)
LGU-B1 nflow, cqr_pool λ 1.0	0.76 [0.67, 0.83] (0.7568 [0.6715, 0.8262])	0.79 [0.70, 0.84] (0.7851)	supported(replication (unblinded))
LGU-B1 cfm, cqr_pool λ 1.0	0.65 [0.56, 0.76] (0.6529 [0.5636, 0.7645])	0.68 [0.60, 0.80] (0.6844)	supported(replication (unblinded))
LGU-B2 interval score nflow – placebo(nflow), confirmatory	+3.29 [–4.17, 8.16]	+0.03 [–6.49, 5.52]	undetermined(blind). verdict_ 3region undetermined, n_ neg_regions 1, coverage_pool 0.89(0.8919)
LGU-B3 nflow – cbq	–7.41 [–52.55, 10.00]	–1.46 [–61.11, 23.49]	undetermined(equivalent not testable)
LGU-B4 nflow – const(as calibrated)	+3.75 [–10.73, 12.91]	+4.06 [–7.02, 15.18]	undetermined
LGU-B4 nflow – const(mean log width matched)	+4.96 [–0.81, 9.00]	+0.72 [–3.80, 6.65]	undetermined
LGU-B4 cbq – const(as calibrated)	+11.16 [–11.20, 53.46]	+5.52 [–24.01, 65.10]	undetermined
LGU-B4 cbq – const(width matched)	+5.24 [–5.34, 18.66]	+0.18 [–9.62, 12.90]	undetermined
LGU-B4 cfm – const(as calibrated)	–1.70 [–17.91, 10.03]	–0.58 [–11.86, 10.63]	undetermined
LGU-B4 cfm – const(width matched)	+2.72 [–1.38, 6.88]	–1.00 [–4.62, 3.66]	undetermined
LGU-B4 phys – const(as calibrated)	–0.83 [–8.74, 2.54]	–4.64 [–12.13, 0.82]	undetermined
LGU-B4 phys – const(width matched)	+0.78 [–4.02, 3.46]	–2.20 [–7.42, 1.80]	undetermined
LGU-B6 nflow – const, n 10 / 40 / 160(range B, Lena Delta and Canada· Alaska)	+10.71 [4.45, 15.57] / +10.81 [2.97, 18.57] / +10.93 [2.83, 18.93]	+3.50 [–3.52, 10.42] / +3.25 [–3.01, 13.03] / +2.55 [–3.04, 12.06]	undetermined(cell-weighted higher error side)
LGU-B6 cbq – const, n 10 / 40 / 160	+8.37 [–8.95, 135.07] / +38.34 [1.30, 272.33] / +43.00 [3.33, 279.22]	–8.86 [–34.22, 135.12] / +18.05 [–24.13, 298.48] / +23.93 [–22.35, 300.43]	undetermined

f. *U4 normalizers at zero labels (SI README 2.5)*

Normalizer	cov10	Two-stage CI	cov10 block- equal	Width (cm)	Interval score (cm)
const	0.86 (0.8640)	[0.83, 0.90]	0.85	81.29	131.21
phys	0.87 (0.8724)	[0.84, 0.90]	0.87	82.98	130.38
nflow	0.89 (0.8919)	[0.86, 0.90]	0.89	96.01	134.96
nflow placebo	0.89 (0.8872)	[0.86, 0.90]	0.88	94.95	131.67
cfm	0.88 (0.8850)	[0.86, 0.90]	0.88	94.34	129.50
cbq	0.95 (0.9533)	[0.91, 0.97]	0.95	132.44	142.36

g. U5 hierarchical predictive distributions with labels and AB10 (LGU-A) (SI README 2.6)

Hypothesis	Columns	Value	Verdict
LGU-A1 (iii) – B4, n 10(confirmatory, AB10)	delta / ci_lo / ci_hi	+9.51 / –0.75 / 22.11	undetermined(blind)
same	delta_beq / ci_lo_beq / ci_hi_beq	+8.64 / –2.55 / 22.04	same
same	verdict_1stage / ci_lo_1stage / ci_hi_1stage	higher error / 4.55 / 13.02	depends on calibration uncertainty
same	eq_state / half_width / ref_score / coverage_pool	not testable(precision insufficient) / 12.29 / 99.03 / 0.90(0.9011)	same
LGU-A1 n 40	delta / ci_lo / ci_hi	–2.80 / –12.50 / 7.89	undetermined
same	delta_beq / ci_lo_beq / ci_hi_beq	–2.48 / –13.71 / 8.58	same
same	verdict_1stage / ci_lo_1stage / ci_hi_1stage	lower error / –4.95 / –0.56	depends on calibration uncertainty
same	eq_state / half_width / ref_score / coverage_pool	not testable(precision insufficient) / 11.15 / 105.46 / 0.90(0.9017)	same
LGU-A1 overall	verdict	undetermined	undetermined
same contrast % converted(figure table)	delta_pct / ci_lo_pct / ci_hi_pct	n 10: 9.60 / –0.76 / 22.33; n 40: –2.66 / –11.85 / 7.48; B2: 2.50 / –3.17 / 6.19	undetermined
AB10 two weighting p	delta / p_cell / p_beq / p_eq / verdict4	9.51 / 0.07 / 0.09 / 0.80 / undetermined	undetermined
AB10 bundle rows	delta / holm_p / holm_p_eq / holm_m / verdict4 / abstract_rule	9.51 / 0.28 / 1.00 / 10 / undetermined / '(d) no difference established'	undetermined(J7)
LGU-A3 (iii) median – P1, RMSE, main 4 regions n 3	delta / ci_lo / ci_hi; delta_beq / ci_lo_beq / ci_hi_beq	+1.43 / 0.68 / 1.81; +1.73 / 0.96 / 2.16	higher error
same n 10	same	+1.60 / 0.57 / 1.88; +1.65 / 0.76 / 2.13	higher error
same W Russia excluded 3 regions n 3 / n 10	delta / ci_lo / ci_hi	+0.90 / 0.41 / 1.68; +1.08 / 0.32 / 1.80	higher error
LGU-A7 (iv) – (iii), interval score, n 10	delta / ci_lo / ci_hi	+25.23 / 17.46 / 34.02	higher error
LGU-A2 infinite interval share, n 3(5rungs)	delta(inf10) maximum	0.00	passed(infinite interval 0)

h. U6 R1 intervals with labels (LGX L25) (SI README 2.7)

Target	n	Coverage	Width (cm)	Width R0, n 0	In band	Narrower
Lena x	40	0.89(0.8940)	67.88	74.62	1	1
Lena x	160	0.90(0.8984)	72.64	74.62	1	1
Canada x	40	0.85(0.8475)	92.39	104.84	0	0
Canada x	160	0.84(0.8351)	86.00	104.84	0	0
Alaska x	40	0.89(0.8914)	56.60	93.20	1	1
Alaska x	160	0.90(0.8982)	51.82	93.20	1	1

Supplementary Table S11. Sub-regions, targets with point estimates only and the North Atlantic group.

Sub-regions of Alaska (AL-1 to AL-6), Canada (CA-2, CA-3) and the Lena Delta (LE-1, LE-2) were defined by k -means clustering of 0.5° block centroids without labels (133 blocks, assignment fixed before the main run). Mode x excludes the parent region from the source; mode i keeps the rest of the parent (simulated new area inside a labelled region). Ranges are over valid splits. E_0 in $\text{cm } (^\circ\text{C d})^{-1/2}$. The North Atlantic group is reported only in the full-data edition (licence-unverified rows; D README E19); n.a., not available. Source: paper/claims/D_data_and_design/tables/Table1_data.csv (copy of the v2 Table 1 data) and README E19; North Atlantic splits, A cells, scored cells and E_0 from the shard records of the full-edition LGD run (data/processed/lgd/NAtlantic/shards, unit.json).

Target	Group	Modes	Rows	1 km loc.	Blocks	Splits	A cells	Scored cells (blocks)	E_0 , x / i	Label type
Russia C	Point estimate only	x	7	7	6	5	3–4	3–4 (2–3)	1.60	CALM site mean
Greenland	Point estimate only	x	3	2	2	0	1–2	0–1 (0–1)	1.59	CALM site mean
AL-1	Sub-region	i, x	5,154	51	10	4	288–4,834	320–4,866 (2–9)	1.50 / 1.54	Point, ABoVE (mostly GPR)
AL-2	Sub-region	i, x	3,491	128	20	5	1,333–1,924	1,567–2,158 (5–17)	1.50 / 1.54	Point, ABoVE (mostly GPR)
AL-3	Sub-region	i, x	651	69	16	5	269–361	290–382 (3–10)	1.50 / 1.61	Point, ABoVE (mostly GPR)
AL-4	Sub-region	i, x	471	44	4	3	139–287	184–332 (2–3)	1.50 / 1.60	Point, ABoVE (mostly GPR)
AL-5	Sub-region	i, x	619	27	17	5	270–454	165–349 (4–14)	1.50 / 1.59	Point, ABoVE (mostly GPR)
AL-6	Sub-region	i, x	3,220	24	7	4	1,516–1,705	1,515–1,704 (2–5)	1.50 / 1.60	Point, ABoVE (mostly GPR)
CA-2	Sub-region	i, x	395	27	14	5	161–237	158–234 (2–8)	1.59 / 1.60	Point, ABoVE
CA-3	Sub-region	i, x	349	53	19	5	113–224	125–235 (3–13)	1.59 / 1.58	Point, ABoVE
LE-1	Sub-region	i, x	1,786	135	14	4	448–1,184	602–1,335 (3–12)	1.62 / 1.62	Point, ALLena (single visits)
LE-2	Sub-region	i, x	1,251	66	6	3	183–1,062	167–1,014 (4–5)	1.62 / 1.62	Point, ALLena (single visits)
North Atlantic	New region, full edition only	x	41	n.a.	13	5	19–24	12–19 (3–7)	1.59	1 km cell means

Supplementary Table S12. Data sources, licences and excluded data.

Licences as recorded at access; rows from sources without a confirmed licence are excluded from the deposited tables and from the licence-verified edition used for the main verdicts. Values from these sources can be obtained from the original providers. Source: docs/MANUSCRIPT_DRAFT_METHODS_INTRO_2026-09-30.md (Data availability, 30 September 2026) and front.tex (Data availability); licence status per data/processed/fidelity_base_v4_meta.json (by_source).

Use	Data set	Repository and identifier	Licence	Status
Labels, main regions	GTN-P/CALM compilation (Streletskiy et al., 2025)	PANGAEA, doi:10.1594/PANGAEA.972777	CC BY 4.0	used
Labels, main regions	ALLena thaw depth, Lena River Delta (Veremeeva et al., 2025)	PANGAEA, doi:10.1594/PANGAEA.973813; 10.1594/PANGAEA.974408	CC BY 4.0	used
Labels, main regions	ABoVE soil moisture and ALT, version 2 (Moore et al., 2025)	ORNL DAAC, doi:10.3334/ORNLDAAC/2369	NASA EOSDIS open data, no restrictions; citation required	used
Labels, new regions	GPR and pit survey points, Tibetan Plateau (Du et al., 2026)	Zenodo, doi:10.5281/zenodo.21999366; field data doi:10.12072/ncdc.permafrost.db7703.2026	MIT (Zenodo record); licence of the field data not yet confirmed	used
Labels, new regions	Ilulissat probing (Scheer et al., 2024)	PANGAEA, doi:10.1594/PANGAEA.964306	CC BY 4.0	used
Labels, new regions	T-MOSaIC myThaw 2021–2023 (Martin et al., 2023; Boike et al., 2024; Hammar et al., 2025)	PANGAEA, doi:10.1594/PANGAEA.956039; .971586; .974461	CC BY 4.0	used
Labels, new regions	Additional CALM events of the PANGAEA compilation	PANGAEA, doi:10.1594/PANGAEA.972777	CC BY 4.0	used
Labels, new regions	Disko Island (Zastruzny et al., 2024)	PANGAEA, doi:10.1594/PANGAEA.967139	CC BY 4.0	used
Labels, new regions	Tavvavuoma (Sannel, 2020)	Bolin Centre Database, doi:10.17043/sannel-2020-temperature-1	CC BY 4.0	used
Labels, new regions	Cape Mamontov Klyk (Grosse, 2007)	PANGAEA, doi:10.1594/PANGAEA.611409	CC BY 3.0	used
Labels, new regions	Pedon data (Palmtag et al., 2022)	Bolin Centre Database, doi:10.17043/palmtag-2022-pedon-1	ODC-By	used
Labels, new regions	FireALT, measured depths only (Talucci et al., 2024)	NSF Arctic Data Center, doi:10.18739/A2RN3092P	CC BY 4.0	used
Labels, new regions	Syrdakh observatory (Pohl et al., 2026)	Zenodo, doi:10.5281/zenodo.19890671	CC BY 4.0	used
Labels, new regions	Kolyma Water-Balance Station (Makarieva et al., 2017)	PANGAEA, doi:10.1594/PANGAEA.881754	CC BY 3.0	used
Labels, new regions	Indigirka tree mounds (Liang et al., 2023)	PANGAEA, doi:10.1594/PANGAEA.961876	CC BY 4.0	used
Labels, new regions	Yamal transect (Walker et al., 2009)	PANGAEA, doi:10.1594/PANGAEA.842711	CC BY 3.0	used (PANGAEA rows)
Label definition test	Temperature-derived labels (Fu, 2025)	figshare, doi:10.6084/m9.figshare.29206613.v1	CC BY 4.0	L39 only
Labels, licence not confirmed	CALM web files, Abisko and Kapp Linné (Åkerman, 1998; Åkerman and Johansson, 2008)	https://www2.gwu.edu/~calm/data/north.htm	not confirmed	full-data edition only
Labels, licence not confirmed	CUSP v1.1 synthesis (Jorgenson and Kanevskiy, 2025; Petrone et al., 2016)	github.com/jonschwenk/cusp v1.1; doi:10.18739/A27P8TG0G (CC0 1.0); doi:10.1594/PANGAEA.845258 (CC BY-NC-SA 3.0)	not confirmed	full-data edition only
Labels, licence not confirmed	NSIDC GGD353 thaw tubes (Nixon, 2003)	doi:10.7265/7m84-k262	provider asks to be consulted	full-data edition only
Labels, licence not confirmed	Yamal transect, ten report-derived rows	report	not confirmed	full-data edition only
Descriptive only	NSIDC GGD402; Yang and Qiu (2026)			parsed, no analysis used
Covariates	ERA5-Land monthly averaged data (Muñoz-Sabater et al., 2021)	Copernicus Climate Data Store, doi:10.24381/cds.68d2bb30	Copernicus licence (CC BY type, CDS page)	
Covariates	ESA CCI Permafrost ALT, version 4.0 (Westermann et al., 2024)	CEDA, doi:10.5285/d34330ce3f604e368c06d76de1987ce5	ESA CCI Permafrost terms of use (registered and unregistered use; citation required)	used
Covariates	SoilGrids 2.0 (Poggio et al., 2021)	ISRIC WCS service	CC BY 4.0 (release 'latest' at access; version tag not recorded)	used

Use	Data set	Repository and identifier	Licence	Status
Covariates	Copernicus DEM GLO-30	doi:10.5270/ESA-c5d3d65	© DLR e.V. 2010–2014 and Airbus 2014– 2018, Copernicus programme	used
Covariates, XM only	ESA WorldCover 10 m 2021 v200 (Zanaga et al., 2022)	Zenodo, doi:10.5281/zenodo.7254221	CC BY 4.0; © ESA WorldCover project 2021 / Contains modified Copernicus Sentinel data (2021) processed by ESA WorldCover consortium	XM only
Covariates, XM only	Sentinel-2 L2A, July and August 2019–2023 (534 scenes)	Earth Search v1 STAC (Element 84), collection sentinel-2-l2a; AWS Open Data sentinel-cogs	Copernicus Sentinel data terms (free, full and open); Contains modified Copernicus Sentinel data [2019–2023]	XM only
Map masks	ESA CCI Permafrost extent (permafrost fraction), version 4.0, 1997–2021 mean	data set DOI not yet confirmed	ESA CCI data policy: free and open access	maps only
Map masks	JRC Global Surface Water occurrence (Pekel et al., 2016)	Pekel et al., Nature 540, 418–422 (2016), doi:10.1038/nature20584		maps only
Optional	KPDC Council case study (KOPRI-KPDC-00002125, -00002707, -00002955)	KPDC	KPDC access conditions	[DECISION: KPDC 자료, SI 포함 여부 와 식별자, 이 용 조건]

Supplementary Table S13. Software and execution environments.

LG main run, LGX-A and LGX-B ran on cloud nodes (Rescale iolite-4: 64 CPU cores, four NVIDIA T4 GPUs). LGT, LGF, LGU, LGD, the local continuation of LGX and the aggregations ran on a shared local server with NVIDIA RTX 3090 GPUs (24 GB) at low priority (nice 10, at most 32 CPU threads, one process per GPU). Package versions of the WF runs and of X job A (XB, XD-alg, XH, XI, XJ; 5 October 2026) are those recorded in results/rescale_wf*/logs_wf/ and results/rescale_xbatch_A/logs_xbatch_A/ (env.log, deps.log); XA, XC, XD-4, XE, XG and XM ran on the local server. Each shard records the code hash, the configuration hash, the package versions and the device. Source: docs/MANUSCRIPT_DRAFT_METHODS_INTRO_2026-09-30.md Table 4 and methods.tex (Code availability).

Item	Local server	Cloud (LG, LGX)	Cloud (WF runs, X job A)
Python	3.9.18	3.11.10	3.11.10
NumPy / pandas / scikit-learn / SciPy	1.26.4 / 2.1.4 / 1.3.0 / 1.11.4	2.4.6 / 2.3.3 / 1.9.1 / 1.17.1	NumPy 2.1.1 (third WF run 2.4.6) / pandas 2.3.3 / scikit-learn 1.9.1 / SciPy 1.17.1
CatBoost / PyTorch / pytabkit	1.2.10 / 2.6.0 (CUDA 12.4) / 1.7.3	1.2.10 / 2.4.1 (CUDA 12.1) / 1.7.3	1.2.10 / 2.4.1 (CUDA build 12.1, run on CPU) / not installed
Foundation models	tabpfn 8.0.7, tabicl 2.0.2	none	none
GPU	NVIDIA RTX 3090, 24 GB	NVIDIA T4	CPU nodes

Supplementary Table S14. Abstract contrast bundle AB1 to AB10.

Four main regions, stratified means; AB1 to AB9 with 10,000 block resamples in both weightings, AB10 (interval score of the hierarchical predictive interval against the calibrated constant-width interval B4; Lena Delta, Canada, Alaska) with 1,000 resamples from LGU-A1, for which the copied bundle row has no CI. P values are uncorrected one-sided bootstrap values; Holm adjustment over the ten contrasts. Common-CI verdict: the verdict under the common resampling of both arms; 'yes' marks verdicts that depend on the split-independence assumption. The abstract uses verdict words only for these contrasts. Source: `paper/claims/C1_label0_safety/tables/lgw_bundle.csv` (copy of `data/processed/lgw/lgw_bundle.csv`), rows scope = MEAN. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: $|x$ parent region excluded from the source, $|i$ rest of the parent kept, $|r$ within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

AB	Contrast	Δ cell-weighted [95% CI] (cm)	Δ block-equal [95% CI] (cm)	P cell	P block	Holm P	Verdict	Common- CI verdict	Split- dependent
AB1	D0-P0 n0	+2.25 [+1.10, +3.42]	+1.62 [+0.52, +2.69]	0.0004	0.005	0.023	higher error	undetermined	yes
AB2	B:ens-P0 n0	-2.73 [-3.33, -0.91]	+0.05 [-0.94, +1.01]	0.002	0.912	1.000	undetermined	undetermined	
AB3	D1-D1@shuffle n0	-1.63 [-1.98, -1.04]	-1.05 [-1.51, -0.57]	0.0001	0.0001	0.001	lower error	lower error	
AB4	P1-P0 n10	-2.45 [-3.04, -1.88]	-2.62 [-3.30, -1.87]	0.0001	0.0001	0.001	lower error	lower error	
AB5	R1-P1 n10	-0.18 [-0.53, -0.01]	-0.29 [-0.51, -0.08]	0.034	0.008	0.136	lower error	undetermined	yes
AB6	R2-R1 n10	-0.13 [-0.31, +0.22]	-0.06 [-0.31, +0.20]	0.604	0.656	1.000	equivalent	equivalent	
AB7	R1-F1k n10	-3.74 [-5.23, -2.75]	-4.07 [-5.14, -2.98]	0.0001	0.0001	0.001	lower error	lower error	
AB8	R1-P0 all	-2.64 [-3.48, -1.85]	-2.64 [-3.51, -1.69]	0.0001	0.0001	0.001	lower error	lower error	
AB9	R1-P1 all	-0.40 [-0.76, -0.22]	-0.51 [-0.74, -0.27]	0.0001	0.0001	0.001	lower error	lower error	
AB10	interval score (α 0.1), rung (iii) – B4, n10	+9.51 [n.a.]	n.a.	0.072	0.092	0.277	undetermined	n.a.	

Supplementary Table S15. Transfer without target labels: learners, physics information and physics baselines.

Direct ML minus the source-coefficient Stefan model for ten learners (four main regions), regional rows, the three-region auxiliary column, new regions, placebo pseudo-labels, low-weight residuals and the physics baselines. Holm families differ by row (abstract bundle, LGX, LGT, LGF). Learners 1 to 4 ran on the cloud platform, the other learners and companion rows on the local GPU server; differences between platforms are not computed. Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: | x parent region excluded from the source, | i rest of the parent kept, | r within-region design. n = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

a. Direct ML minus source-coefficient Stefan, zero labels, four main regions (C1 README 2.1)

#	Learner	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Holm P	Common CI (cell / block), verdict	Dependence
1	CatBoost main learner(catboost_lo)	+2.25 [1.10, 3.42]	+1.62 [0.52, 2.69]	higher error	0.023(abstract bundle m = 10)	[0.31, 4.19] / [−0.21, 3.43], undetermined	depends on split independence
2	CatBoost higher capacity(catboost)	+1.94 [0.88, 3.43]	+1.56 [0.30, 2.76]	higher error	0.0032	[0.24, 4.32] / [−0.45, 3.57], undetermined	depends on split independence
3	CatBoost source-CV tuning(catboost_tuned)	+1.59 [0.65, 2.69]	+1.63 [0.56, 2.64]	higher error	0.0032	[0.07, 3.37] / [−0.08, 3.33], undetermined	depends on split independence
4	random forest(rf)	+2.63 [1.47, 3.96]	+2.03 [0.99, 3.06]	higher error	0.0003	[0.72, 4.75] / [0.31, 3.75], higher error	
5	TabPFN v2	+1.02 [0.23, 2.35]	+1.22 [0.27, 2.15]	higher error(holm_note = 'significant before correction')	0.1168	[−0.27, 3.13] / [−0.34, 2.75], undetermined	depends on split independence
6a	TabICL v2, context cap 10,000 rows	+0.93 [0.28, 3.12]	+2.43 [1.35, 3.55]	higher error	0.0232	[−0.36, 4.20] / [0.72, 4.20], undetermined	depends on split independence
6b	TabICL v2, full source context	+0.85 [0.07, 3.19]	+2.22 [1.09, 3.45]	higher error	0.0374	[−0.60, 4.26] / [0.53, 4.10], undetermined	depends on split independence
7	MLP(leave-one-source-region-out tuning)	+2.45 [1.62, 4.11]	+2.05 [0.95, 3.19]	higher error	0.0004	[0.95, 4.90] / [0.25, 3.94], higher error	
8	multi-head MLP(tabm*, not the public TabM)	+2.44 [1.58, 4.73]	+3.94 [2.68, 5.29]	higher error	0.0004	[0.87, 5.98] / [1.92, 6.33], higher error	
9	reduced FT-Transformer(ftt*)	+4.13 [3.28, 5.63]	+4.58 [3.44, 5.76]	higher error	0.0004	[2.62, 6.52] / [2.65, 6.59], higher error	
10	RealMLP(realmlp*)	+5.21 [4.07, 7.17]	+5.91 [4.60, 7.25]	higher error	0.0004	[3.18, 8.31] / [3.73, 8.19], higher error	
auxiliary	physics-output input direct ML(F1k)	+2.62 [1.38, 4.36]	+2.73 [1.57, 3.85]	higher error	0.0023	[0.40, 5.53] / [0.82, 4.62], higher error	
companion	same context CatBoost(local, D0[C])	+2.64 [1.36, 3.72]	+1.79 [0.69, 2.81]	higher error	none	[0.59, 4.39] / [−0.005, 3.51], undetermined	depends on split independence

b. AB1 regional rows (C1 README 2.2)

Region	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Common CI (cell / block), verdict	Dependence
Lena	+4.02 [1.98, 5.62]	+0.20 [−1.66, 1.97]	undetermined	[0.61, 6.58] / [−3.00, 3.18], undetermined	
Delta					
Canada	−1.56 [−2.70, 1.41]	−0.01 [−1.61, 1.62]	undetermined	[−3.27, 3.23] / [−2.85, 2.63], undetermined	

Region	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Common CI (cell / block), verdict	Dependence
W Russia	+2.05 [−0.94, 4.49]	+2.78 [−0.03, 5.28]	undetermined	[−3.28, 6.35] / [−2.11, 7.13], undetermined	
E Russia	+4.48 [1.41, 6.51]	+3.51 [1.14, 5.91]	higher error	[−0.32, 7.50] / [−0.10, 7.20], undetermined	depends on split independence

c. New independent regions, licence-verified edition (L1e, PE2, zero labels) (C1 README 2.2)

Region	Δ cell-weighted [CI]	Δ block-equal [CI]	sig	sig (common)	Flags	Cross-environment
Central Russia(Russia_C x)	+2.75 [2.11, 4.09]	+4.83 [3.17, 6.60]	worse	worse	none	cross-environment; (i) CatBoost not passed
Tibetan Plateau(Tibet_LGD x)	−59.98 [−62.90, −57.15]	−62.91 [−65.38, −60.32]	improve	improve	predicted outcome(unblinded); unblinded(label statistics viewed)	cross-environment; (i) CatBoost not passed

d. Three-region auxiliary column (Lena Delta, Canada, Alaska) (C1 README 2.2)

Learner	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Common CI, verdict	Dependence	Four-region Δ
catboost_lo	+7.75 [5.49, 10.11]	+6.23 [5.04, 7.40]	higher error	[4.11, 11.45] / [4.23, 8.17], higher error		+2.25
catboost	+8.09 [5.89, 9.44]	+5.69 [4.63, 6.74]	higher error	[4.95, 10.28] / [3.87, 7.46], higher error		+1.94
rf	+6.91 [5.37, 8.07]	+5.17 [4.16, 6.16]	higher error	[4.51, 8.85] / [3.39, 6.88], higher error		+2.63
catboost_tuned	+4.06 [2.62, 4.85]	+2.61 [1.79, 3.41]	higher error	[2.09, 5.30] / [1.22, 3.94], higher error		+1.59
TabPFN v2	+7.83 [5.40, 8.66]	+4.96 [3.94, 5.98]	higher error	[4.67, 9.05] / [3.23, 6.62], higher error		+1.02
TabICL v2, context cap 10,000 rows	+12.25 [7.55, 14.21]	+7.91 [6.55, 9.26]	higher error	[5.53, 15.38] / [5.72, 10.15], higher error		+0.93
TabICL v2, full source context	+0.31 [−0.68, 2.37]	+0.70 [−0.35, 1.78]	undetermined	[−1.41, 3.45] / [−1.01, 2.58], undetermined		+0.85
mlp*	+9.28 [6.30, 10.46]	+6.82 [5.75, 7.90]	higher error	[5.45, 11.12] / [5.01, 8.63], higher error		+2.45
tabm*	+9.26 [6.64, 10.77]	+7.91 [6.61, 9.18]	higher error	[5.29, 11.77] / [5.78, 10.06], higher error		+2.44
ftt*	+1.17 [0.48, 2.14]	+0.94 [0.32, 1.55]	higher error	[0.03, 2.76] / [−0.14, 2.04], undetermined	depends on split independence	+4.13
realmlp*	+6.54 [4.67, 7.37]	+6.10 [5.07, 7.14]	higher error	[3.78, 8.01] / [4.36, 7.82], higher error		+5.21
auxiliary F1k	+7.69 [5.48, 10.33]	+7.05 [5.78, 8.33]	higher error	[3.96, 11.97] / [4.92, 9.12], higher error		+2.62
companion D0[C]	+8.00 [5.55, 10.36]	+6.01 [4.88, 7.15]	higher error	[4.28, 11.50] / [4.12, 7.82], higher error		+2.64

e. Limit row: TabICL v2 with ten labels (C1 README 2.2)

Filter	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Common CI, verdict	Dependence
contrast=D0[I]-P0 n10, scope=MEAN	−2.91 [−3.58, −1.11]	−1.49 [−2.65, −0.21]	lower error	[−4.07, −0.31] / [−3.15, 0.39], undetermined	depends on split independence
contrast=D0[I]-P0 n10, scope=MEAN3	+5.98 [2.26, 7.22]	+3.46 [2.52, 4.43]	higher error	[1.85, 7.53] / [2.02, 4.99], higher error	

f. Source of the physics information: placebo pseudo-labels (AB3, L15) (C1 README 2.3)

Contrast	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Holm P
AB3 D1 − shuffled pseudo-labels, n 0	−1.63 [−1.98, −1.04]	−1.05 [−1.51, −0.57]	lower error	0.001(abstract bundle m = 10)
L15 shuffle, n 0	−1.63 [−1.99, −1.04]	−1.05 [−1.52, −0.58]	lower error	0.003
L15 shuffle, n 10	−1.15 [−1.43, −0.73]	−0.68 [−1.06, −0.29]	lower error	0.003

Contrast	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Holm P
L15 const_t, n 0	-1.68 [-2.06, -1.12]	-1.06 [-1.55, -0.56]	lower error	0.003
L15 const_t, n 10	-1.16 [-1.46, -0.76]	-0.74 [-1.11, -0.37]	lower error	0.003
L15 const_src, n 0	-3.63 [-4.33, -2.44]	-2.81 [-3.49, -2.11]	lower error	0.003
L15 const_src, n 10	-2.21 [-2.89, -1.27]	-1.54 [-2.23, -0.86]	lower error	0.003
L15 tddlin, n 0	-0.48 [-0.62, -0.27]	-0.41 [-0.61, -0.21]	lower error(size below 0.5 cm)	0.0034
L15 tddlin, n 10	-0.08 [-0.18, 0.05]	-0.03 [-0.15, 0.09]	equivalent	1
L15 shuffle, n 0, 3 regions	-3.78 [-4.31, -1.97]	-1.19 [-1.78, -0.61]	lower error	none

g. Low-weight residuals and combination structure at zero labels (C1 README 2.4)

Contrast	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Common CI, verdict	Dependence	Note
R0[T] - P0(TabPFN residual, λ 0.25)	-0.31 [-0.48, 0.05]	-0.21 [-0.43, 0.02]	equivalent	[-0.59, 0.23] / [-0.58, 0.15], undetermined	depends on split independence	p_eq 0.0163, holm_p_eq none(before correction)
same contrast, 3 regions	-0.13 [-0.31, 0.27]	+0.38 [0.10, 0.66]	undetermined	[-0.44, 0.47] / [-0.07, 0.84], undetermined		
R0[C] - P0(same context CatBoost residual)	-0.05 [-0.42, 0.16]	-0.21 [-0.48, 0.04]	equivalent	[-0.62, 0.31] / [-0.64, 0.20], undetermined	depends on split independence	p_eq 0.0145
same contrast, 3 regions	+0.02 [-0.29, 0.53]	+0.39 [0.04, 0.73]	undetermined	[-0.46, 0.76] / [-0.22, 0.98], undetermined		
R0[I] - P0(TabICL residual)	-0.27 [-0.48, 0.25]	+0.01 [-0.28, 0.30]	equivalent	[-0.67, 0.54] / [-0.48, 0.49], undetermined	depends on split independence	eq_p 0.0191, holm_p 1
same contrast, 3 regions	+0.64 [0.17, 1.12]	+0.71 [0.25, 1.18]	higher error	[-0.11, 1.43] / [-0.08, 1.49], undetermined	depends on split independence	
R0 - P0(Rescale catboost_lo, λ 0.25), stage T0	-0.10 [-0.44, 0.14]	CI [-0.48, 0.03]	equivalent	summary table common CI column none. same contrast fixed-composition pool curve rows(below) common-CI verdict undetermined	unknown(summary main4_noninf True. table column none)	m3_delta -0.08(m3_verdict4 undetermined). independent 5 regions: lower error 1, equivalent 1, undetermined 3, higher error 0, non-inferior 2. worst Russia_E x +0.51 [-0.587, 1.098]. 'equivalent 1'(Lena Delta) tables/lgw_scenarios.csv common-CI verdict undetermined(depends on split independence), 'lower error 1'(Canada) common-CI verdict lower error noninf False
same contrast, Alaska cell	+0.42 [-0.30, 1.92]	+2.23 [1.29, 3.13]	undetermined	[-0.72, 2.67] / [0.56, 3.82], undetermined		
R0 - P0, λ 0.25(fixed-composition pool curve)	-0.10 [-0.45, 0.13]	-0.22 [-0.48, 0.03]	equivalent	[-0.65, 0.29] / [-0.66, 0.20], undetermined	depends on split independence(two verdict columns compared)	role = 'descriptive(verdict not used)'
R2 - P0, λ 0.25(fixed-composition pool curve)	-0.08 [-0.26, 0.01]	-0.29 [-0.43, -0.14]	equivalent	[-0.37, 0.10] / [-0.53, -0.06], undetermined	depends on split independence(two verdict columns compared)	descriptive
R0 - F1k(residual structure vs physics-input structure)	-2.72 [-4.34, -1.73]	-2.95 [-3.93, -1.97]	lower error	[-5.37, -0.92] / [-4.59, -1.34], lower error		Holm 0.0023

h. Target curves at zero labels (LG main run; 1,000 resamples) (C1 README 2.6)

Target	Method	RMSE	RMSE P0	Δ cell-weighted [CI]	Δ block-equal [CI]	sig
Alaska	D0	35.33	14.54	+20.79 [14.14, 26.92]	+18.50 [16.09, 20.91]	worse
Alaska	D1	16.33	14.54	+1.79 [0.37, 2.57]	+1.04 [0.34, 1.85]	worse
Alaska	R0 λ 0.25	14.96	14.54	+0.42 [-0.31, 1.90]	+2.23 [1.30, 3.12]	ns
Alaska	R2 λ 0.25	14.13	14.54	-0.41 [-0.60, 0.10]	+0.19 [0.01, 0.40]	ns
Lena	D1	21.88	21.69	+0.18 [-0.21, 0.42]	-0.45 [-0.99, 0.01]	ns

Target	Method	RMSE	RMSE P0	Δ cell-weighted [CI]	Δ block-equal [CI]	sig
Canada	D1	27.94	28.16	-0.22 [-1.13, 0.40]	-1.53 [-2.52, -0.58]	ns
Russia_W	D1	41.16	42.98	-1.82 [-3.75, -0.27]	-1.45 [-3.09, 0.09]	ns
Russia_E	D1	30.43	29.72	+0.70 [0.17, 1.36]	+0.69 [-0.02, 1.43]	ns

i. Physics baselines at zero labels (C1 README 2.7)

Contrast	Δ cell-weighted [CI]	Δ block-equal [CI]	Verdict	Holm P	Common CI, verdict	Dependence
P* – P0(year-matched degree-days Stefan)	-1.00 [-1.35, -0.32]	-0.41 [-0.74, -0.08]	lower error	0.0056	[-1.61, 0.06] / [-0.98, 0.19], undetermined	depends on split independence
B:ens – P0(physics-model and product ensemble)	-2.73 [-3.33, -0.91]	+0.05 [-0.94, 1.01]	undetermined	1.0(abstract bundle)	[-4.07, 0.15] / [-1.64, 1.63], undetermined	
D0 – P*(direct ML catboost_lo vs year-matched degree-days Stefan)	+3.24 [1.86, 4.29]	+2.03 [0.90, 3.12]	higher error	0.0050(X9 bundle)	[0.99, 4.99] / [0.17, 3.88], higher error	

Supplementary Table S16. Deployment scenarios SC1w to SC3w and related wrap-up tests.

Registered in WRAPUP (316714c, 30 September 2026, 03:02:42), partly unblinded (earlier shrinkage curves were known). The verdict column translates the registered verdict string; values are copied from the stat column. Source: paper/claims/C1_label0_safety/tables/lgw_tests.csv (copy of data/processed/lgw/lgw_tests.csv) and docs/MANUSCRIPT_DRAFT_RESULTS_ABSTRACT_2026-09-30.md (Deployment scenarios). Method codes (internal identifiers allowed in the Supplementary Information): P0 source-coefficient Stefan model; P* year-matched Stefan model; P1 recalibrated Stefan model ($\kappa = 10$); P2 local least-squares Stefan model; P1* or P1@ed soil-property recalibrated Stefan model; R0 source anchor plus residual ML; R1 recalibrated anchor plus residual ML; R2 augmentation plus anchor plus residual; D0 direct ML; D1 physics pseudo-label augmentation; F1a, F1k, F1n physics-input ML; RM multiplicative residual; Re Stefan-CCI mean anchor plus residual; W selection rule (cross-validation within the target labels); S1 random cells, S2 block stratification, S3 covariate-cluster stratification, S4 covariate max-min distance, S5 source-extrapolation first, S6 sequential selection by prediction variance, S7 $\sqrt{\text{TDD}}$ first. Target suffixes: |*x* parent region excluded from the source, |*i* rest of the parent kept, |*r* within-region design. *n* = number of target labels; all = all labels of half A. Verdicts: lower error, higher error, equivalent (± 0.5 cm), undetermined; both 95% block-bootstrap CIs (cell-weighted and block-equal) must agree.

ID	Role	Question	Registered verdict	Values
SC1w	main	Few-label recipe R1 non-inferior to P0 (0.5 cm) at 3 and 10 labels in five regions	safety not confirmed (non-inferior cells 4/10); no higher-error cell	non-inferior: Lena Delta <i>n</i> 3, Canada <i>n</i> 3, W Russia <i>n</i> 3 and 10; not confirmed: Lena Delta <i>n</i> 10, Canada <i>n</i> 10, E Russia <i>n</i> 3 and 10, Alaska <i>n</i> 3 and 10
SC1w-d10	auxiliary (δ 1.0 cm)	Same with a 1.0 cm margin	safety not confirmed (non-inferior cells 5/10)	not confirmed: Canada <i>n</i> 10, E Russia <i>n</i> 3 and 10, Alaska <i>n</i> 3 and 10
SC1w-P	main	Recalibration alone (P1) non-inferior to P0	rejected	Canada <i>n</i> 3: $\Delta +0.97$ [$+0.26, +1.30$], Holm <i>P</i> 0.021; Canada <i>n</i> 10: $+2.26$ [$+0.71, +2.93$], Holm <i>P</i> 0.008
SC2w	main	Residual learning versus recalibrated Stefan at 40 labels (Lena Delta, Canada, Alaska)	sentence A: lower error in 2 of 3 regions	Lena Delta lower; Canada lower; Alaska undetermined
SC2w-n160	auxiliary	Same at 160 labels	sentence A: lower error in 2 of 3 regions	Lena Delta lower; Canada lower; Alaska undetermined
SC3w	main	Sequential stopping rule (τ 0.05) halves labels against always 40 with error increase ≤ 0.3 cm and no higher-error region	rule not operating (label ratio 0.91); Δ described only	label ratio 0.909; three-region mean $\Delta -0.048$ [$-0.079, -0.002$], block-equal -0.077 [$-0.105, -0.049$]; condition (i) not met, (ii) to (iv) met
SC3w, τ 0.025	descriptive	τ sensitivity	rule not operating (label ratio 0.99)	label ratio 0.988; $\Delta -0.001$ [$-0.006, +0.004$], block-equal -0.009 [$-0.014, -0.003$]
SC3w, τ 0.1	descriptive	τ sensitivity	rejected (condition (i) not met)	label ratio 0.542; $\Delta -0.177$ [$-0.291, -0.055$], block-equal -0.310 [$-0.394, -0.225$]
SC3w, τ 0.2	descriptive	τ sensitivity	all four conditions met (not the main verdict)	label ratio 0.185; $\Delta -0.677$ [$-0.909, -0.353$], block-equal -0.845 [$-1.041, -0.640$]
S-a	sensitivity (interpretation rule fixed before results)	Label unit: point labels versus 1 km location means, P1 – P0	with point labels the few- label recalibration gain was smaller (Canada, <i>n</i> 10, -0.86 cm); deployment label unit is the 1 km location mean	median Δ_{unit} : Lena Delta <i>n</i> 3 $+0.31$, <i>n</i> 10 $+0.34$; Canada <i>n</i> 3 -0.47 , <i>n</i> 10 -0.86 ; Alaska <i>n</i> 3 $+0.07$, <i>n</i> 10 $+0.09$
S-b	sensitivity	Label year: single-year versus multi-year mean labels (W and E Russia)	recalibration gain with 3 to 10 labels maintained	median Δ_{year} : W Russia <i>n</i> 3 $+0.02$, <i>n</i> 10 $+0.05$; E Russia <i>n</i> 3 $+0.01$, <i>n</i> 10 $+0.04$
L43	main	Block-dispersed versus cell-random placement for P1	placement effect not established	P1 <i>n</i> 10 undetermined (regions 4/4, Holm <i>P</i> 1); <i>n</i> 40 undetermined (regions 2/4, Holm <i>P</i> 1)
AK1w	main	Combinations that beat the physics model in Alaskan sub-regions with an Alaskan source also do so with an outside-only source	majority (<i>r</i> 0.57, 68 pairs, 5 targets)	<i>r</i> 0.574; <i>r'</i> 0.779; target-level majority 2/5; sensitivity <i>r</i> 0.619

Supplementary Table S17. Selection and withdrawal history and earlier negative results.

Part a lists claims, hypotheses and rules that were dropped, withdrawn, replaced or corrected, with the commit that recorded the action and its timing relative to the first retrieval of LG-family results (T_{res}, 30 September 2026, 07:20:22). Part b lists earlier negative results reclassified under the four-way verdict rule (WRAPUP 1.4 (b)); original verdicts are not changed. Its CI-type, reclassification and value columns count the stored rows of the reclassification table data/processed/lgw/lgw_rescore.csv (WRAPUP execution step C7, 1 October 2026) by CI type and by four-way verdict (tools/build_reclass_summary.py; no value recomputed); many rows are curve points of one experiment, so the counts describe the stored tables and are not tests. Audit entries: docs/AUDIT_2026-09-26.md. Source: docs/MANUSCRIPT_DRAFT_SUPPORT_2026-09-30.md M4.5 and M4.4 (draft of 30 September 2026).

a. Selection and withdrawal history

#	Item	Type	Origin (commit, KST)	Action and reason	Recorded in (commit, KST)	Timing relative to LG-family results
1	"Component decomposition: 72–95 % of the few-label gain in level regions is coefficient recalibration" (NOVELTY N3; FINAL plan §10.3)	dropped claim	FINAL plan and session records, 2026-09-26	Rejected as a contribution claim (2 reviewers reject, 1 conditional). Reasons: the residual CI includes 0 only in Russia W; CA-2 degrades under recalibration (+7.7 cm); the share keeps only targets where recalibration worked; the classification uses scored labels; at $\alpha = 1$ the residual does not respond to target labels. Remaining form: case values in cm, no percentage	NOVELTY §1, §5, §10 ('d8355d2' 2026-09-29 18:18:36)	before (no LG result)
2	"The ranking of ML and physics reverses with the validation scheme" (NOVELTY N8)	dropped claim	earlier internal answers	Rejected. The paired Alaskan block contrast failed the two-weighting rule (cell-weighted +1.66 [−1.32, 3.93], block-equal −1.87 [−3.15, −0.72] cm); direct ridge was below Stefan in every scheme; no CI; pattern already in Gautam 2025. Remaining form: methods justification. The registered test is L19, whose rule forbids the word "reversal" unless direct ML wins under random splits	NOVELTY §5 ('d8355d2' 2026-09-29 18:18:36); L19 ('8513cc2' 18:17:29)	before
3	NOVELTY N5 placement clause (dispersed versus concentrated labels)	dropped clause	NOVELTY draft	Deleted: no block-equal CI on mean rows; sign of the $n = 160$ value depends on the target set. A placement sentence is allowed only through L43	NOVELTY §5 ('d8355d2'; L43 ('316714c' 2026-09-30 03:02:42)	before
4	NOVELTY N2 (two baselines) as an independent contribution	demoted	NOVELTY draft	Moved to Methods; depends on N1 and has precedents	NOVELTY §5 ('d8355d2')	before
5	"Break-even n predicted by $\text{abs}(\log E \text{ ratio})$, $\rho = -0.84$ "; "H24 k-center adopted"; "Lena and Canada cannot exceed at $n \leq 320$ "; contest result "CCI combination cancels bias"; control ladder "all rungs significant"	dropped claims	H24, H25, H27, contest 2026-07	Discarded or withdrawn, no reuse	RESEARCH_FRAME A.4 ('6830277' 2026-09-29 13:18:02); AUDIT ('f098031' 2026-09-26 21:42:16)	before
6	SC1–SC3 (NEXT P3; RESEARCH_FRAME §5.1 rank 5, §5.2)	withdrawn hypotheses	NEXT 'f227fc0' 2026-09-28 17:40:56; RESEARCH_FRAME '6830277'	Withdrawn and replaced by SC1w–SC3w. Reasons: the level/structure classification became a post hoc variable; the 3-split h25b protocol was replaced by the LG grid. Never computed	WRAPUP 3.1 ('316714c' 2026-09-30 03:02:42); NEXT revision line ('ff1be02' 11:44:30)	before (withdrawal); the NEXT record line after T _{res}
7	AK1 (NEXT §8.3)	withdrawn hypothesis	NEXT '6830277' 2026-09-29 13:18:02	Withdrawn and replaced by AK1w. Reason: stage 2 re-scores the same Alaskan sub-regions with another source, so "transfers to new regions" was broader than the design. Never computed	WRAPUP 5 ('316714c'); NEXT revision line ('ff1be02')	before; NEXT line after T _{res}

#	Item	Type	Origin (commit, KST)	Action and reason	Recorded in (commit, KST)	Timing relative to LG-family results
8	SC1w decision rule "no inferior cell and ≥ 8 decided cells"	replaced rule	WRAPUP initial draft (uncommitted, file hash 5976615c...)	Replaced by cell-wise non-inferiority before commit; the draft rule counted undetermined cells toward support	WRAPUP revision 1 (2) ('316714c')	before
9	Three-way classification in WRAPUP 7.2 (a)3	withdrawn rule	WRAPUP initial draft	Withdrawn before commit; L38 follows the LG 6B.5 wording	WRAPUP revision 1 (3) ('316714c')	before
10	Abstract bundle selection: the review example (L1, L2, L4 n = 10, L8, P1 – P0 n = 10, L10, L15, L29, L30) became AB1–AB10	selection	WRAPUP_ REVIEW item 4 ('422c34d' 2026-09-30 01:45:19)	L30 removed (learner robustness is written as a qualifier); AB9 (second baseline) and AB10 (uncertainty) added. Selection made with knowledge of earlier experiments (M1, P2, W3, E1–E3, h25b, a2, b1, b2)	WRAPUP 1.1 (('316714c'))	before
11	LGX X7: existing product comparison and CALM trend reproduction	excluded analysis	LG 6A.10 (('8513cc2' 2026-09-29 18:17:29))	Excluded; product comparison later registered only as a conditional descriptive SI table (WRAPUP 10, executed only if the map is in the main text)	WRAPUP 10 (('316714c'))	before
12	LG 6A.10 exclusions of the TabPFN label grid and of independent-region expansion	exclusions replaced	LG 6A.10 (('8513cc2'))	Replaced by LGT (LG revision 9) and LGD (LG revision 10); the Mongolia and Central Asia exclusion is kept	'6799b74' 2026-09-29 22:43:12	before
13	WRAPUP 11.1: local re-run of running Rescale jobs; LG refit with site-level draws; time-axis validation, campaign hold-out, CALM trends; site-scale GIPL2 or CryoGrid runs; selection rules at $n \geq 80$ and a measurement-priority map; maps for several regions; Ran 2022 ESSD and 15 pending data sets; ERA5-Land 1990–2009 year matching; residual kriging baseline (conditional on LGU- C1); $n = 80$ grid point (conditional on L4 minimum $n = 160$); LGF lower-priority tiers; a new ML model of E; Mongolia in the confirmatory pool	excluded analyses	WRAPUP 11.1 (('316714c' 2026-09-30 03:02:42))	Not performed; reasons as in WRAPUP 11.1. The two conditional items (execution plan C12, C13) follow the registration format of LG revision 15 (n)	WRAPUP 11.1 (('316714c')); LG revision 15 (n) (('ff1be02'))	before
14	Exception to row 13, first item: remaining FT-Transformer shards of the stopped Qjpbep job	scoped exception	user directive 2026-09-30 08:45	Run locally as a cross-environment auxiliary only; the primary verdict closes on Rescale shards; auxiliary verdicts are not used in confirmatory sentences or the abstract	WRAPUP revision 2 (d); LG revision 15 (c)(d)(f) ('ff1be02' 11:44:30)	after T_res, bundles not unpacked
15	Covariate-space uniform selection rule (RESEARCH_ FRAME 5.2, FAILURE_ ANALYSIS 6), Gautam 2025 set-up reproduction, R0 covariate-support check rule	excluded analyses	RESEARCH_ FRAME 5.2 (('6830277'))	Not performed in this paper; LGX X3a covers random versus region hold-out; map panel (e) marks extrapolation	WRAPUP revision 2 (e) ('ff1be02' 11:44:30)	after T_res, not unpacked

#	Item	Type	Origin (commit, KST)	Action and reason	Recorded in (commit, KST)	Timing relative to LG-family results
16	NOVELTY 7.4 option 3 (exclusion of ALT values below 20 cm; 138 Alaskan and 89 Lena rows) and option 4 (source-target TDD range overlap table)	excluded analyses	NOVELTY 7.4 ('d8355d2' 2026-09-29 18:18:36)	Not registered by the user; sensitivity analyses unrelated to confirmatory verdicts	WRAPUP revision 2 (e); LG revision 15 (o) ('ff1be02')	after T_res; not unpacked
17	Region-equal-weight E0 sensitivity	not registered	WRAPUP REVIEW item 11 ('422c34d' 2026-09-30 01:45:19)	Not registered and not performed; only the E interpretation paragraph is written (M3)	execution plan decision 9; WRAPUP revision 2 (e); LG revision 15 (o) ('ff1be02')	after T_res; would have required registration before LG P0 retrieval
18	LGX-8.6 options: (a) Rescale continuation and (e) Rescale summary job	removed options	WRAPUP 8.6 ('316714c')	Removed by the user directive of 2026-09-30 08:45 (no further Rescale cost); option (b) local merge into the primary verdict was already withdrawn by LG revision 14 (4) after cross-environment check (i)	LG revision 15 (a) ('ff1be02'); revision 14 ('028900a' 04:19:28)	(a), (e) after T_res; (b) before
19	AB10 resampling count "2,000"	corrected	WRAPUP 1.1 ('316714c')	Corrected to 1,000 (LGU revision 2, p resolution 1e-3); the WRAPUP 1.1 body keeps the old text	LGU revision 2 ('582be9a' 03:36:00); WRAPUP revision 2 (b) ('ff1be02')	LGU change before; WRAPUP record after T_res
20	LGU candidate sentence "labels correct the interval location but do not reduce its width"	dropped sentence	LGU review document	Not used; width statements belong to LGX L25	LGU initial ('a112cef' 2026-09-29 23:30:02)	before
21	Label "revision after retrieval, bundles not unpacked and not viewed"	status label	WRAPUP 0.3 fourth item	Four amendments (LG revision 15, LGF revision 6, LGU revision 4, WRAPUP revision 2) were committed after T_res (07:20:22) in 'ff1be02' (11:44:30). WRAPUP 0.3 would label result-dependent rules "registered after viewing results". The amendments change only technical corrections, a map-mask interpretation, the status of auxiliary local runs and the licence-verified data rule; at commit time no bundle was unpacked and no result table was created or opened. They therefore carry the label above instead. The user has not yet confirmed this judgement (execution plan decision 11); if the user decides otherwise, the affected rows carry "registered after viewing results"	LG revision 15 (l); WRAPUP revision 2 (g) ('ff1be02')	after T_res; LGX bundles unpacked at 11:44:39
22	LG revision 15 (b) statement of the remaining nn FT-Transformer shards (16 main plus 35 others; Greenland has no unit)	corrected record	LG revision 15 ('ff1be02' 11:44:30)	After unpacking, the 51 shards were 16 main, AL-1 4, six targets $\times$ 5 = 30, and Greenland split 2 = 1; the Greenland sentence was wrong	LG revision 15 post-unpack note ('1a86df0' 2026-09-30 11:46:13)	after unpacking (file names and status fields only)

b. Past negative results, reclassified

Past ID	Source table	CI type	Original resamples	Reclassification	Value	Related registered test	Audit and withdrawal history
M1 H-series	'm1/m1_sc_tests.csv'	no CI (252 rows); two weightings (910 rows)	not recorded	reclassified (unblinded), four-way rule	lower 60, higher 189, equivalent 24, undetermined 637, no CI 252	H13 to L8 and AB8; H3 to L15 and AB3; H15, H17 to LGU-B1	AUDIT stats:F4 ('f098031' 2026-09-26 21:42:16): the nested recipe rests on the pre-registered H13 (Holm P 0.004 and 0.008); its 'all target regions' wording does not hold in Canada (-0.01) and E Russia ($+0.33$); M1 Holm family $m = 2$ (H13 rows)
FP-F4	'h4/b1_summary.csv' (d_phys columns), 'b1_protocol.csv'	two weightings (378 rows)	1,000	reclassified (unblinded), four-way rule; summary tables: descriptive	lower 109, higher 84, equivalent 28, undetermined 157; b1_protocol.csv: number of targets with higher error and oracle-branch gap	SC1w, SC3w (b1 "always shrink plus residual")	FINAL verdict rejected: the 3-label two-stage protocol was not better than always shrink plus residual ('f098031' 2026-09-26 21:42:16)
FP-F5, FP-F6	'h4/b2_curve.csv' (d_phys, d_shrink columns), 'b2_regional.csv' (cell-weighted only), 'b2_f5.csv', 'b2_theory_corr.csv'	mixed	1,000	reclassified (unblinded), four-way rule; cell-weighted rows: four-way verdict not possible (CI protocol differs); summary tables: descriptive	lower 533, higher 2,424, equivalent 256, undetermined 3,069, no CI 684, one weighting only 1,000; b2_f5.csv: F5 summary (wide table); b2_theory_corr.csv: Spearman correlation	AB4, L28 (κ), P3	FINAL verdicts rejected: adaptive pooling did not beat $\kappa = 10$ (F5); theoretical label count uncorrelated (F6)
FP-F7	'h4/b3_summary.csv' (blk_d_phys, blk_d_rand columns), 'b3_region.csv', 'b3_f7.csv'	no CI (198 rows); two weightings (3,834 rows)	1,000	reclassified (unblinded), four-way rule; summary tables: descriptive	lower 666, higher 639, equivalent 371, undetermined 2,158, no CI 198; b3_f7.csv: variance ratio of the E estimate	none (selection rules at $n \geq 80$ excluded, LG 6A.10)	FINAL verdict rejected: no gain from D-optimal selection
FP-F8	'h4/b4_tests.csv'	no CI (33 rows); two weightings (473 rows)	not recorded	reclassified (unblinded), four-way rule	lower 56, higher 65, equivalent 14, undetermined 338, no CI 33	L32, L33 (TabPFN)	FINAL verdict supported; "TabPFN on par" wording to be corrected (J10)
FP-F9	'h4/c1_tests.csv'	no CI (27 rows); two weightings (45 rows)	1,000	reclassified (unblinded), four-way rule	lower 1, equivalent 1, undetermined 43, no CI 27	L29, AB2 (physics and product baselines)	FINAL verdict rejected: no gain from multi-layer CCI combination
H18-H22	'h2/h_tests_all.csv' (block-equal CIs present, 1,055 rows); predictions 'm1/h1819_shard*_preds.npz', 'h2/h21_preds.npz' (includes H20), 'h2/h22_preds.npz'	two weightings	not recorded	reclassified (unblinded), four-way rule; stored predictions re-scored (unblinded)	stored CIs: lower 47, higher 160, equivalent 140, undetermined 708, no CI 144; re-scored (9 tests, 68 rows): lower 1, higher 12, equivalent 6, undetermined 41, fewer than 8 blocks 8	none	WRAPUP 1.4 (b) body text corrected by revision 2 (c)

Past ID	Source table	CI type	Original resamples	Reclassification	Value	Related registered test	Audit and withdrawal history
H23, H24	‘h2/h23_tests.csv’, ‘h2/h24_tests.csv’	row resampling	not recorded	protocol differs, descriptive	row-resampling CIs, not reclassified (771 rows)	L43 (placement)	H24 verdict "A (adopted)" withdrawn (AUDIT action 4, ‘f098031’ 2026-09-26 21:42:16); "H24 k-center adopted" discarded (RESEARCH_FRAME A.4, ‘6830277’ 2026-09-29 13:18:02)
H25	‘h3/h25b_curve.csv’, ‘h3/h25x_curve.csv’ (two weightings); ‘h3/h25_curve.csv’ (row CI); break-even tables (descriptive)	mixed	1,000 (h25b, h25x); not recorded (h25)	reclassified (unblinded), four-way rule; row-resampling rows: protocol differs, descriptive; summary tables: descriptive	lower 231, higher 442, equivalent 63, undetermined 1,056, descriptive 2,178; h25b_breakeven.csv: break-even n; h25x_breakeven.csv: break-even n	L4, AK1w	"break-even n set by $\text{abs}(\log E \text{ ratio})$, $\rho - 0.84$ " redefined (AUDIT action 1, ‘f098031’) and discarded (RESEARCH_FRAME A.4, ‘6830277’)
H26	registered paired test not produced	none	none	source missing	none	none	rejected without the registered paired CI; criterion unattainable at registration (AUDIT workflow:F5, ‘f098031’)
H27	‘h3/h27*_rule.csv’ (4 files)	rule tables	not recorded	protocol differs, descriptive	h27_rule.csv: LOO R ² and Spearman; h27b_rule.csv: Spearman, Kendall and Mann–Whitney; h27c_rule.csv: Spearman, Kendall and Mann–Whitney (audit-corrected edition); h27x_rule.csv: Spearman, Kendall and Mann–Whitney	none	redefined with H25 (AUDIT action 1)
H28	‘h3/h28_tests.csv’	two weightings (42 rows)	not recorded	reclassified (unblinded), four-way rule	lower 7, undetermined 35	L3 (nested α selection)	H28 = rejected (AUDIT; RESEARCH_FRAME A.4)
H29	‘h3/h29_summary.csv’, ‘h3/h29b_summary.csv’	summary	not recorded	protocol differs, descriptive	h29_summary.csv: share of harmful blocks and correlation; h29b_summary.csv: share of harmful blocks and correlation (unique blocks, audit edition)	L43 (h29b harmful single-block share)	block-CI re-run h29b (‘f098031’ 2026-09-26 21:42:16)
H30	‘h3/h30_deploy_tests.csv’	no CI (14 rows); two weightings (35 rows)	not recorded	reclassified (unblinded), four-way rule	lower 2, higher 2, equivalent 2, undetermined 29, no CI 14	SC1w to SC3w (deployment scenarios)	Holm column added to the H30 family; aoa_resid stays significant (0.001 to 0.007) (AUDIT stats:F4, ‘f098031’ 2026-09-26 21:42:16)

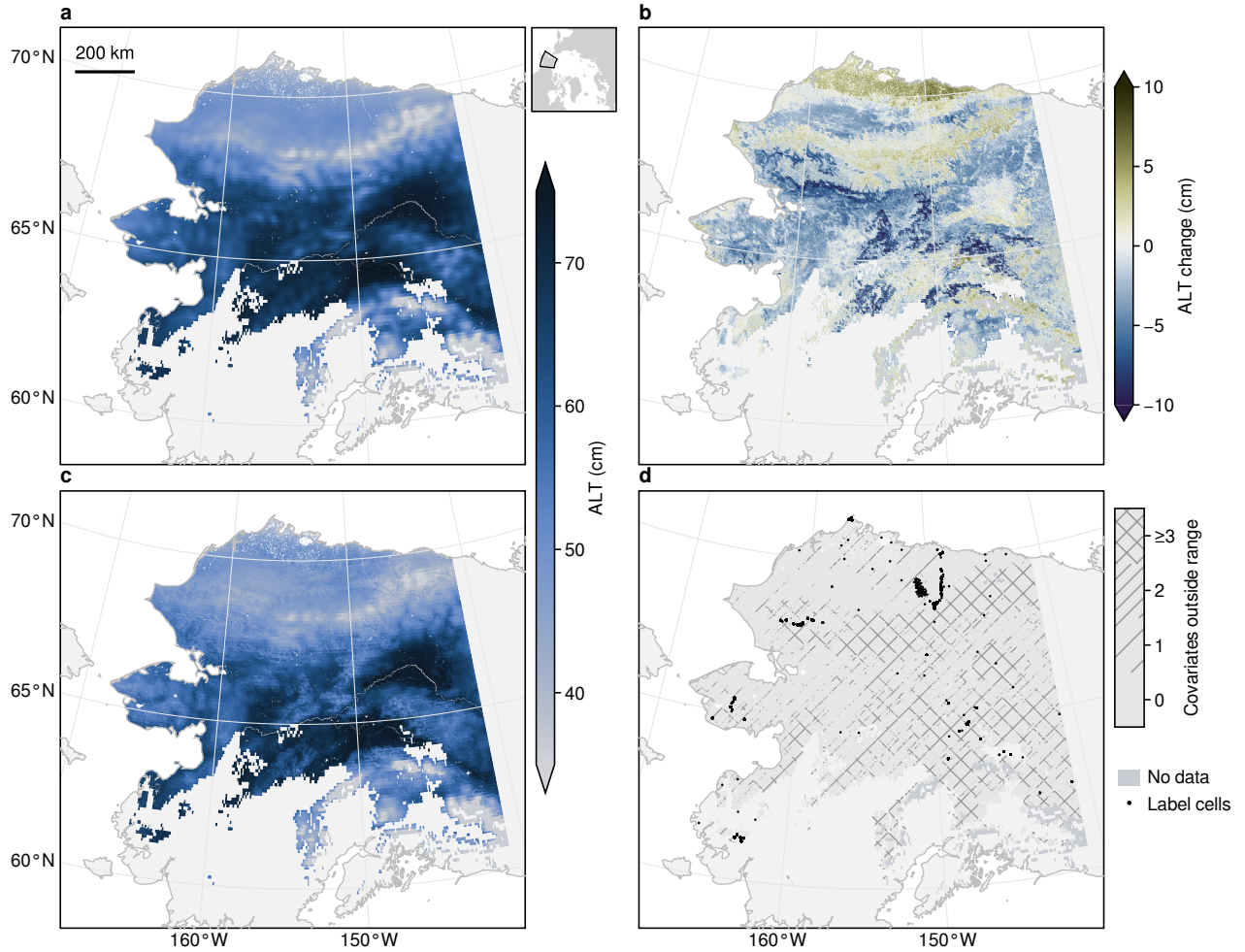
Supplementary Table S18. Survey of public data for independent regions (XF).

Public field ALT data sets surveyed for independent regions (survey of 4 October 2026, docs/research/2026-10-04/open_regions.md) and the outcome of the eligibility rules of Supplementary Methods 2 applied blind in XF. Rows are filled after the XF run (placement fixed before results: Supplementary Information). Pending: registered data deadline 11 October 2026 (XF).

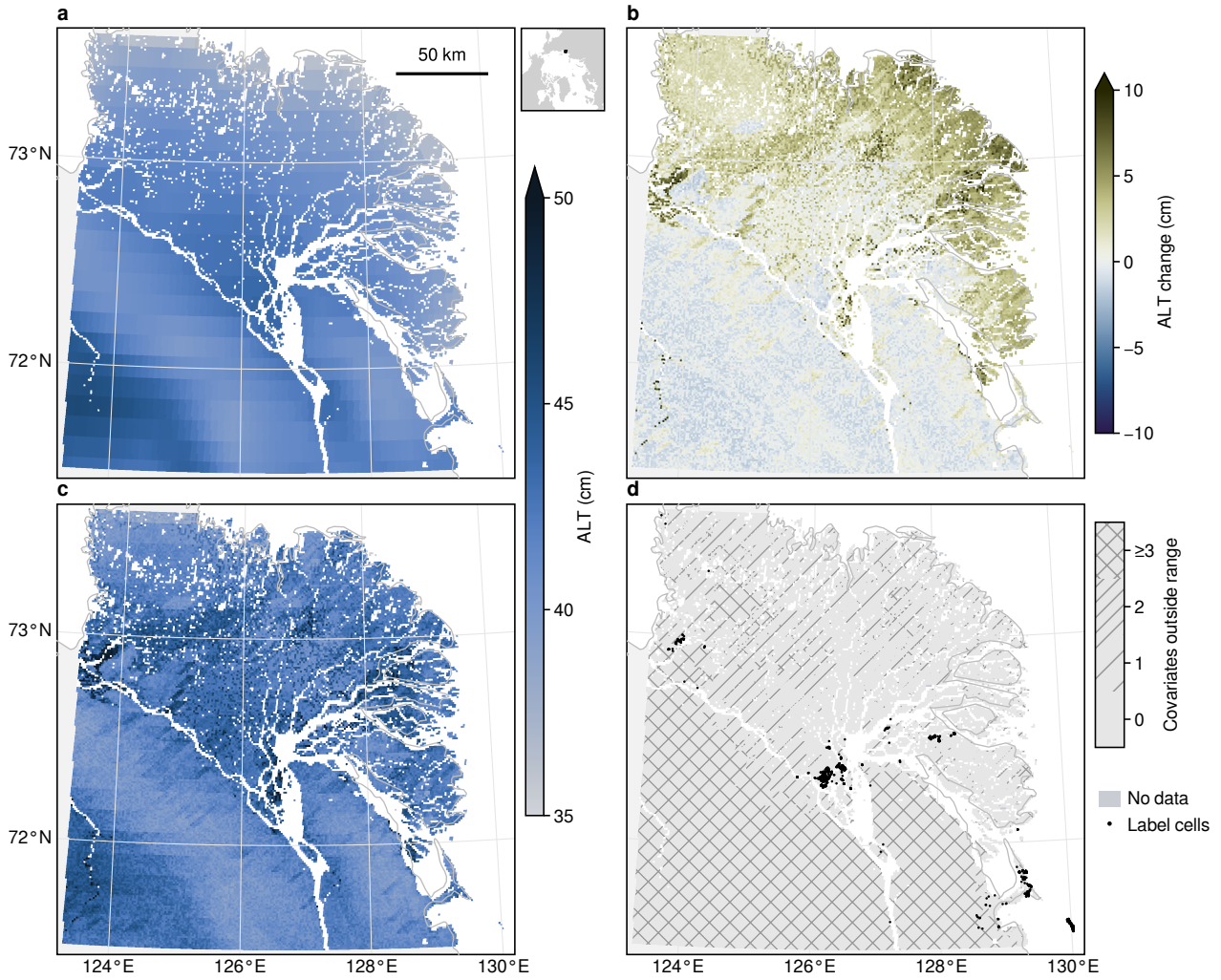
Data set	Identifier	Licence	Method and period	Size and relation to existing regions	Survey decision	XF eligibility
Pending: registered data deadline 11 October 2026 (XF; survey rows of docs/research/2026-10-04/open_regions.md 4.1–4.4 with their XF eligibility).						

Supplementary Figures

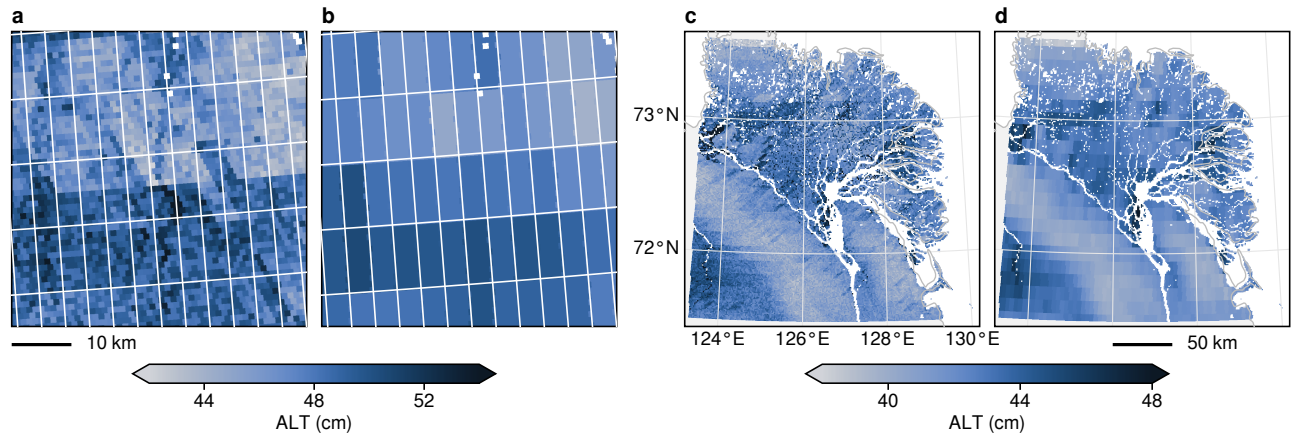
Supplementary Figs S1 to S6 are the 1 km maps and the grid-size figure of the added registrations (XK, XL). Supplementary Fig. S7 shows the scoring-scheme comparison of Supplementary Note 2, and Supplementary Figs S8 to S14 show result maps and panels moved out of the main figures.



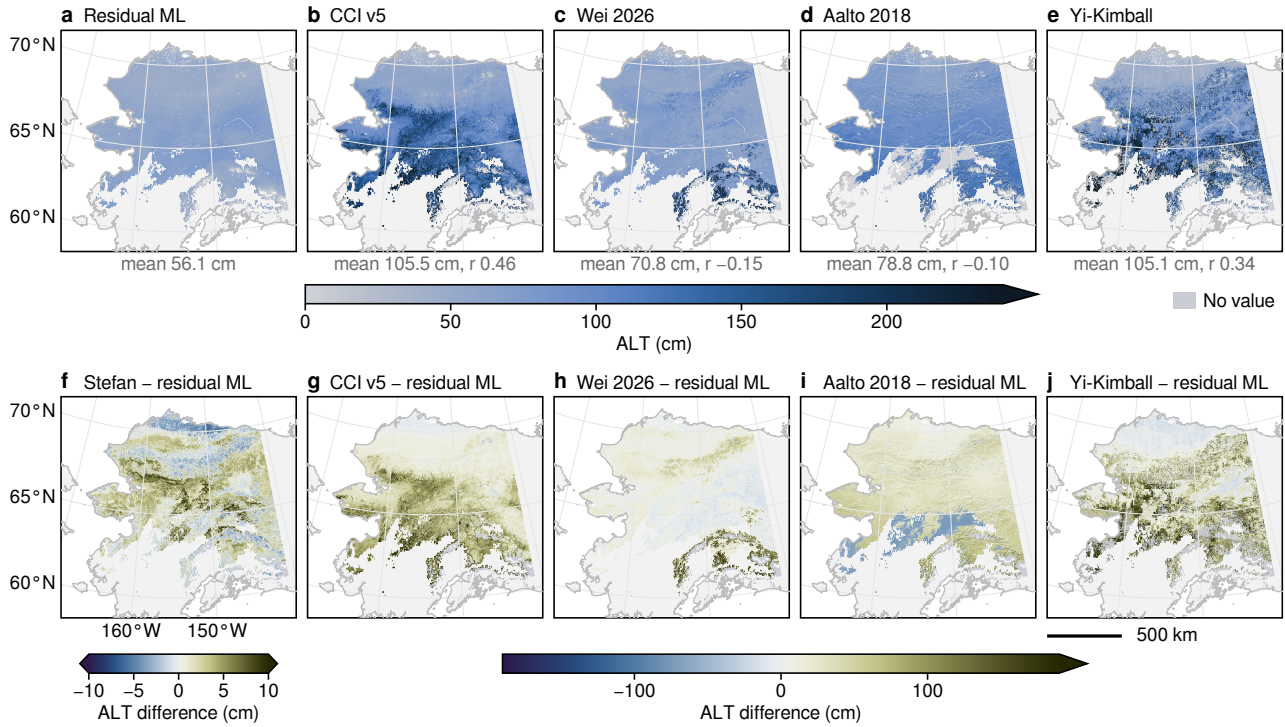
Supplementary Fig. S1 Active-layer thickness (ALT) of the Alaskan permafrost zone at 1 km display resolution: recalibrated Stefan model, residual machine-learning correction and final map. a, Recalibrated Stefan ALT, $E \cdot \sqrt{TDD}$ with $E = 1.617$ (least squares on all 13,606 label cells, shrunk towards the source coefficient). b, Residual ML correction $\lambda \cdot g$ (final minus recalibrated Stefan; positive, deeper), where g is a CatBoost residual model on 25 covariates (mean of two seeds) and $\lambda = 0.5$ was selected from 0.25, 0.5, 1.0 by 0.5° block five-fold cross-validation. c, Final ALT, a plus b. a and c share one colour scale (35–75 cm, 1st–99th percentile of both maps). d, Number of covariates outside the 0.5–99.5 percentile range of the training labels (hatching density), with label cells as dots; 73% of mapped cells have at least one, mostly terrain roughness, slope and topographic position. The 90% prediction interval of c is ± 20.2 cm everywhere (constant-width split conformal on block cross-validation residuals; cross-validated RMSE 13.8 cm, recalibrated Stefan 14.3 cm; lower bound truncated at 0 cm in 5,384 cells). Its nominal coverage is an average over cells similar to the labels and is not guaranteed in hatched cells, so the width is not mapped. The 1 km grid is a display resolution: air thawing index and climate covariates come from ERA5-Land (0.1° , about $4\text{--}6 \times 11$ km here), soil covariates from SoilGrids windows at about 5 km, and the residual correction acts mainly between climate-grid cells (main text, Fig. 6c). Cells are drawn as their native 0.009° rectangles. Domain: land (Natural Earth 10 m) with ESA CCI permafrost fraction of at least 50% (1997–2021 mean), $59\text{--}71.5^\circ$ N, $168\text{--}141^\circ$ W; 858,517 of 899,633 cells mapped. White, sea or water (Global Surface Water occurrence of at least 50%); grey, missing CCI ALT or soil thaw index. NAD83 Alaska Albers (EPSG:3338); inset, location. Data: ABoVE ALT v2 (Moore et al. 2025, ORNL DAAC) and GTN-P/CALM (Streletskiy et al. 2025) labels; ERA5-Land (Copernicus C3S); SoilGrids 2.0 (ISRIC); Copernicus DEM GLO-30 (ESA, AWS Open Data); ESA CCI Permafrost v4.0; JRC Global Surface Water; Natural Earth; Cartopy 0.23.



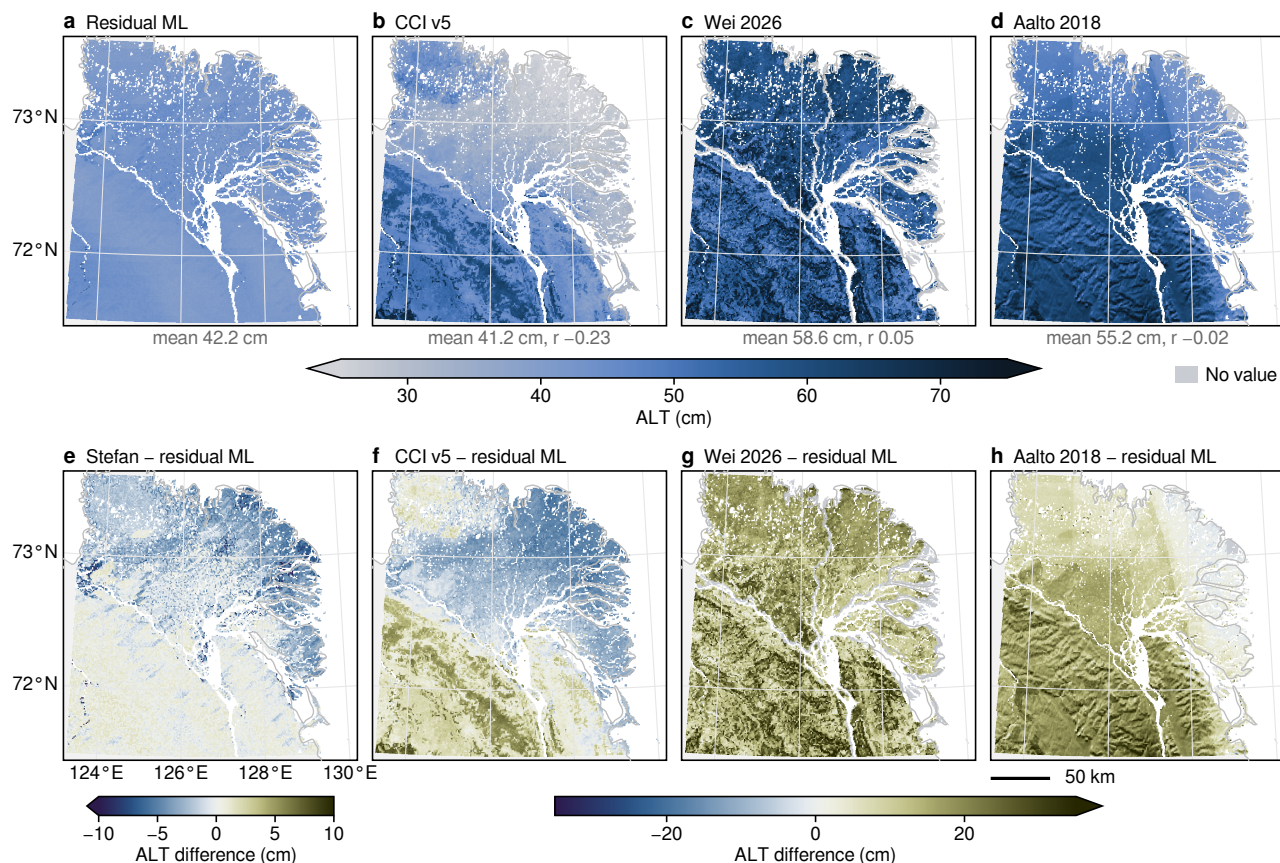
Supplementary Fig. S2 Active-layer thickness (ALT) of the Lena River Delta at 1 km display resolution: recalibrated Stefan model, residual machine-learning correction and final map. a, Recalibrated Stefan ALT, $E \cdot \sqrt{TDD}$ with $E = 1.435$ (least squares on all 3,037 Lena label cells, shrunk towards the source coefficient 1.619). b, Residual ML correction $0.25 \cdot g$ (final minus recalibrated Stefan; positive, deeper), where g is a CatBoost residual model (mean of two seeds) fitted to residuals of source-region cells beyond a 100 km buffer and of the Lena labels. c, Final ALT, a plus b. a and c share one colour scale (35–50 cm, 1st–99th percentile of both maps). d, Number of covariates outside the 0.5–99.5 percentile range of the training rows of c (source cells and Lena labels), with label cells as dots; 84% of mapped cells have at least one, mostly ERA5-Land coldest-month temperature, mean annual air temperature and freezing index. Relative to source cells alone, all have three or more. The 90% prediction interval of c is ± 25.4 cm everywhere (constant-width split conformal on 0.5° block five-fold cross-validation residuals within the Lena labels; cross-validated RMSE 20.8 cm, recalibrated Stefan 21.5 cm). Its nominal coverage is an average over cells similar to the labels and is not guaranteed in hatched cells, so the width is not mapped. The 1 km grid is a display resolution: air thawing index and climate covariates come from ERA5-Land (0.1° , about 3×11 km here), soil covariates from SoilGrids windows at about 5 km, and the residual correction acts mainly at the climate-grid scale (main text, Fig. 6c); the horizontal bands in a follow ERA5-Land rows. Cells are drawn as their native 0.009° rectangles. Grid: $71.5\text{--}73.6^\circ$ N, $123.3\text{--}130.1^\circ$ E; 38,086 mapped cells. White, sea or water (Global Surface Water occurrence of at least 50%); grey, missing CCI ALT. Polar stereographic projection (central meridian 126.7° E, true scale at 70° N); inset, location. Data: ALLena thaw-depth compilation (Veremeeva et al. 2025, PANGAEA); ERA5-Land (Copernicus C3S); SoilGrids 2.0 (ISRIC); Copernicus DEM GLO-30 (ESA, AWS Open Data); ESA CCI Permafrost v4.0; JRC Global Surface Water; Natural Earth 10 m; Cartopy 0.23.



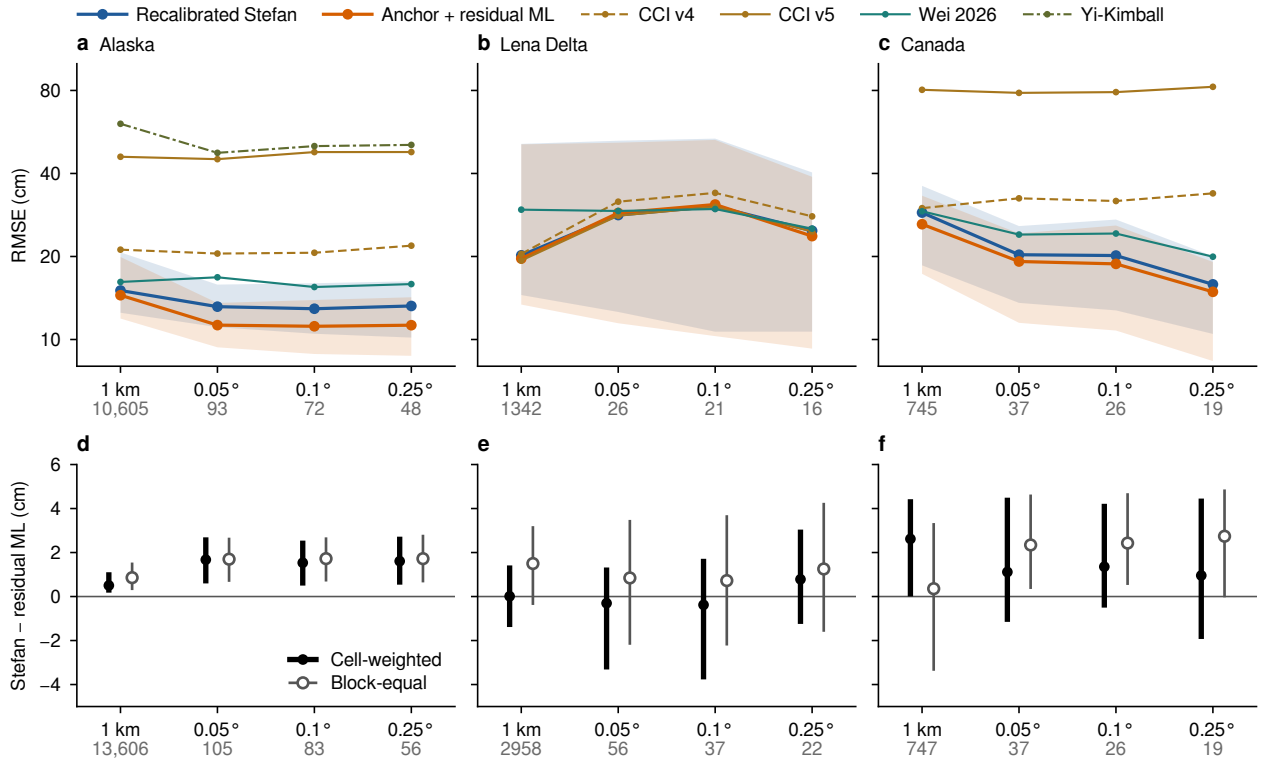
Supplementary Fig. S3 The same residual ML ALT map at its 1 km cells and as ERA5-Land 0.1° cell means. a, b, Alaska, 50 × 50 km around the 0.5° block with the most label cells (69.1° N, 148.8° W; 2,892 cells); white lines, ERA5-Land 0.1° cell boundaries. c, d, Lena Delta, full grid. a and c show the 1 km cells; b and d colour each cell with the mean of the mapped cells in its ERA5-Land 0.1° cell (boundaries at grid point $\pm 0.05^\circ$). All cells are drawn as their native 0.009° rectangles without resampling. One colour scale per region from the 1st–99th percentile of both panels (Alaska 42–54 cm, Lena Delta 38–48 cm), narrower than in Supplementary Figs S1 and S2 so that within-cell detail is visible. The 0.1° cell means explain 97% (Alaska, whole domain) and 61% (Lena Delta) of the variance of the 1 km map; the standard deviation of the 1 km values around their 0.1° cell mean is 1.8 cm in Alaska (total 9.5 cm) and 1.1 cm in the Lena Delta (total 1.8 cm). The 1 km map is a display resolution; its information resolution is bound to the climate input (ERA5-Land, about 10 km) and the soil input (SoilGrids, about 5 km). Within-cell detail comes from the terrain (Copernicus DEM 30 m windows) and CCI ALT inputs. Projections as in Supplementary Figs S1 and S2 (NAD83 Alaska Albers; polar stereographic, central meridian 126.7° E). Data as in Supplementary Figs S1 and S2; Cartopy 0.23.



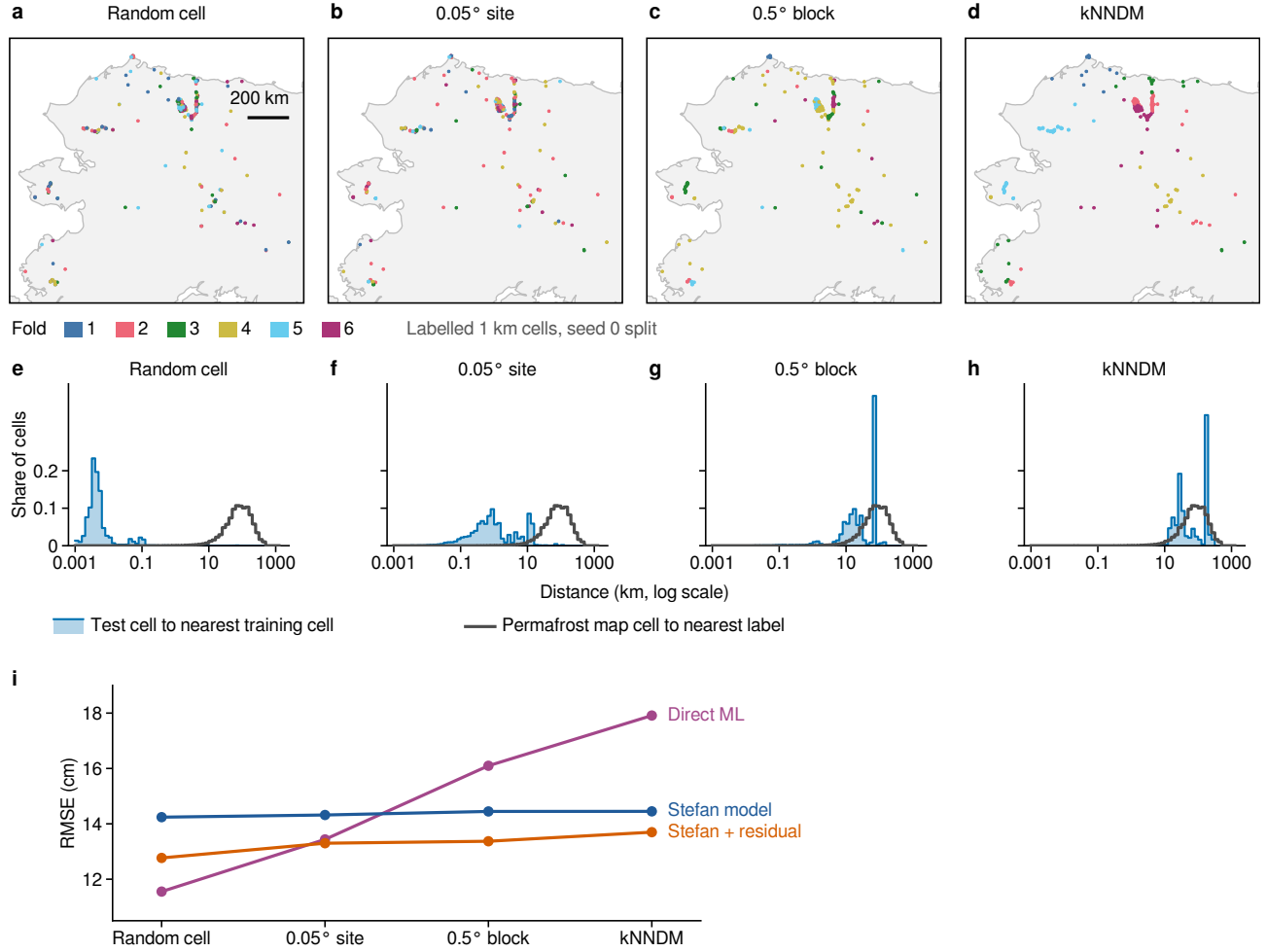
Supplementary Fig. S4 The 1 km Alaska ALT map compared with four ALT products and with the recalibrated Stefan model. a–e, ALT of the residual ML map (a, as in Supplementary Fig. S1) and of CCI v5 (1997–2023 mean, b), Wei 2026 (2000–2024 mean, c), Aalto 2018 (2000–2014 baseline, d) and Yi-Kimball (2001–2015 mean, e) on one colour scale (0–240 cm, 2nd–98th percentile of all five maps over mapped cells). Grey text, area mean over mapped cells with a product value and Pearson correlation r with the residual ML map. f, Recalibrated Stefan – residual ML on its own scale (± 10 cm, the scale of panel b of Supplementary Figs S1 and S2 with the sign reversed). g–j, Product – residual ML (positive, product deeper) on one scale (± 190 cm, 98th percentile of the absolute differences of g–j). Products are read at the 1 km cell centres (nearest pixel, period mean). Mean differences are +49.4 cm (CCI v5), +14.7 (Wei 2026), +22.6 (Aalto 2018), +49.3 (Yi-Kimball) and +1.4 cm (Stefan). Large differences with CCI v5 and Yi-Kimball are widespread in Interior Alaska; the largest differences with Wei 2026 lie at about 61–63° N, 141–150° W; Aalto 2018 is shallower than the map at about 62–64° N west of 148° W. These maps carry no accuracy statement; accuracy is compared only at label cells (Supplementary Table S2 and Supplementary Fig. S6). The 1 km map is a display resolution; its information resolution is bound to the climate input (ERA5-Land, about 10 km) and the soil input (SoilGrids, about 5 km). Grey cells, no product value (Yi-Kimball covers 86% of mapped cells). NAD83 Alaska Albers (EPSG:3338). Data: CCI v5 (ESA CCI Permafrost, CEDA, doi:10.5285/a6fbedd8ee5b472c8e84e55f746c1704); Wei et al. 2026 (Zenodo 21667583, CC BY 4.0); Aalto et al. 2018 (Zenodo 5003804, CC0); Yi and Kimball (ORNL DAAC 1760); covariates and labels as in Supplementary Fig. S1; Natural Earth 10 m; Cartopy 0.23.



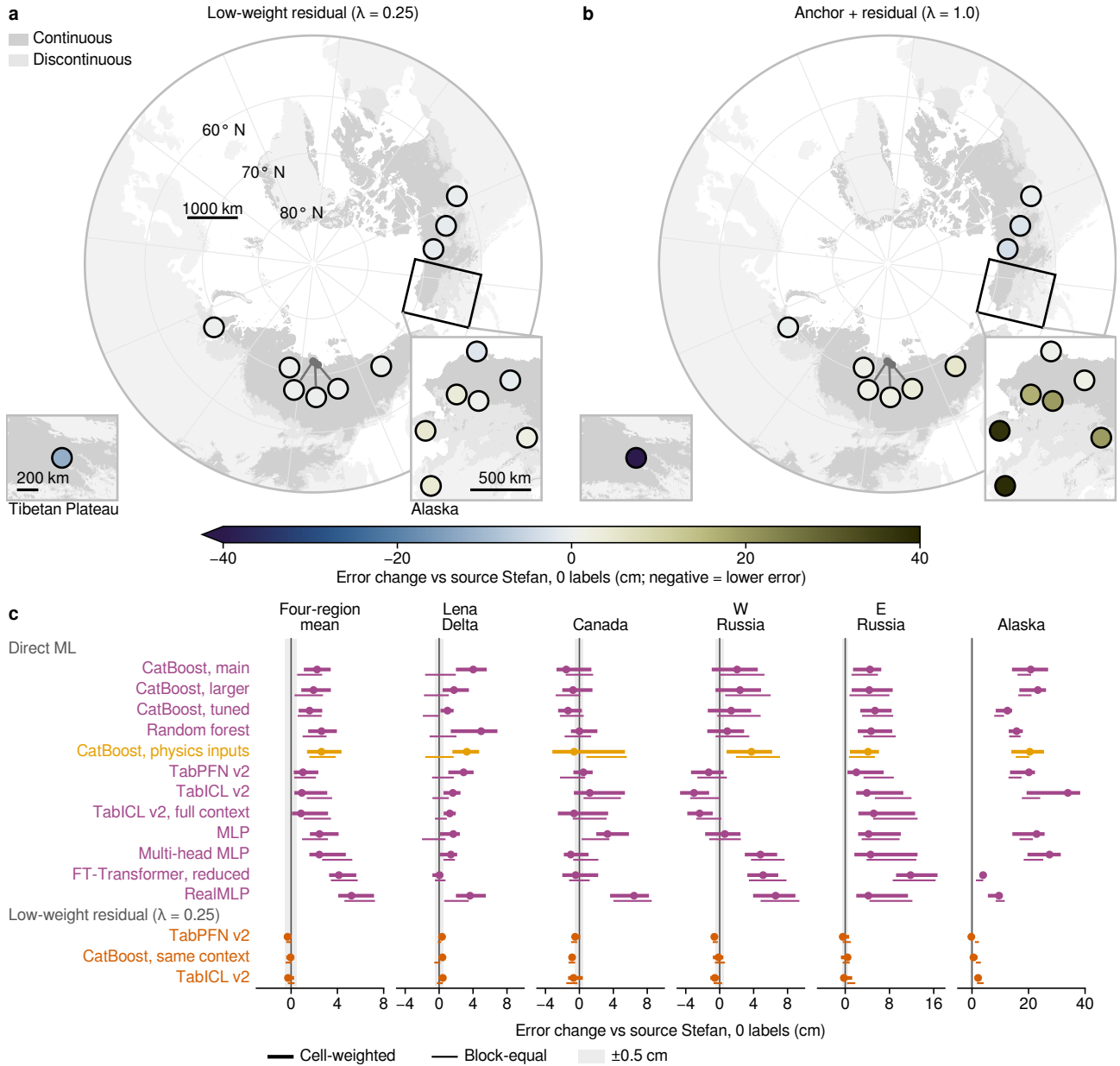
Supplementary Fig. S5 The 1 km Lena Delta ALT map compared with three ALT products and with the recalibrated Stefan model. a–d, ALT of the residual ML map (a, as in Supplementary Fig. S2) and of CCI v5 (1997–2023 mean, b), Wei 2026 (2000–2024 mean, c) and Aalto 2018 (2000–2014 baseline, d) on one colour scale (25–75 cm, 2nd–98th percentile of all four maps over mapped cells). Grey text, area mean over mapped cells with a product value and Pearson correlation r with the residual ML map. e, Recalibrated Stefan – residual ML on its own scale (± 10 cm, the scale of panel b of Supplementary Figs S1 and S2 with the sign reversed). f–h, Product – residual ML (positive, product deeper) on one scale (± 35 cm, 98th percentile of the absolute differences of f–h). Products are read at the 1 km cell centres (nearest pixel, period mean). Mean differences are -0.9 cm (CCI v5), $+16.5$ (Wei 2026, 94% of cells with values), $+13.0$ (Aalto 2018) and -1.1 cm (Stefan). CCI v5 is shallower than the map in the northern and eastern delta and deeper in the south-western part of the grid; Wei 2026 and Aalto 2018 are deeper over most of the grid. These maps carry no accuracy statement; accuracy is compared only at label cells. The 1 km map is a display resolution; its information resolution is bound to the climate input (ERA5-Land, about 10 km) and the soil input (SoilGrids, about 5 km). Grey cells, no product value. Polar stereographic projection (central meridian 126.7° E, true scale at 70° N). Data: CCI v5 (ESA CCI Permafrost, CEDA); Wei et al. 2026 (Zenodo 21667583, CC BY 4.0); Aalto et al. 2018 (Zenodo 5003804, CC0); covariates and labels as in Supplementary Fig. S2; Natural Earth 10 m; Cartopy 0.23.



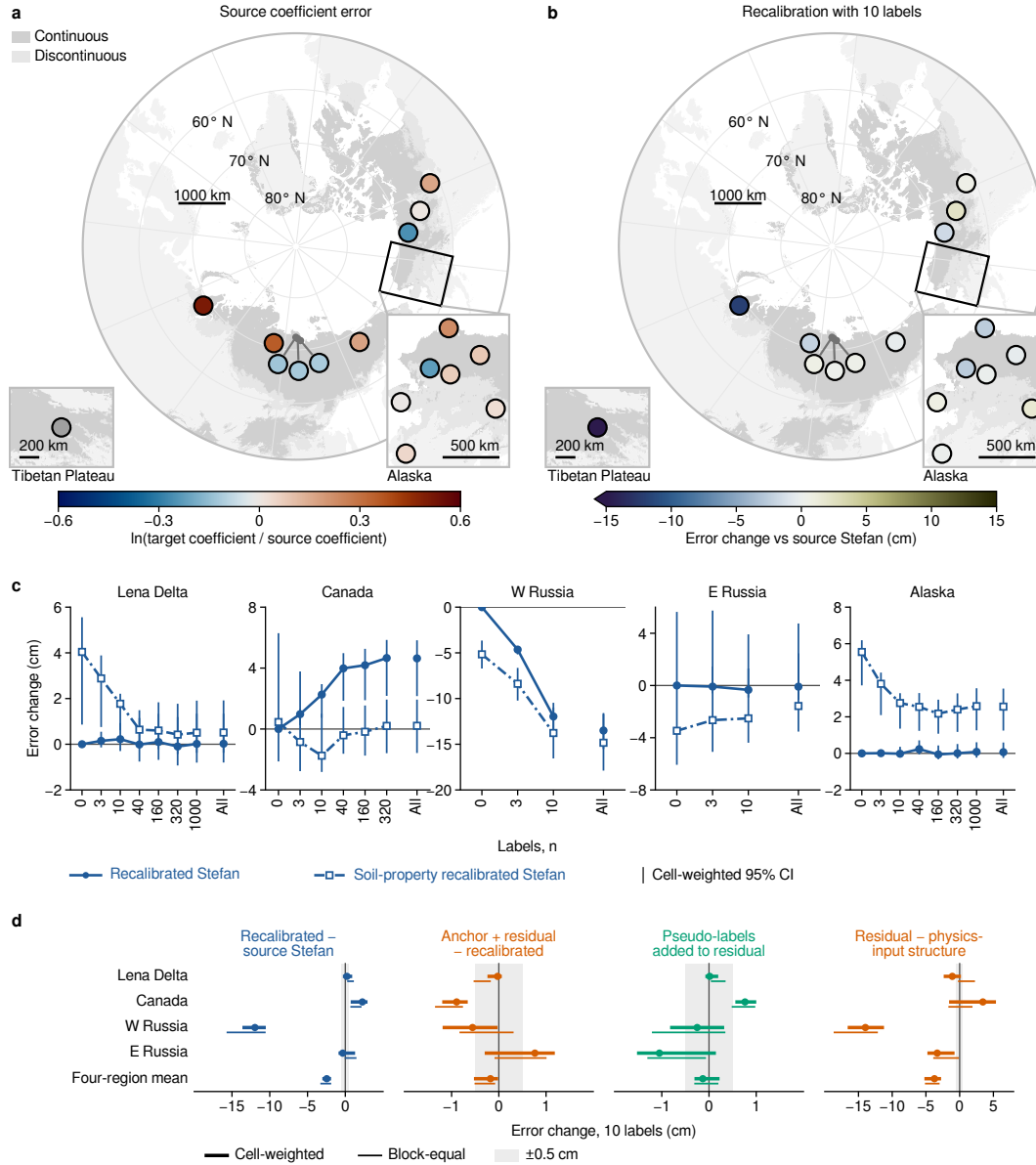
Supplementary Fig. S6 Prediction error of the recalibrated Stefan model, the residual ML model and four ALT products after aggregation to larger grid cells. Model values are within-region out-of-block predictions from 0.5° block five-fold cross-validation; the Stefan coefficient is refitted on the training folds, and λ of the residual model is selected by an inner block cross-validation within the training folds. Product values are period means at the label cells: CCI v4 (1997–2021), CCI v5 (1997–2023), Wei 2026 (2000–2024) and, for Alaska, Yi-Kimball (2001–2015). Labels and predictions are averaged in grid cells of 0.05°, 0.1° and 0.25° whose boundaries follow the ERA5-Land cells (0.1° = ERA5-Land cell); grid cells with fewer than three label cells are dropped; 1 km denotes single label cells. Grey numbers, grid cells per size. a–c, Cell-weighted RMSE (logarithmic axis) on the cells where all shown values exist (Alaska 10,605 cells in 60 blocks, Lena Delta 1,342 in 15, Canada 745 in 34); bands, 95% block bootstrap intervals for the two models (1,000 resamples of 0.5° blocks). d–f, RMSE of the recalibrated Stefan model minus RMSE of the residual ML model on all label cells with model values (Alaska 13,606, Lena Delta 2,958, Canada 747), cell-weighted (filled) and block-equal (open) with 95% block bootstrap intervals. In Alaska the cell-weighted difference is 0.50 cm (0.18 to 1.10) for single cells and 1.54 to 1.67 cm (lower bounds 0.50 to 0.60) at 0.05°, 0.1° and 0.25°. In Canada it is 2.62 cm (0.00 to 4.42) for single cells and 0.96 to 1.35 cm at larger sizes, with intervals that include zero. In the Lena Delta all intervals include zero. Removing cells within 5 km of Wei 2026 training sites changes the Alaska differences by at most 0.17 cm (data/processed/xbatch/XK_support_scale/). Products are compared only as RMSE differences at the same grid size.



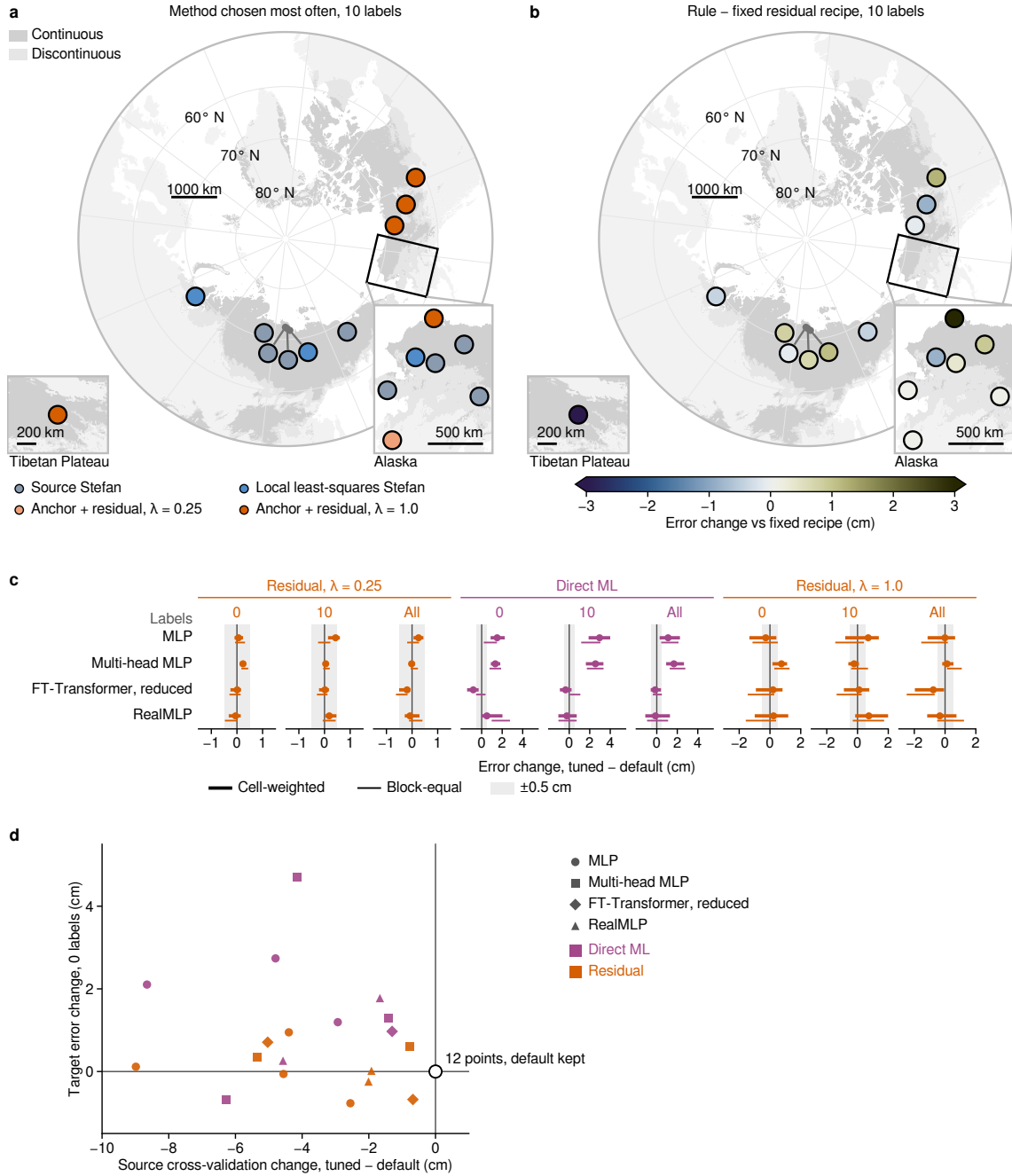
Supplementary Fig. S7 Four scoring schemes in Alaska (M1 six-fold comparison): split maps, distances and RMSE. This is the comparison cited in Supplementary Note 2, not the five-stage XH validation ladder of Fig. 1e. **a–d**, Fold assignment of the 13,606 labelled 1 km cells (six folds, seed 0) for random cells, 0.05° sites, 0.5° blocks and k -fold nearest-neighbour distance matching (kNNDM; 13 k -means clusters merged into six folds). **e–h**, Distances from test cells to the nearest training cell (blue) and from 20,000 sampled permafrost cells of the 0.02° prediction grid (mean annual air temperature below 0 °C) to the nearest label (grey), share of cells per logarithmic bin (seed 0); medians (mean of three seeds), 0.004, 1.11, 25.8 and 53.5 km test-to-training and 81.6 km map-to-label. **i**, RMSE (mean of three seeds) of direct ML (CatBoost), the Stefan model fitted to the training folds and the Stefan model plus a ridge residual (weight 0.75), rising from random cells to kNNDM from 11.6 to 17.9 cm, 14.2 to 14.5 cm and 12.77 to 13.70 cm. Alaska Albers equal-area projection; coastlines, Natural Earth 50 m.



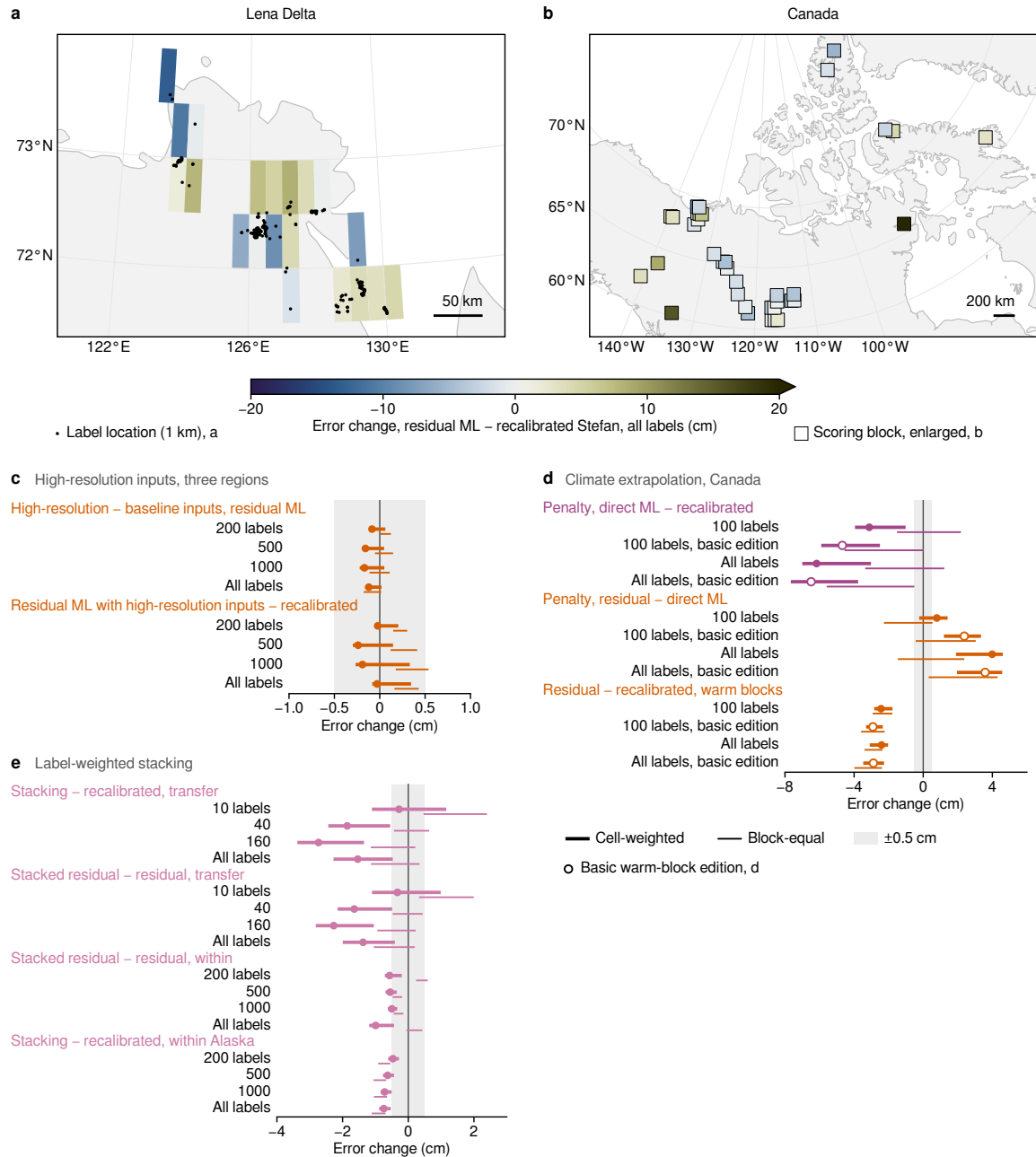
Supplementary Fig. S8 Transfer without target labels: residual models by target and learner detail. Error change relative to the source-coefficient Stefan model without target labels. **a, b**, Source anchor plus residual ML (CatBoost) with residual weight 0.25 (low-weight residual, **a**) and 1.0 (**b**) at each target's labelled-cell centre (17 targets; source excludes the parent region; post hoc re-analysis of the label-grid curves); below -40 cm, end colour (Tibetan Plateau in **b**, -47.34 cm); Alaska targets enlarged, Lena Delta targets offset with leader lines. **c**, Direct ML (ten learners, TabICL v2 with two context settings, CatBoost with physics inputs) and low-weight residual (three learners), four-region stratified mean and region rows, one axis per column; the main CatBoost uses the abstract-bundle contrast for the four regions and the label-grid curve for Alaska. Intervals, 95% block-bootstrap confidence intervals (CIs) weighted by cell (thick) and by 0.5° block (thin); band, ±0.5 cm. Polar stereographic projections (central meridian 127° E; Alaska enlargement, 157° W), scale bars true at 70° N; Tibetan Plateau inset, Lambert azimuthal equal-area. Permafrost zones, ESA CCI Permafrost fraction v4.0 (1997–2021 mean); coastlines, Natural Earth 50 m.



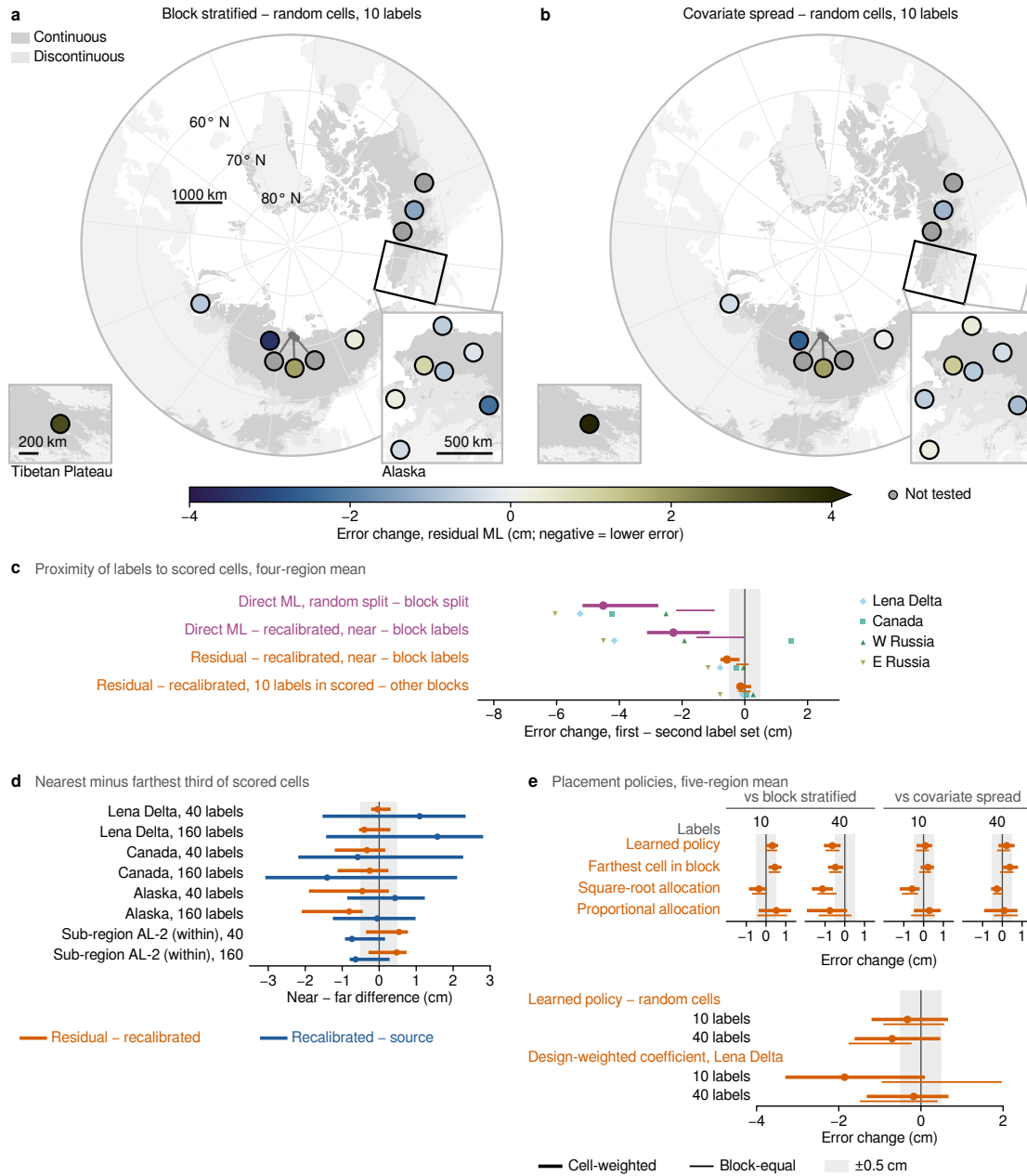
Supplementary Fig. S9 Coefficient recalibration: coefficient error and ten-label gain by target, soil-property Stefan curves and regional contrasts. **a**, Source-coefficient error $\ln(E_{\text{own}}/E_{\text{source}})$, E_{own} fitted by least squares to all labels of the target (label-grid target table; positive, target coefficient larger; Tibetan Plateau not in the table, grey). **b**, Error change of the Stefan model recalibrated with ten target labels relative to the source coefficient (post hoc re-analysis of the label-grid curves); below -15 cm, end colour (Tibetan Plateau, -96.37 cm). **c**, Recalibrated and soil-property recalibrated Stefan models minus the source-coefficient Stefan model by number of labels, each region a new target (cell-weighted 95% intervals). **d**, Four ten-label contrasts by region and four-region stratified mean (abstract contrast bundle): recalibrated minus source Stefan (-2.45 cm; 95% CI -3.04 to -1.88), anchor plus residual ML minus recalibrated Stefan (-0.18 ; -0.53 to -0.01), pseudo-label augmentation of the anchor plus residual (-0.13 ; -0.31 to 0.22) and residual minus physics-input structure (-3.74 ; -5.23 to -2.75); 95% block-bootstrap CIs weighted by cell (thick) and block (thin); band, ± 0.5 cm. Map conventions as in Supplementary Fig. S8.



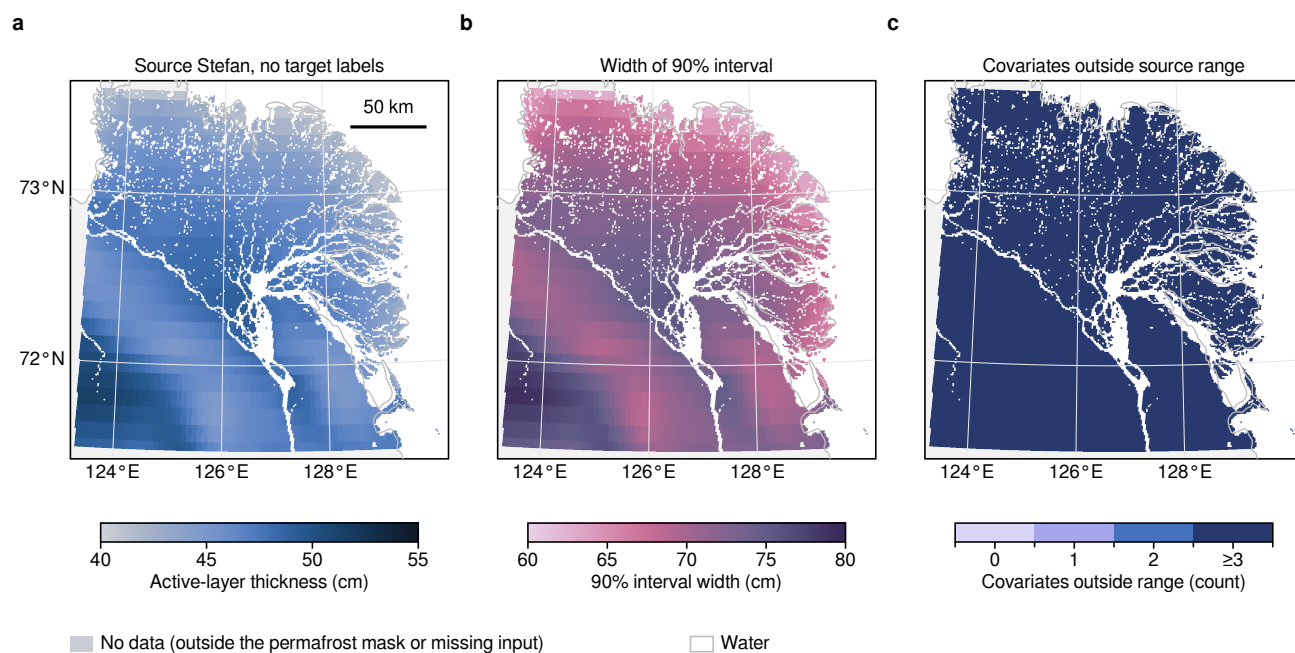
Supplementary Fig. S10 Method selection: choices of the cross-validation rule by target and source cross-validation tuning of neural networks. **a**, Method chosen most often by five-fold block cross-validation within ten target labels (25 selections per target and scoring mode, 5 splits $\times$ 5 draws; scoring modes pooled for sub-regions); the rule was designed after the label-grid results had been viewed. **b**, RMSE of the rule minus the fixed residual recipe (anchor plus residual ML, residual weight 0.25), ten labels, point estimates; beyond ± 3 cm, end colours (Tibetan Plateau, -29.53 cm). **c**, Neural networks configured by leave-one-source-region-out cross-validation minus their default configuration, four-region stratified mean, for residual ML (weights 0.25 and 1.0) and direct ML, without labels and with ten and all labels; one axis per column. **d**, Change in source cross-validation error against change in target error without labels (32 points: four networks, two model kinds, four regions; Spearman $\rho = -0.32$, descriptive); open circles, the 12 points where tuning kept the default. Intervals, 95% block-bootstrap CIs weighted by cell (thick) and block (thin); band, ± 0.5 cm. Map conventions as in Supplementary Fig. S8.



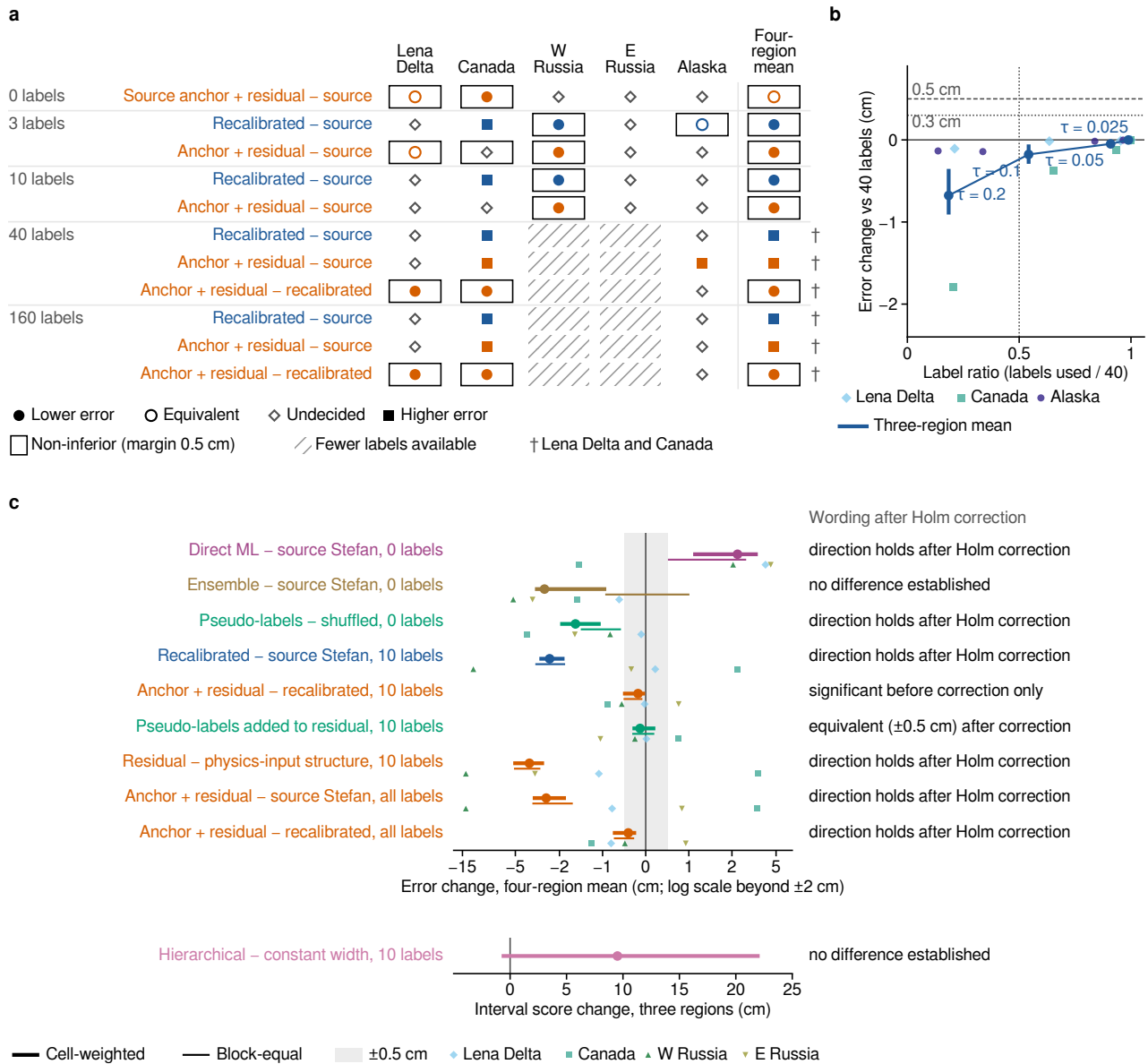
Supplementary Fig. S11 Gains within label-rich regions: error change by scoring block, high-resolution inputs, climate extrapolation and stacking. All analyses were designed after earlier results had been viewed. **a**, **b**, Residual ML (cross-validated residual weight, mean of two seeds) minus the Stefan model recalibrated on the same labels, by 0.5° scoring block, all labels within regions (half of the blocks scored per split; Lena Delta 20 blocks over 24 splits, Canada 36 over 25; split means +0.47 and +0.45 cm); in **b** blocks are enlarged squares at their centres, and one Canadian block (+23.77 cm) takes the end colour. **c**, Residual ML with ten added within-grid inputs minus residual ML with the baseline inputs and minus recalibrated Stefan, three-region stratified mean (500 and 1000 labels: Alaska and Lena Delta only; auxiliary contrast, partly unblinded). **d**, Spatial proxy for climate extrapolation in Canada (warm blocks scored; not a true extrapolation test): penalty of direct ML relative to recalibrated Stefan, of residual ML relative to direct ML, and residual ML minus recalibrated Stefan in warm blocks; filled, main edition (extrapolation width 0.97 in $\sqrt{\text{TDD}}$ units); open, basic edition (width –0.002). **e**, Label-weighted stacking of physics models and products minus recalibrated Stefan, and stacked residual minus residual, transfer pool and within regions (partly unblinded). Intervals, 95% block-bootstrap CIs weighted by cell (thick) and block (thin); band, ± 0.5 cm. Polar stereographic projections; coastlines, Natural Earth 50 m.



Supplementary Fig. S12 Placement of new labels: strategy contrasts by target, label proximity and learned placement policies. **a, b**, Error change of residual ML (anchor plus residual, weight 0.25) with ten transfer labels placed by block stratification (**a**) or covariate maximin-distance spreading (**b**) instead of random cells (13 targets with a placement test; grey, not tested); above 4 cm, end colour (Tibetan Plateau in **b**, +12.71 cm). **c**, Proximity contrasts, four-region stratified mean (symbols, regions): direct ML on a random minus a block split; error relative to recalibrated Stefan with labels near the scored cells minus labels in held-out blocks, for direct and residual ML; residual ML with ten labels inside minus outside the scored blocks. **d**, Nearest minus farthest third of scored cells, for residual ML minus recalibrated Stefan and recalibrated minus source Stefan (cell-weighted 95% intervals). **e**, Learned placement policy and three variants minus block stratification and minus covariate spreading (five-region mean, 10 labels; Lena Delta and Canada, 40 labels), learned policy minus random cells, and design-weighted coefficient minus learned policy in the Lena Delta (two-stage 95% intervals; XD, exploratory). Panels **a, b** and **e** were designed after earlier results had been viewed. Intervals in **c** and **e**, 95% CIs weighted by cell (thick) and block (thin); band, ± 0.5 cm. Map conventions as in Supplementary Fig. S8.



Supplementary Fig. S13 Prediction intervals for the Lena Delta without target labels. **a**, ALT predicted by the source-coefficient Stefan model (source regions exclude the Lena Delta and a 100 km buffer; ERA5-Land 2015–2020 climatology). **b**, Width of the 90% prediction interval of **a**: hierarchical conformal quantile of leave-one-region-out log errors ($q = 0.71$; calibration regions Alaska, Canada, E Russia and W Russia), so the width is proportional to the prediction; with four calibration regions the interval carries no finite-sample guarantee. **c**, Number of covariates outside the 0.5–99.5 percentile range of the source rows, missing values included; all mapped cells have three or more (most often mean annual air temperature, freezing degree-days and coldest-month temperature); the count is not an error-risk indicator. 38,086 of 53,011 grid cells shown; 87 of 38,173 land cells (0.2%) lack inputs. 1 km display grid; north polar stereographic projection centred at 126.7° E; coastlines, Natural Earth 10 m.



Supplementary Fig. S14 Deployment scenarios: scenario matrix, sequential stopping rule and abstract contrast bundle. **a**, Four-way verdict by stage (labels in the new region) and contrast for five regions and the four-region stratified mean (95% block-bootstrap intervals weighted by cell and by block; equivalence margin ± 0.5 cm); frames, non-inferior (margin 0.5 cm); hatching, more labels than the region provides; dagger, Lena Delta and Canada pool; source and recalibrated, source-coefficient and recalibrated Stefan models. **b**, Sequential stopping rule for recalibration (3, 10, then 40 labels; stop when the jackknife standard error of the log coefficient is at most τ): RMSE change relative to always using 40 labels against the mean share of labels used, three-region mean (Lena Delta, Canada, Alaska; cell-weighted 95% intervals) and regional values; with $\tau = 0.05$ the rule used 0.91 of the labels and changed RMSE by -0.05 cm (95% CI -0.08 to -0.002). **c**, Abstract contrast bundle in the order of Supplementary Table S14: four-region stratified mean (95% block-bootstrap intervals by cell, thick, and block, thin), regional values (symbols) and wording after Holm correction over ten contrasts; ensemble, mean of scale-calibrated Stefan, Kudryavtsev and ESA CCI anchors; last row, interval scores in the Lena Delta, Canada and Alaska ($+9.51$ cm; 95% CI -0.75 to 22.11). Horizontal axis linear within ± 2 cm, logarithmic beyond.

Supplementary References

- [1] Streletskiy, D. A. *et al.* GTN-P CALM: 35 years of Active Layer Thickness (ALT) across latitudinal and elevational gradients in the Northern Hemisphere. PANGAEA (2025). URL <https://doi.org/10.1594/PANGAEA.972777>.
- [2] Veremeeva, A. *et al.* ALLena: Thaw depth measurements of the active layer in the Lena River Delta region from 1998 to 2022, Northeastern Siberia. PANGAEA (2025). URL <https://doi.org/10.1594/PANGAEA.973813>.
- [3] Moore, M. A. *et al.* ABoVE: Soil moisture and active layer thickness in Alaska, USA and Canada, 2005–2024, version 2. ORNL Distributed Active Archive Center (2025). URL <https://doi.org/10.3334/ORNLDAAC/2369>.
- [4] Kudryavtsev, V. A., Garagula, L. S., Kondrat'yeva, K. A. & Melamed, V. G. *Osnovy Merzlotnogo Prognoza [Fundamentals of Frost Forecasting in Geological Engineering Investigations]* (Moscow State University, 1974). In Russian. English translation: CRREL Draft Translation 606, US Army Cold Regions Research and Engineering Laboratory, Hanover, NH, 1977.
- [5] Gneiting, T. & Raftery, A. E. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* **102**, 359–378 (2007).
- [6] Dunn, R., Wasserman, L. & Ramdas, A. Distribution-free prediction sets for two-layer hierarchical models. *Journal of the American Statistical Association* **118**, 2491–2502 (2023).
- [7] Hollmann, N. *et al.* Accurate predictions on small data with a tabular foundation model. *Nature* **637**, 319–326 (2025).
- [8] Qu, J., Holzmüller, D., Varoquaux, G. & Le Morvan, M. TabICLv2: a better, faster, scalable, and open tabular foundation model. Preprint at <https://arxiv.org/abs/2602.11139> (2026).
- [9] Bergstra, J. & Bengio, Y. Random search for hyper-parameter optimization. *Journal of Machine Learning Research* **13**, 281–305 (2012).
- [10] Cawley, G. C. & Talbot, N. L. C. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research* **11**, 2079–2107 (2010).
- [11] Riseborough, D., Shiklomanov, N., Etzelmüller, B., Gruber, S. & Marchenko, S. Recent advances in permafrost modelling. *Permafrost and Periglacial Processes* **19**, 137–156 (2008).
- [12] Klene, A. E., Nelson, F. E., Shiklomanov, N. I. & Hinkel, K. M. The n-factor in natural landscapes: variability of air and soil-surface temperatures, Kuparuk River Basin, Alaska, U.S.A. *Arctic, Antarctic, and Alpine Research* **33**, 140–148 (2001).
- [13] Nelson, F. E. *et al.* Estimating active-layer thickness over a large region: Kuparuk River basin, Alaska, U.S.A. *Arct. Alp. Res.* **29**, 367–378 (1997).
- [14] Shiklomanov, N. I. & Nelson, F. E. Active-layer mapping at regional scales: a 13-year spatial time series for the Kuparuk region, north-central Alaska. *Permafrost and Periglacial Processes* **13**, 219–230 (2002).
- [15] Romanovsky, V. E. & Osterkamp, T. E. Thawing of the active layer on the coastal plain of the Alaskan Arctic. *Permafrost and Periglacial Processes* **8**, 1–22 (1997).